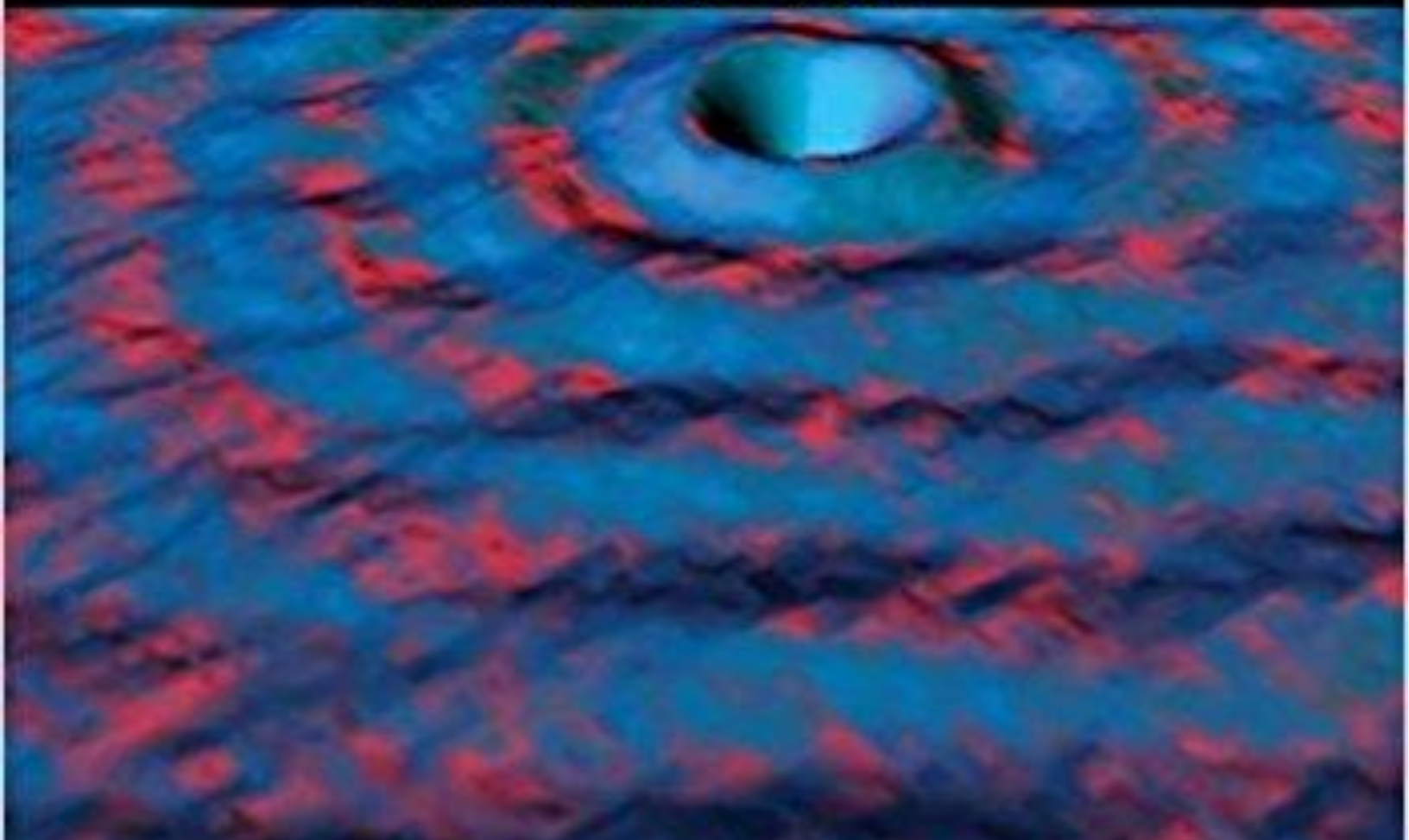


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B.Sc. Physics
WAVES and
OSCILLATIONS

Notes (HRK)



by MUHAMMAD ALI MALIK

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Chapter # 15: Oscillations

Each day we encounter many kinds of oscillatory motion. Common examples include the swinging pendulum of a clock and vibrating guitar string. Examples on the microscopic scale are vibrating atoms in quartz crystal of a wristwatch and vibrating molecules of air that transmit sound waves. The above cases are mechanical oscillations. We are also familiar with electromagnetic oscillations, such as electrons surging back and forth in circuits that are responsible for transmitting and receiving radio or TV signals.

One common feature of all these systems is the mathematical formulation used to describe their oscillations. In all cases, the oscillating quantity, whether it be the displacement of a particle or the magnitude of an electric field, can be described in terms of sine or cosine functions.

In this chapter we concentrate on mechanical oscillations and their description.

15.1 The Simple Harmonic Oscillator (SHO)

“An oscillator whose acceleration is directly proportional to the displacement and acceleration is directed towards the mean position is called simple harmonic oscillator.”

Derivation of the Equation of Simple Harmonic Oscillator

Consider an oscillating system consisting of a particle of mass m attached to a spring of spring constant ' k ' that is free to move over a frictionless horizontal surface is subjected to restoring force F such that:

$$F \propto -x$$

$$F = -kx \quad \text{--- (1)}$$

where x is the displacement of the particle from its equilibrium position. k is also called the stiffness factor and its value depends on the nature of spring. It is large for a hard spring and small for a soft spring.

From 2nd law of motion

$$F = ma$$

$$\Rightarrow F = m\ddot{x} \quad \text{--- (2)} \quad \because a = \ddot{x}$$

From (1) and (2), we have:

$$-kx = m\ddot{x}$$

$$\Rightarrow \ddot{x} = -\frac{k}{m}x$$

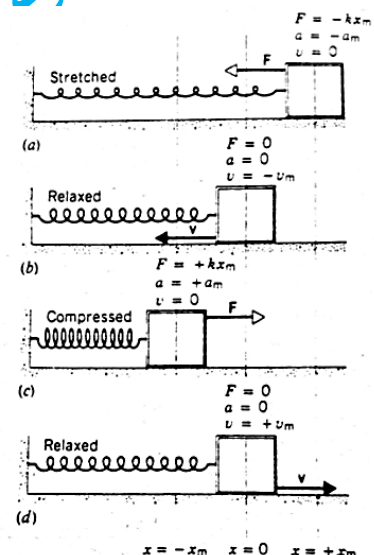
$$\Rightarrow \ddot{x} + \frac{k}{m}x = 0 \quad \because \ddot{x} = \frac{d^2x}{dt^2}$$

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

Putting $\frac{k}{m} = \omega^2$ we get

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \quad \text{--- (3)}$$

This is the equation of motion of simple harmonic oscillator. This is second order differential equation. This equation gives a relation between x and its 2nd derivative $\ddot{x}$.



A simple harmonic oscillator, consisting of a spring acting on a body that slides on a frictionless horizontal surface. In (a), the spring is stretched so that the body has its maximum displacement from equilibrium. In (c) the spring is fully compressed. In (b) and (d), the body is passing through equilibrium with its maximum speed, and the spring is relaxed.

Solution of Equation of Simple Harmonic Oscillator

The motion of simple harmonic oscillator is described by formula:

$$x'' + \omega^2 x = 0$$

Multiplying both sides by $\dot{x}$ we get

$$\dot{x}\ddot{x} + \omega^2 x\dot{x} = 0$$

Integrating both sides with respect to time t , we have:

$$\int \dot{x}\ddot{x} dt + \omega^2 \int x\dot{x} dt = \int 0 dt$$

$$\Rightarrow \frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2 = \text{constant}$$

Say constant = B

$$\Rightarrow \frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2 = B$$

$$\Rightarrow \dot{x}^2 + \omega^2 x^2 = 2B$$

$$\Rightarrow \dot{x}^2 = 2B - \omega^2 x^2$$

$$\Rightarrow \dot{x}^2 = \omega^2 \left(\frac{2B}{\omega^2} - x^2 \right)$$

$$\Rightarrow \dot{x} = \omega \sqrt{\frac{2B}{\omega^2} - x^2}$$

$$\Rightarrow \frac{dx}{dt} = \omega \sqrt{\frac{2B}{\omega^2} - x^2}$$

$$\Rightarrow \frac{dx}{\sqrt{\frac{2B}{\omega^2} - x^2}} = \omega dt$$

Integrating both sides, we have:

$$\int \frac{dx}{\sqrt{\frac{2B}{\omega^2} - x^2}} = \int \omega dt$$

$$\Rightarrow \sin^{-1} \left(\frac{x}{\sqrt{2B/\omega^2}} \right) = \omega t + c$$

At $t = 0, x = x_0$

$$c = \sin^{-1} \left(\frac{x_0}{\sqrt{2B/\omega^2}} \right)$$

So the above expressions becomes

$$\sin^{-1} \frac{x}{\sqrt{2B/\omega^2}} = \omega t + \sin^{-1} \left(\frac{x_0}{\sqrt{2B/\omega^2}} \right)$$

$$\text{Let } \sin^{-1} \left(\frac{x_0}{\sqrt{2B/\omega^2}} \right) = \varphi$$

$$\sin^{-1} \frac{x}{\sqrt{2B/\omega^2}} = (\omega t + \varphi)$$

$$\Rightarrow \frac{x}{\sqrt{2B/\omega^2}} = \sin(\omega t + \varphi)$$

$$\Rightarrow x = \sqrt{2B/\omega^2} \sin(\omega t + \varphi)$$

$$\text{Let } \sqrt{\frac{2B}{\omega^2}} = x_m$$

$$x = x_m \sin(\omega t + \varphi) \text{ --- (4)}$$

This is the solution of differential equation of simple harmonic oscillator. This solution can also be put in the form as:

$$x = x_m \cos(\omega t + \varphi) \text{ --- (5)}$$

Equation (5) is also solution of the equation of simple harmonic oscillator. So we find that equation (4) and (5) both are the solutions of equation (3) because both the solution satisfy equation (3) as shown below:

If $x = x_m \sin(\omega t + \varphi)$ is solution, then $\dot{x} = \omega x_m \cos(\omega t + \varphi)$ $\ddot{x} = -\omega^2 x_m \sin(\omega t + \varphi) \Rightarrow \ddot{x} = -\omega^2 x$ Now take equation (3) $\ddot{x} + \omega^2 x = 0$ Consider L.H.S = $\ddot{x} + \omega^2 x$ $= -\omega^2 x + \omega^2 x = 0$ $= R.H.S$ So $x = x_m \sin(\omega t + \varphi)$ is solution of equation of SHO	If $x = x_m \cos(\omega t + \varphi)$ is solution, then $\dot{x} = -\omega x_m \sin(\omega t + \varphi)$ $\ddot{x} = -\omega^2 x_m \cos(\omega t + \varphi) \Rightarrow \ddot{x} = -\omega^2 x$ Now take equation (3) $\ddot{x} + \omega^2 x = 0$ L.H.S = $\ddot{x} + \omega^2 x$ $= -\omega^2 x + \omega^2 x = 0$ $= R.H.S$ So $x = x_m \cos(\omega t + \varphi)$ is solution of equation of SHO
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Various Parameters of Oscillator:

- (i) **Amplitude:** It is the maximum displacement from the mean position. The displacement will be maximum when $\sin(\omega t + \varphi) = \pm 1$ and $\cos(\omega t + \varphi) = \pm 1$. Hence,

$$\text{Maximum Displacement} = \pm x_m$$

So the amplitude of the oscillator is $+x_m$ and $-x_m$. Now,

$$x_m = \sqrt{\frac{2B}{\omega^2}}$$

$$\Rightarrow x_m = \frac{\sqrt{2B}}{\omega}$$

- (ii) **Time Period:** Time taken to complete one vibration is called time period. As ω has the dimensions of angular velocity, so time period T is given by:

$$T = \frac{2\pi}{\omega}$$

$$\Rightarrow T = \frac{2\pi}{\left(\sqrt{\frac{k}{m}}\right)} \quad \because \omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{k}}$$

This shows that time period depends only on the mass of oscillator and spring constant and does not depend on the amplitude.

(iii) The Displacement of SHO is periodic as the function of its Time: In equation (3) if we increase the time t by $T = \frac{2\pi}{\omega}$ then the value of x remain unchanged. So

$$[\text{Displacement at time } t] = [\text{Displacement at time } t + T] \Rightarrow x_t = x_{t+T}$$

Consider $R.H.S = x_{t+T}$

$$\begin{aligned} &= x_m \cos[\omega(t) + \varphi] \\ &= x_m \cos[\omega(t + T) + \varphi] \\ &= x_m \cos\left[\omega\left(t + \frac{2\pi}{\omega}\right) + \varphi\right] \\ &= x_m \cos[\omega t + 2\pi + \varphi] \\ &= x_m \cos(\omega t + \varphi) \\ &= x_t = L.H.S \end{aligned}$$

Hence proved that Displacement of SHO is periodic as function of time.

(iv) Frequency: The reciprocal of time period is called frequency. It gives the number of vibrations per second. Thus frequency is given by

$$\begin{aligned} f &= \frac{1}{T} \\ \Rightarrow f &= \frac{\omega}{2\pi} & \because T = \frac{2\pi}{\omega} \Rightarrow \frac{1}{T} = \frac{\omega}{2\pi} \\ \Rightarrow f &= \frac{1}{2\pi} \sqrt{\frac{k}{m}} & \because \omega = \sqrt{\frac{k}{m}} \end{aligned}$$

This shows that frequency is independent of amplitude.

(v) Phase: The term $(\omega t + \varphi)$ is called the phase of oscillator. It determines the displacement as well as the direction of motion for SHO.

Important Note:

At $t = 0$ the value of $\text{phase} = \varphi$. So φ is called initial phase. It gives the position of start of oscillator.

Problem: A SHM takes 12 s to undergo 5 complete vibrations. Find (a) Time Period (b) Frequency and (c) Angular Frequency of oscillator.

Given Data: Total Time $t = 12$ s, Number of Vibrations $n = 5$

To Determine: (a) Time Period $T = ?$, (b) Frequency $f = ?$, (c) Angular Frequency $\omega = ?$

Calculations: (a) $T = \frac{t}{n} = \frac{12}{5} = 2.4$ s

(b) $f = \frac{1}{T} = \frac{1}{2.4} = 0.42$ s

(c) $\omega = 2\pi f = 2 \times 3.14 \times 0.42 = 2.62 \frac{\text{rad}}{\text{s}}$

Problem: A load of 15 g elongates the spring by 2 cm. Find spring constant.

Given Data: Mass $m = 15\text{ g} = 0.015\text{ kg}$, Elongation $x = 2\text{ cm} = 0.02\text{ m}$
To Determine: Spring constant $k = ?$
Calculations: For Present Case: $F = kx \Rightarrow mg = kx \Rightarrow k = \frac{mg}{x} = \frac{0.015 \times 9.8}{0.02} = 7.35 \frac{\text{N}}{\text{m}}$

Problem. A block weighing 4.0 kg extends a spring by 0.16 m from its un-stretched position. The block is removed and a 0.50 kg body is hung from the same spring. If the spring is now stretched and then released, what is the period of vibration?

Given Data: Mass of First Object $m_1 = 4\text{ kg}$, Elongation $x = 0.16\text{ m}$ Mass of Second Object $m_2 = 0.50\text{ kg}$
To Determine: Time Period $T = ?$ For time period, we have to find the Spring Constant k
Calculations: <u>By Hook's Law</u> $F = kx \Rightarrow m_1g = kx \Rightarrow k = \frac{m_1g}{x} = \frac{4 \times 9.8}{0.16} = 245\text{ Nm}^{-1}$ As $T = 2\pi \sqrt{\frac{m_2}{k}} = 2 \times 3.14 \times \sqrt{\frac{0.50}{245}} = 0.28\text{ s}$

Problem. Find the amplitude, frequency and period of an object vibrating at the end of a spring, if the equation from its position, as a function of time is $x = 0.25 \cos\left(\frac{\pi}{8}\right)t$. What is the displacement of the object after 2.0 s?

Given Data: Equation of Motion of body executing SHM $x = 0.25 \cos\left(\frac{\pi}{8}\right)t$ — (1)
To Determine: Amplitude $x_0 = ?$, Frequency $f = ?$, Time Period $T = ?$ Displacement $x = ?$ at $t = 2\text{ s}$
Calculation: General Equation of Motion of body executing SHM $x = x_0 \cos \omega t$ — (2) (i) <u>Comparing (1) and (2):</u> $x_0 = 0.25$ $\omega = \frac{\pi}{8} \Rightarrow 2\pi f = \frac{\pi}{8} \Rightarrow f = \frac{1}{16}\text{ Hz}$ $T = \frac{1}{f} = \frac{1}{(1/16)} = 16\text{ s}$ (ii) <u>Putting $t = 2\text{ s}$ in equation (1):</u> $x = 0.25 \cos\left[\left(\frac{\pi}{8}\right) \times 2\right] = 0.18\text{ m}$

Sample Problem # 1. A spring hangs vertically when a body of mass $M = 1.65\text{ kg}$ is suspended from it. Its length increases by 7.33 cm . The spring is then mounted horizontally and a block of mass $m = 2.43\text{ kg}$ is attached to the spring. The block is free to slide along a frictionless horizontal surface.
(a) What is the force constant k of the spring? (b) How much horizontal force is required to stretch the spring by a distance of 11.6 cm ? (c) When the block is displaced a distance of 11.6 cm and released, with what period will it oscillate?

Given Data: Mass of First Object $M = 1.65\text{ kg}$, Elongation $x_1 = 7.33\text{ cm}$, Mass of Second Object $m = 2.43\text{ kg}$, Displacement as Result of Horizontal Force $x_2 = 11.6\text{ cm}$
To Determine: (a) Spring Constant $k = ?$, (b) Horizontal Force (for Displacement of 11.6 cm) $F = ?$ (c) Time Period $T = ?$

Calculations: (a) When suspended body is in equilibrium the spring force kx balances weight mg

$$kx = mg \Rightarrow k = \frac{mg}{x} = \frac{1.65 \times 9.8}{0.0733} = 221 \text{ N/m}$$

(b) As by Hook's Law: $F = kx = 221 \times 0.116 = 25.6 \text{ N}$

(c) As $T = 2\pi \sqrt{\frac{m}{k}} = 2 \times 3.14 \sqrt{\frac{2.43}{221}} = 0.659 \text{ s}$

Problem: A 7 kg mass is hung from the bottom of vertical spring. The mass is set to vertical oscillation with period of 2.6 s. Find force constant of spring.

Given Data: Mass $m = 7 \text{ kg}$, Time Period $T = 2.6 \text{ s}$

To Determine: Force Constant $k = ?$,

Calculation: As $T = 2\pi \sqrt{\frac{m}{k}} \Rightarrow T^2 = 4\pi^2 \frac{m}{k} \Rightarrow k = 4\pi^2 \frac{m}{T^2} = 4 \times (3.14)^2 \times \frac{7}{(2.6)^2} = 40.9 \frac{\text{N}}{\text{m}}$

Relation between Displacement and Velocity:

The displacement of simple harmonic oscillator is given by:

$$x = x_m \cos(\omega t + \phi)$$

Velocity is the time derivative of displacement, therefore,

$$v = \frac{dx}{dt}$$

$$\Rightarrow v = \frac{d}{dt} [x_m \cos(\omega t + \phi)]$$

$$\Rightarrow v = -\omega x_m \sin(\omega t + \phi)$$

$$\Rightarrow v = -\omega x_m \sqrt{1 - \cos^2(\omega t + \phi)} \quad \because \sin(\omega t + \phi) = \sqrt{1 - \cos^2(\omega t + \phi)}$$

$$\Rightarrow v = -\omega x_m \sqrt{1 - \frac{x^2}{x_m^2}} \quad \because \frac{x}{x_m} = \cos(\omega t + \phi)$$

$$\Rightarrow v = -\omega x_m \sqrt{\frac{x_m^2 - x^2}{x_m^2}} = -\omega x_m \left(\frac{\sqrt{x_m^2 - x^2}}{x_m} \right)$$

$$\Rightarrow v = -\omega \sqrt{x_m^2 - x^2}$$

This is the relation between displacement and velocity of SHO. From this relation we find that velocity of SHO is maximum when displacement x is minimum and velocity is minimum when displacement x is maximum.

Relation between Displacement and Acceleration of SHO

As acceleration is described as the time derivative of velocity, so

$$a = \frac{dv}{dt}$$

$$\Rightarrow a = \frac{d}{dt} [-\omega x_m \sin(\omega t + \phi)] \quad \because \text{Velocity of SHO } v = -\omega x_m \sin(\omega t + \phi)$$

$$\Rightarrow a = -\omega^2 x_m \cos(\omega t + \phi)$$

$$\Rightarrow a = -\omega^2 x$$

$$\therefore x = x_m \cos(\omega t + \varphi)$$

This is the relation between displacement and acceleration. From this relation we find that:

- Acceleration is maximum when displacement is maximum but direction of acceleration is opposite to the displacement. Thus acceleration is maximum at extreme position because elastic restoring force is maximum at extreme position.
- Acceleration is zero when displacement is zero. The acceleration is zero at the mean position because elastic restoring force is zero at mean position.

Problem. A load of 15.0 g elongates a spring by 2.0 cm. If body of mass 294 g is attached to the spring and is set into vibration with an amplitude of 10.0 cm, what will be the (i) period (ii) Spring Constant (iii) maximum speed of its vibration.

Given Data: Mass of First Object $m_1 = 15 \text{ g} = 0.015 \text{ kg}$, Elongation $x = 2 \text{ cm} = 0.02 \text{ m}$ Mass of Second Object $m_2 = 294 \text{ g} = 0.294 \text{ kg}$, Amplitude $x_0 = 10 \text{ cm} = 0.10 \text{ m}$
To Determine: (i) Period $T = ?$ (ii) Spring Constant $k = ?$ (iii) Maximum Speed $v_0 = ?$
Calculations: By Hook's Law (ii) $F = kx \Rightarrow m_1 g = kx \Rightarrow k = \frac{m_1 g}{x} = \frac{0.015 \times 9.8}{0.02} = 7.35 \text{ Nm}^{-1}$
(i) As $T = 2\pi \sqrt{\frac{m_2}{k}} = 2 \times 3.14 \times \sqrt{\frac{0.294}{7.35}} = 1.26 \text{ s}$
(iii) As Maximum Speed $v_0 = x_0 \sqrt{\frac{k}{m_2}} = 0.10 \times \sqrt{\frac{7.35}{0.294}} = 0.05 \text{ ms}^{-1}$

Problem: A spring, whose spring constant is 80 Nm^{-1} vertically supports a mass of 1.0 kg in the rest position. Find the distance by which the mass must be pulled down, so that on being released, it may pass the mean position with a velocity of 1.0 ms^{-1} .

Given Data: Spring Constant $k = 80 \text{ Nm}^{-1}$, Mass $m = 1.0 \text{ kg}$, Maximum Speed $v_0 = 1.0 \text{ ms}^{-1}$
To Determine: Amplitude $x_0 = ?$
Calculation: As $v_0 = x_0 \sqrt{\frac{k}{m}} \Rightarrow x_0 = \frac{v_0}{\left(\sqrt{\frac{k}{m}}\right)} = v_0 \sqrt{\frac{m}{k}} = 1 \times \sqrt{\frac{1}{80}} = 0.11 \text{ m}$

Problem. A block of mass 1.6 kg is attached to a spring with spring constant 1000 Nm^{-1} . The spring is compressed through a distance of 2.0 cm and the block is released from rest. Calculate the velocity of the block as it passes through equilibrium position, $x = 0$, if the surface is frictionless.

Given Data: Spring Constant $k = 1000 \text{ Nm}^{-1}$, Mass $m = 1.6 \text{ kg}$, Amplitude $x_0 = 2 \text{ cm} = 0.02 \text{ m}$
To Determine: Maximum Velocity $v_0 = ?$,
Calculation: $v_0 = x_0 \sqrt{\frac{k}{m}} = 0.02 \times \sqrt{\frac{1000}{1.6}} = 0.5 \text{ ms}^{-1}$

Problem: A particle executes SHM with amplitude of 3 cm. At what distance from mid-point of its position, does its speed equal to one half of its maximum speed.

Given Data: Amplitude $x_m = 3 \text{ cm} = 0.03 \text{ m}$,
To Determine: Displacement $x = ?$ When Velocity $v = \frac{v_{\max}}{2}$

Calculations: As $v = \omega \sqrt{(x_m)^2 - (x)^2} \Rightarrow \frac{v_{max}}{2} = \sqrt{\frac{k}{m}} \times \sqrt{(x_m)^2 - (x)^2}$

$$\Rightarrow \frac{1}{2} \sqrt{\frac{k}{m}} x_m = \sqrt{\frac{k}{m}} \times \sqrt{(x_m)^2 - (x)^2} \Rightarrow \frac{1}{2} x_m = \sqrt{(x_m)^2 - (x)^2} \Rightarrow \frac{(x_m)^2}{4} = (x_m)^2 - (x)^2$$

$$\Rightarrow (x)^2 = (x_m)^2 - \frac{(x_m)^2}{4} \Rightarrow x = \sqrt{\frac{3(x_m)^2}{4}} = \sqrt{\frac{3(0.03)^2}{4}} = 0.026 \text{ m}$$

Energy Consideration in Simple Harmonic Motion:

In simple harmonic motion, the total energy of the oscillator remains constant.

The energy of oscillator at any displacement “x” from the mean position is partly K.E and partly P.E.

The instantaneous displacement of SHO is described by formula:

$$x = x_m \cos(\omega t + \varphi)$$

Potential Energy of SHO:

The work done in stretching or compressing a spring will store in the form of potential energy. The instantaneous potential energy at any instant is given by:

$$U = \int_0^x F dx = \int_0^x kx dx = k \int_0^x x dx$$

$$\Rightarrow U = k \left[\frac{x^2}{2} \right]_0^x = \frac{1}{2} kx^2$$

$$\Rightarrow U = \frac{1}{2} k [x_m^2 \cos^2(\omega t + \varphi)] \quad \because x = x_m \cos(\omega t + \varphi)$$

$$\Rightarrow U = \frac{1}{2} kx_m^2 \cos^2(\omega t + \varphi) \quad \text{----- (1)}$$

So P.E varies with time and has maximum value.

$$U_{max} = \frac{1}{2} kx_m^2 \quad \text{----- (4)} \quad \because \text{Maximum Value of } \cos(\omega t + \varphi) = 1$$

Kinetic Energy of SHO

The kinetic energy at any instant is given by

$$K.E = \frac{1}{2} mv^2$$

$$\Rightarrow K.E = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2$$

$$\Rightarrow K.E = \frac{1}{2} m \left[\frac{d}{dt} \{x_m \cos(\omega t + \varphi)\} \right]^2 \quad \because x = x_m \cos(\omega t + \varphi)$$

$$\Rightarrow K.E = \frac{1}{2} m [(-\omega x_m) \sin(\omega t + \varphi)]^2 \quad \because \frac{d}{dt} \{x_m \cos(\omega t + \varphi)\} = -\omega x_m \sin(\omega t + \varphi)$$

$$\Rightarrow K.E = \frac{1}{2} m \omega^2 x_m^2 \sin^2(\omega t + \varphi)$$

$$\Rightarrow K.E = \frac{1}{2} kx_m^2 \sin^2(\omega t + \varphi) \quad \text{----- (2)} \quad \because \omega^2 = \frac{k}{m} \quad \& \quad m\omega^2 =$$

So kinetic energy varies with time and has maximum value.

$$K.E_{max} = \frac{1}{2} kx_m^2 \quad \text{----- (B)} \quad \because \text{Maximum Value of } \sin(\omega t + \varphi)_{max} = 1$$

Total Instantaneous Energy of SHO

Total mechanical energy E of SHO at any displacement x is given by adding (1) and (2):

$$E = K.E + P.E$$

$$\Rightarrow E = \frac{1}{2} kx_m^2 \cos^2(\omega t + \phi) + \frac{1}{2} kx_m^2 \sin^2(\omega t + \phi) = \frac{1}{2} kx_m^2 [\sin^2(\omega t + \phi) + \cos^2(\omega t + \phi)]$$

$$\Rightarrow E = \frac{1}{2} kx_m^2 \quad \text{--- (C)}$$

From equations A, B & C, we find that:

$$(P.E)_{\max} = (K.E)_{\max} = E$$

Hence total energy remains conserved.

- At the mean position the energy is entirely K.E and P.E is zero.
- At the extreme position the energy is entirely potential and K.E is zero.
- At any other position the energy is partly kinetic and partly potential.

From equations A, B and C, We also note that energy of oscillator is directly proportional to the square of amplitude of motion.

Problem: A 0.5 kg cube connected to a spring for which force constant is 20 N/m oscillate on a horizontal frictionless track. (a) Calculate total energy of system and maximum speed of cube if amplitude of motion is 3 cm. (b) Find velocity, Determine instantaneous P.E and K.E when displacement is 2 cm.

Given Data: Mass $m = 0.5 \text{ kg}$, Force Constant $k = 20 \text{ N/m}$, Amplitude $x_m = 3 \text{ cm} = 0.03 \text{ m}$
To Determine: (a) Total Energy $E = ?$, Maximum Speed $v_{\max} = ?$ (b) Instantaneous Velocity $v = ?$, $P.E = ?$, $K.E = ?$ when $x = 2 \text{ cm} = 0.02 \text{ m}$
Calculations: (a) $E = \frac{1}{2} kx_m^2 = 0.5 \times 20 \times (0.03)^2 = 9 \times 10^{-3} \text{ J}$ $v_{\max} = \omega x_m = \sqrt{\frac{k}{m}} x_m = \sqrt{\frac{20}{0.5}} \times 0.03 = 0.1897 \frac{\text{m}}{\text{s}}$ (b) As $v = \omega \sqrt{(x_m)^2 - (x)^2} = \sqrt{\frac{k}{m}} \times \sqrt{(x_m)^2 - (x)^2} = \sqrt{\frac{20}{0.5}} \times \sqrt{(0.03)^2 - (0.02)^2} = 0.1414 \frac{\text{m}}{\text{s}}$ $P.E = \frac{1}{2} kx^2 = 0.5 \times 20 \times (0.02)^2 = 0.04 \text{ J}$ $K.E = \frac{1}{2} mv^2 = 0.5 \times 0.5 \times (0.1414)^2 = 0.005 \text{ J}$

Problem: A mass spring system oscillates with amplitude 3.5 cm. if the spring constant is 250 N/m, and mass attached to the spring is 0.5 kg, find (a) mechanical energy of the system, (b) maximum speed of system and (c) maximum acceleration of system.

Given Data: Mass $m = 0.5 \text{ kg}$, Force Constant $k = 250 \text{ N/m}$, Amplitude $x_m = 3.5 \text{ cm} = 0.035 \text{ m}$
To Determine: (a) Mechanical Energy $E = ?$, (b) Max Speed $v_{\max} = ?$ (c) Max Acceleration $a_{\max} = ?$
Calculations: (a) $E = \frac{1}{2} kx_m^2 = 0.5 \times 250 \times (0.035)^2 = 0.153 \text{ J}$ (b) $v_{\max} = \omega x_m = \sqrt{\frac{k}{m}} x_m = \sqrt{\frac{250}{0.5}} \times 0.035 = 0.7826 \frac{\text{m}}{\text{s}}$ (c) $a_{\max} = \omega^2 x_m = \left(\frac{k}{m}\right) x_m = \frac{250}{0.5} \times 0.035 = 17.5 \frac{\text{m}}{\text{s}^2}$

Show that $\langle K.E \rangle = \langle P.E \rangle = \frac{1}{2}E$, where E is total energy of SHO

As kinetic energy at any instant is described by formula:

$$K.E = \frac{1}{2}kx_m^2 \sin^2(\omega t + \varphi)$$

And instantaneous Potential energy is given by:

$$P.E = \frac{1}{2}kx_m^2 \cos^2(\omega t + \varphi)$$

Now the average K.E of SHO will be:

$$\begin{aligned} \langle K.E \rangle &= \left\langle \frac{1}{2}kx_m^2 \sin^2(\omega t + \varphi) \right\rangle \\ \Rightarrow \langle K.E \rangle &= \frac{1}{2}kx_m^2 \langle \sin^2(\omega t + \varphi) \rangle \quad \because \langle \sin^2(\omega t + \varphi) \rangle = \frac{1}{2} \\ \Rightarrow \langle K.E \rangle &= \frac{1}{2}kx_m^2 \cdot \left(\frac{1}{2}\right) \\ \Rightarrow \langle K.E \rangle &= \frac{1}{4}kx_m^2 \quad \text{----- (a)} \end{aligned}$$

The average Potential Energy of SHO will be:

$$\begin{aligned} \langle P.E \rangle &= \left\langle \frac{1}{2}kx_m^2 \cos^2(\omega t + \varphi) \right\rangle \\ \Rightarrow \langle P.E \rangle &= \frac{1}{2}kx_m^2 \langle \cos^2(\omega t + \varphi) \rangle \quad \because \langle \cos^2(\omega t + \varphi) \rangle = \frac{1}{2} \\ \Rightarrow \langle P.E \rangle &= \frac{1}{2}kx_m^2 \cdot \frac{1}{2} \\ \Rightarrow \langle P.E \rangle &= \frac{1}{4}kx_m^2 \quad \text{----- (b)} \end{aligned}$$

Now total energy of oscillator is given by

$$\begin{aligned} E &= K.E + P.E \\ \Rightarrow E &= \frac{1}{2}kx_m^2 \sin^2(\omega t + \varphi) + \frac{1}{2}kx_m^2 \cos^2(\omega t + \varphi) \\ \Rightarrow E &= \frac{1}{2}kx_m^2 [\sin^2(\omega t + \varphi) + \cos^2(\omega t + \varphi)] \\ \Rightarrow E &= \frac{1}{2}kx_m^2 \quad \text{----- (c)} \end{aligned}$$

From (a) and (b)

$$\begin{aligned} \langle K.E \rangle &= \langle P.E \rangle = \frac{1}{4}kx_m^2 \\ \Rightarrow \langle K.E \rangle &= \langle P.E \rangle = \frac{1}{2} \left(\frac{1}{2}kx_m^2 \right) \\ \Rightarrow \langle K.E \rangle &= \langle P.E \rangle = \frac{1}{2}E \quad \because \text{From equation (c) } \frac{1}{2}kx_m^2 = E \end{aligned}$$

Question. From the expression $E = K.E + P.E$ obtain the expression for velocity for SHO.

The kinetic energy of SHO is described by formula:

$$K.E = \frac{1}{2}mv^2$$

The instantaneous P.E. of SHO is given by:

$$P.E = \frac{1}{2} kx^2$$

The total energy of simple harmonic oscillator is described as:

$$E = \frac{1}{2} kx_m^2$$

Total energy of SHO is related to instantaneous P.E and instantaneous K.E as:

$$E = K.E + P.E$$

$$\Rightarrow \frac{1}{2} kx_m^2 = \frac{1}{2} mv^2 + \frac{1}{2} kx^2 \Rightarrow \frac{1}{2} kx_m^2 - \frac{1}{2} kx^2 = \frac{1}{2} mv^2$$

$$\Rightarrow mv^2 = kx_m^2 - kx^2 \Rightarrow mv^2 = k(x_m^2 - x^2)$$

$$\Rightarrow v^2 = \frac{k}{m} (x_m^2 - x^2) \Rightarrow v = \pm \sqrt{\frac{k}{m} (x_m^2 - x^2)}$$

$$\Rightarrow v = \pm \sqrt{\omega^2 (x_m^2 - x^2)} \quad \because \text{For Horizontal Mass Spring System } \frac{k}{m} = \omega^2$$

$$\Rightarrow v = \pm \omega \sqrt{(x_m^2 - x^2)}$$

This is the expression for velocity of SHO. From this expression we find that velocity is maximum at the mean position and velocity is zero at extreme position.

Problem. A block of mass 4.0 kg is dropped from a height of 0.80 m on to a spring of spring constant $k = 1960 \text{ Nm}^{-1}$. Find the maximum distance through which the spring will be compressed.

Given Data: Spring Constant $k = 1960 \text{ Nm}^{-1}$, Mass $m = 4.0 \text{ kg}$, Height $h = 0.80 \text{ m}$

To Determine: Maximum Distance $x_0 = ?$

Calculation: For Present Case: Elastic P.E. = Gravitational P.E. $\Rightarrow \frac{1}{2} kx_0^2 = mgh$

$$\Rightarrow \frac{1}{2} kx_0^2 = mgh \Rightarrow x_0 = \frac{2mgh}{k} = \frac{2 \times 4 \times 9.8 \times 0.80}{1960} = 0.42 \text{ m}$$

Problem. An 8.0 kg body executes SHM with amplitude 30 cm. The restoring force is 60 N when the displacement is 30 cm. Find (i) Period (ii) Acceleration, speed, kinetic energy and potential energy when the displacement is 12 cm.

Given Data: Restoring Force $|F_r| = 60 \text{ N}$, when Displacement $x_1 = 30 \text{ cm} = 0.3 \text{ m}$

Mass $m = 8 \text{ kg}$, Amplitude $x_0 = 30 \text{ cm} = 0.3 \text{ m}$

To Determine: (i) Time Period $T = ?$

(ii) Acceleration $a = ?$, Speed $v = ?$, K.E. = ?, P.E. = ? when displacement $x = 12 \text{ cm} = 0.12 \text{ m}$

Calculations: (i) As Restoring Force $|F_r| = kx_1 \Rightarrow k = \frac{|F_r|}{x_1} = \frac{60}{0.3} = 200 \text{ Nm}^{-1}$

- Time Period $T = 2\pi \sqrt{\frac{m}{k}} = 2 \times 3.14 \times \sqrt{\frac{8}{200}} = 1.3 \text{ s}$
- (ii) Acceleration $a = -\frac{k}{m}x = -\frac{200}{8} \times 0.12 = -3 \text{ ms}^{-2}$
- Speed $v = \sqrt{\frac{k}{m}(x_0^2 - x^2)} = \sqrt{\frac{200}{8}[(0.3)^2 - (0.12)^2]} = 1.33 \text{ ms}^{-1}$
- K.E. $= \frac{1}{2} k(x_0^2 - x^2) = \frac{1}{2} \times 200 \times [(0.3)^2 - (0.12)^2] = 7.6 \text{ J}$
- P.E. $= \frac{1}{2} kx^2 = \frac{1}{2} \times 200 \times (0.12)^2 = 1.44 \text{ J}$

Problem. A car of mass 1300 kg is constructed using a frame supported by four springs. Each spring has a spring constant 20000 Nm^{-1} . If two people are riding in the car have a combined mass of 160 kg, find the frequency of vibration of the car, when it is driven over a pot hole in the road. Assume the weight is evenly distributed.

Given Data: Mass of Car $m_1 = 1300 \text{ kg}$, Mass of Two Persons $m_2 = 160 \text{ kg}$, Total Mass $m = m_1 + m_2 = 1300 + 1600 = 1460 \text{ kg}$ Spring Constant of One Spring $k' = 20000 \text{ Nm}^{-1}$, Total Spring Constant $k = 4k' = 80000 \text{ Nm}^{-1}$
To Determine: Frequency $f = ?$,
Calculation: As $f = \frac{1}{T} = \frac{1}{2\pi\sqrt{\frac{m}{k}}} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2 \times 3.14} \times \sqrt{\frac{80000}{1460}} = 1.18 \text{ Hz}$

Sample Problem 2: The block spring combination is stretched in the positive x direction at distance of 11.6 cm from equilibrium and released. (a) What is the total energy stored in the spring? (b) What is the maximum velocity of block? (c) What is the maximum acceleration? (c) If the block is released at $t = 0$, what are its position, velocity and acceleration at $t = 0.125 \text{ s}$? Given $k = 221 \text{ N/m}$ for the spring and mass attached to the spring $m = 2.43 \text{ kg}$.

Given Data: Amplitude $x_m = 11.6 \text{ cm} = 0.116 \text{ m}$, Spring Constant $k = 221 \text{ N/m}$, Mass $m = 2.43 \text{ kg}$
To Determine: (a) Total Energy $E = ?$ (b) Maximum Velocity $v_{max} = ?$, (c) Maximum Acceleration $a_{max} = ?$ (c) Position $x = ?$, Velocity $v = ?$, Acceleration $a = ?$ At $t = 0.125 \text{ s}$
Calculations: (a) $E = \frac{1}{2}kx_m^2 = \frac{1}{2} \times 221 \times (0.116)^2 = 1.49 \text{ J}$ (b) As $(K.E)_{max} = E \Rightarrow \frac{1}{2}mv_{max}^2 = 1.49 \Rightarrow v_{max}^2 = \frac{2 \times 1.49}{m} = \frac{2 \times 1.49}{2.43} = 1.226 \Rightarrow v_{max} = 1.11 \frac{\text{m}}{\text{s}}$ (c) As $a_{max} = \omega^2 x_m = \frac{k}{m} x_m = \frac{221}{2.43} \times 0.116 = 10.55 \frac{\text{m}}{\text{s}^2}$ (d) As $x = x_m \cos \omega t = x_m \cos \left(\sqrt{\frac{k}{m}} t \right) = 0.116 \times \cos \left(\sqrt{\frac{221}{2.43}} \times 0.215 \right) = 0.116 \times \cos(2.05 \text{ rad}) = -0.0534 \text{ m}$ As $v = -\omega x_m \sin \omega t = -\sqrt{\frac{k}{m}} x_m \sin \left(\sqrt{\frac{k}{m}} t \right) = -\sqrt{\frac{221}{2.43}} \times 0.116 \sin \left(\sqrt{\frac{221}{2.43}} \times 0.215 \text{ rad} \right) = -0.981 \frac{\text{m}}{\text{s}}$ As $a = -\omega^2 x = -(9.53)^2 (-0.0534) = 4.85 \frac{\text{m}}{\text{s}^2}$

Sample Problem 3. The block of block-spring system of sample problem 1 is pushed from equilibrium by an external force in the positive x direction. At $t = 0$, when the displacement of the block is $x = 0.0624 \text{ m}$ and its velocity is $v = 0.847 \frac{\text{m}}{\text{s}}$, the external force is removed and the block begins to oscillate. Write an equation for $x(t)$ during the oscillation.

Given Data: From S.P.1 Mass $m = 2.43 \text{ kg}$, Spring Constant $k = 221 \frac{\text{N}}{\text{m}}$, $\omega = 9.53 \frac{\text{rad}}{\text{s}}$ At $t = 0$, Displacement $x = 0.0624 \text{ m}$, Velocity $v = 0.847 \frac{\text{m}}{\text{s}}$
To Determine: Equation for $x(t) = x_m \cos(\omega t + \phi) \quad \dots (1) \quad x_m = ?$, $\phi = ?$
Calculations: As $E = K + U \Rightarrow \frac{1}{2}kx_m^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \Rightarrow kx_m^2 = mv^2 + kx^2 \Rightarrow x_m^2 = \frac{m}{k}v^2 + x^2$

$$\Rightarrow x_m = \sqrt{\frac{m}{k} v^2 + x^2} = \sqrt{\frac{2.43}{221} (0.847)^2 + 0.0624^2} = 0.1085 \text{ m}$$

To find phase ϕ Consider $x(t) = x_m \cos(\omega t + \phi)$, At $t = 0$ $x(0) = x_m \cos(\omega \times 0 + \phi)$

$$\Rightarrow x(0) = x_m \cos \phi \Rightarrow \cos \phi = \frac{x(0)}{x_m} = \frac{0.0624}{0.1085} \Rightarrow \phi = \cos^{-1}(0.5751) = 54.9 \text{ rad}$$

Equation (1) becomes: $x(t) = 0.1085 \cos[(9.53 \times t + 54.9) \text{ rad}]$

Application of SHM

(1) The Torsional Oscillator (Pendulum)

“Torsional oscillator consists of a disk suspended by a wire attached to the center of mass of the disk. The other end of a wire supporting the disk is fixed to a rigid support.”

When the disk is in equilibrium, radial line is drawn from its centre to point P on its rim as shown. If the disk is rotated in a horizontal plane such that line OP moves to the new position OQ, the wire will be twisted.

The twisted wire exerts a restoring torque on the disk which tends to bring the reference line back to its mean position. For small twist the restoring torque is given by

$$T \propto \theta$$

$$\Rightarrow T = -k\theta \text{ --- (1)}$$

where θ is angular displacement and k is constant of proportionality. Its value depends on the nature of suspension wire. It is called torque constant. The negative sign shows that the torque is directed opposite to angular displacement θ . According to angular form of Newton's 2nd law:

$$T = I\alpha \text{ --- (2)}$$

From (1) and (2)

$$-k\theta = I\alpha$$

$$-k\theta = I \frac{d^2\theta}{dt^2} \quad \therefore \alpha = \frac{d^2\theta}{dt^2}$$

$$\frac{d^2\theta}{dt^2} = -\frac{k}{I} \theta \text{ --- (3)}$$

Equation (3) represents angular S.H.M. This equation is similar to linear S.H.M given by:

$$\frac{d^2x}{dt^2} = -\frac{k}{m} x \text{ ... (4)}$$

In equation (3) and (4)

x corresponds to θ .

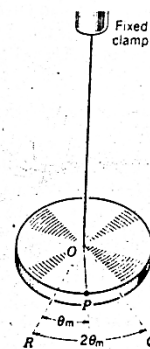
m corresponds to I .

k corresponds to k .

Therefore, the solution of equation (3) can be written as:

$$\theta = \theta_m \cos(\omega t + \phi)$$

where θ_m is maximum angular displacement and ω is angular frequency.



The torsional oscillator. The line drawn from O to P oscillates between OQ and OR, sweeping out an angle $2\theta_m$, where θ_m is the angular amplitude of the motion.

$$\omega = \sqrt{\frac{k}{I}}$$

Time Period. The time period is given by

$$T = 2\pi \sqrt{\frac{I}{k}}$$

The torsional oscillator is also called torsional pendulum.

Uses of Torsional Pendulum

- It is often used for time keeping e.g in the balance wheel of a mechanical watch in which the restoring torque is provided by a spiral hair spring.
- It was used in Cavendish method to find the value of 'G'.

(ii) The Simple Pendulum.

"A simple pendulum consists of a point mass suspended from a frictionless support by a light, inextensible cord (string)."

The point mass which is a small metallic sphere is called bob of the pendulum. When the bob is displaced to one side of its mean position and released, the pendulum begins to oscillate in a vertical plane under the action of gravity. The motion of the pendulum is periodic and oscillatory.

Figure shows a pendulum of length L and mass m in the displaced position. Let x and θ be the linear and angular displacements. The forces acting on the pendulum in the displaced position are

- (a) Weight mg of the bob acting vertically downward and
- (b) The tension T in the string upward.

$mg \cos \theta$ and $mg \sin \theta$ are the rectangular components of weight along the string and $\perp$ to it.

- $mg \cos \theta$ is along the radius L and is called radial component.
- $mg \sin \theta$ is along the tangent to arc and is called tangential component.

As there is no motion along T and $mg \cos \theta$ so they remain equal. So they cancel away. The only force on the pendulum is $mg \sin \theta$ which bring it back and so is the restoring force.

$$F = -mg \sin \theta \quad \text{--- (1)}$$

where negative sign shows that the restoring force F is opposite to the direction of increasing θ . It should be noted that restoring force is proportional to $\sin \theta$. So motion is not S.H.M. For S.H.M θ should be very small ($< 5^\circ$). If θ is quite small then $\sin \theta \approx \theta$ and motion becomes S.H.M.

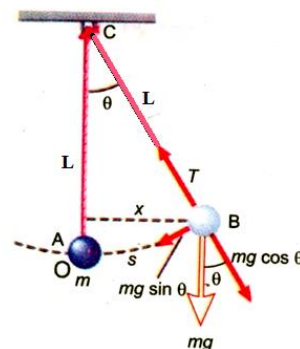
$$F = -mg\theta$$

Now displacement of the bob along the arc is given by

$$x = L\theta \quad \therefore S = r\theta$$

$$\Rightarrow \theta = \frac{x}{L}$$

So the equation (1) becomes



$$F = -mg \frac{x}{L}$$

$$\Rightarrow F = -\frac{mg}{L}x \quad \text{-----} \quad (2)$$

Since $\frac{mg}{L} = \text{Constant}$ for a given pendulum. So force F is proportional to displacement and oppositely directed which is the necessary condition for a body to perform S.H.M. In case of mass attached to a spring the elastic restoring force is given by

$$F = -kx \quad \text{-----} \quad (3)$$

Comparing (2) and (3)

$$-kx = -\frac{mg}{L}x$$

$$\Rightarrow k = \frac{mg}{L} \quad \text{-----} \quad (4)$$

Now time period of the object attached to a horizontal mass spring system is:

$$T = 2\pi \sqrt{\frac{m}{k}}$$

From (3) $\frac{m}{k} = \frac{L}{g}$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

This shows that time period of a simple pendulum is independent of mass and amplitude.

Uses

- We can find the value of g by simple pendulum.
- We can find the length of wire.

Problem: Find the frequency of second pendulum.

Given Data: Time Period of Second Pendulum $T = 2 \text{ s}$

To Determine: Frequency of Second Pendulum $f = ?$

Calculations: $f = \frac{1}{T} = 0.5 \text{ Hz}$

Problem: A simple pendulum is 50 cm long. What is frequency of vibration?

Given Data: Length of Second Pendulum $l = 50 \text{ cm} = 0.5 \text{ m}$

To Determine: Frequency of Second Pendulum $f = ?$

Calculations: $f = \frac{1}{2\pi} \sqrt{\frac{g}{l}} = \frac{1}{2 \times 3.14} \sqrt{\frac{9.8}{0.5}} = 0.7 \text{ Hz}$

Problem: What should be the length of a simple pendulum whose period is 1 s at a place where $g = 9.8 \text{ ms}^{-2}$? What is the frequency of such pendulum?

Given Data: Time Period $T = 1 \text{ s}$, Gravitational Acceleration $g = 9.8 \text{ ms}^{-2}$

To Determine: Length of Pendulum $l = ?$, Frequency $f = ?$

Calculations: For Simple Pendulum $T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow T^2 = 4\pi^2 \frac{l}{g} \Rightarrow l = \frac{gT^2}{4\pi^2} = \frac{9.8 \times (1)^2}{4 \times (3.14)^2} = 0.25 \text{ m}$

As Frequency $f = \frac{1}{T} = \frac{1}{1} = 1 \text{ Hz}$

Problem: A simple pendulum is 50.0 cm long. What will be its frequency of vibration at the place where $g = 9.8 \text{ ms}^{-2}$?

Given Data: Length of Pendulum $l = 50 \text{ cm} = 0.5 \text{ m}$, Gravitational Acceleration $g = 9.8 \text{ ms}^{-2}$

To Determine: Frequency $f = ?$

Calculations: For Simple Pendulum $f = \frac{1}{T} = \frac{1}{2\pi\sqrt{\frac{l}{g}}} = \frac{1}{2\pi\sqrt{\frac{0.5}{9.8}}} = \frac{1}{2 \times 3.14} \times \sqrt{\frac{9.8}{0.5}} = 0.51 \text{ Hz}$

(iii) The Physical Pendulum.

“Any rigid body which can swing in a vertical plane about some axis passing through it is called physical pendulum.”

Figure shows a rigid body of an irregular shape pivoted about a horizontal frictionless axis passing through P displaced from the mean position at an angle θ .

The rigid body is in equilibrium. When center of mass ‘c’ of the body lies vertically below P then distance of P from the center of mass ‘c’ is ‘d’.

In the displaced position weight mg acts vertically downward at an angle θ with vertical. The restoring torque on the pendulum in displaced position is:

$$\tau = (\text{weight of pendulum}) \times (\text{moment arm})$$

$$\tau = -mg \times d \sin \theta$$

$$\tau = -mgd \sin \theta \quad \text{--- (1)}$$

As the restoring torque is proportional to $\sin \theta$ and not to θ . So motion is not S.H.M. So for S.H.M θ should be quite small, then $\sin \theta \approx \theta$. Equation (1) becomes

$$\tau = -mgd(\theta) \quad \text{--- (2)}$$

Restoring torque is also given as

$$\tau = -\kappa\theta \quad \text{--- (3)}$$

From (2) and (3), we have:

$$-\kappa\theta = -mgd\theta$$

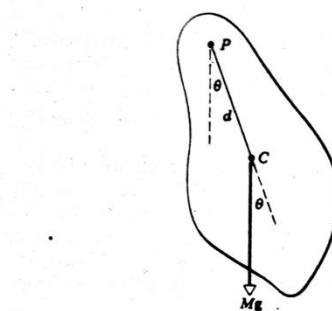
$$\Rightarrow \kappa = mgd$$

Now time period T of physical pendulum, in comparison with the time period of mass spring system, is given by:

$$T = 2\pi \sqrt{\frac{I}{\kappa}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{I}{mgd}} \quad \text{--- (4)}$$

This gives the time period of physical pendulum.



A physical pendulum. The center of mass is at C, and the pivot is at point P. The pendulum is displaced by an angle θ from its equilibrium position, which occurs when C hangs directly below P. The weight Mg provides the restoring torque.

Determination of Moment of Inertia of a Rigid Body

Now squaring both sides of equation (4) we get

$$T^2 = 4\pi^2 \frac{I}{mgd}$$

$$\Rightarrow I = \frac{mgdT^2}{4\pi^2} \quad \text{----- (5)}$$

Equation (5) gives the rotational inertia of physical pendulum.

Simple Pendulum is a special case of Physical Pendulum

The simple pendulum is a special case of physical pendulum in which a light string holds a single particle. If the body is suspended by a long weightless string of length L then $I = mL^2$ and $d = L$. The equation (4) becomes:

$$T = 2\pi \sqrt{\frac{mL^2}{mgL}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{L}{g}} \quad \text{----- (6)}$$

which is the time period of single pendulum.

Equivalent Length of Simple Pendulum

If the mass of a physical pendulum were concentrated at distance L from the pivot then the resulting simple pendulum will have the same time period as original physical pendulum. So comparing equation (4) and (6), we have:

$$2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{I}{mgd}}$$

$$\Rightarrow L = \frac{I}{md}$$

This means that mass of physical pendulum may be connected at a point O distance $L = \frac{I}{md}$ from the pivot point P. The point is called centre of oscillator of the physical pendulum.

Sample Problem 4. A thin uniform rod of mass $M = 0.112 \text{ kg}$ and length $L = 0.096 \text{ m}$ is suspended by a wire that passes through its center and is perpendicular to its length. The wire is twisted and the rod set oscillating. The period is found to be 2.14 s . When a flat body in the shape of an equilateral triangle is found to be 5.83 s . Find the rotational inertia of the triangle about this axis.

Given Data: Mass of the rod $M = 0.112 \text{ kg}$, Length of the rod $L = 0.096 \text{ m}$

Time period of the rod $T_{rod} = 2.14 \text{ s}$, Time period of triangular body $T_{triangle} = 5.838 \text{ s}$

To Determine: Rotational Inertia of triangular body $I_{triangle} = ?$

Calculations: As $I_{rod} = \frac{ML^2}{12} = \frac{0.112 \times 0.096 \times 0.096}{12} = 8.60 \times 10^{-5} \text{ kgm}^2$

As we know that $T = 2\pi \sqrt{\frac{I}{K}}$. This implies $\frac{T_{rod}}{T_{triangle}} = \sqrt{\frac{I_{rod}}{I_{triangle}}} \Rightarrow \left(\frac{T_{rod}}{T_{triangle}}\right)^2 = \frac{I_{rod}}{I_{triangle}}$

$\Rightarrow I_{triangle} = \left(\frac{T_{rod}}{T_{triangle}}\right)^2 (I_{rod}) = \left(\frac{2.14}{5.83}\right)^2 \times 8.60 \times 10^{-5} = 7.422 \times 8.60 \times 10^{-5} = 63.8 \times 10^{-5} \text{ kgm}^2$

Sample problem # 5: A uniform disc is pivoted at its rim. Find its period on small oscillations and the length of the equivalent simple pendulum.

To Determine: Time period of physical pendulum $T = ?$, Length of Equivalent Simple Pendulum $L = ?$

Calculations: Rotational inertia of the disk about axis through its center is $I_{cm} = \frac{MR^2}{2}$
 Rotational inertia about an axis at the rim is obtained by using parallel axis theorem $I = I_{cm} + MR^2$

$$\Rightarrow I = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$$

Time period of physical pendulum with $d = R$ is given by $T = 2\pi \sqrt{\frac{I}{MgR}} = 2\pi \sqrt{\frac{3MR^2}{2MgR}} = \sqrt{\frac{3R}{2g}}$

The length of equivalent simple pendulum is given by $L = \frac{I}{Md} = \frac{I}{MR} = \frac{3}{2}MR^2 \times \frac{1}{MR} = \frac{3}{2}R$

Sample problem # 7: The period of radius 10.2cm executing small oscillations about a pivot its rim is measured to be 0.784 s. Find the value of 'g'.

Given Data: Radius of disk $R = 10.2 \text{ cm} = 0.102 \text{ m}$, Time period of disk $T = 0.784 \text{ s}$

To Determine: Gravitational Acceleration $g = ?$

Calculations: As time period of disk is given by $T = 2\pi \sqrt{\frac{3R}{2g}} \Rightarrow T^2 = 4\pi^2 \left(\frac{3R}{2g}\right) \Rightarrow g = 4\pi^2 \left(\frac{3R}{2T^2}\right) = \frac{6\pi^2 R}{T^2}$
 $\Rightarrow g = \frac{6 \times 3.14 \times 3.14 \times 0.102}{0.784 \times 0.784} = 9.82 \frac{\text{m}}{\text{s}^2}$

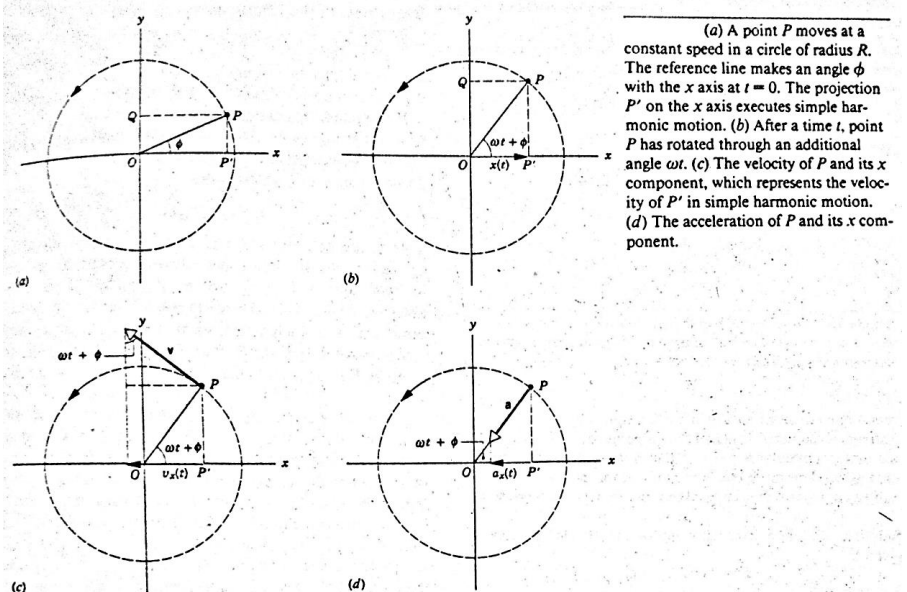
S.H.M and Uniform Circular Motion

Consider a particle is revolving in a circle of radius R with constant angular velocity ω . At $t = 0$, the radius vector $\vec{OP}$ makes an angle ϕ with x-axis. ϕ is called initial phase or phase constant. At later time t , the radius vector $\vec{OP}$ makes an angle $(\omega t + \phi)$ with x-axis.

Draw perpendicular PQ from point P on x-axis. The projection of OP along x-axis is called x-component of radius vector $\vec{OP}$ is described by formula:

$$x(t) = R \cos(\omega t + \phi) \quad (1)$$

We see that as the particle P moves along the circle, its projection Q perform S.H.M on the diameter along x-axis. The tangential velocity of the particle P moving in a circle is described by formula $v = R\omega$. The velocity of Q is the x-component of velocity of P , which is obtained by time derivative of equation (1):



$$v_x(t) = \frac{d}{dt}[x(t)] = \frac{d}{dt}[R \cos(\omega t + \phi)]$$

$$\Rightarrow v_x(t) = -R\omega \sin(\omega t + \phi) \quad \text{--- (2)}$$

The centripetal acceleration of P is given by $a(t) = R\omega^2$ which is directed towards the center of the circle. The acceleration of Q is the x-component of acceleration of P, which is obtained by time derivative of equation (2):

$$a_x(t) = \frac{d}{dt}[v_x(t)] = \frac{d}{dt}[-R\omega \sin(\omega t + \phi)]$$

$$\Rightarrow a_x(t) = -R\omega^2 \cos(\omega t + \phi) \quad \text{--- (3)}$$

We can also show that projection of P on y-axis also performs S.H.M. The projection of P along y-axis is M.

Now the projection of radius vector $\vec{OP}$ along y-axis or y-component of radius vector $\vec{OP}$ is given by

$$y(t) = R \sin(\omega t + \phi) \quad \text{--- (4)}$$

The velocity of projection M is given by

$$v_y(t) = R \cos(\omega t + \phi) \quad \text{--- (5)}$$

The acceleration of projection M is given by

$$a_y(t) = -R\omega^2 \sin(\omega t + \phi) \quad \text{--- (6)}$$

Superposition of two perpendicular SHMs having same Amplitude and Phase

Squaring and adding (1) & (4) we get:

$$x^2 + y^2 = R^2 \cos^2(\omega t + \phi) + R^2 \sin^2(\omega t + \phi)$$

$$\Rightarrow x^2 + y^2 = R^2(1)$$

$$\Rightarrow x^2 + y^2 = R^2 \quad \text{--- (a)}$$

which is equation of circle. Thus the superposition of two perpendicular SHMs having same phase and amplitude will result in circular motion. Now squaring and adding (2) & (5) we get

$$v_x^2 + v_y^2 = R^2 \omega^2 \cos^2(\omega t + \phi) + R^2 \omega^2 \sin^2(\omega t + \phi)$$

$$\Rightarrow v_x^2 + v_y^2 = R^2 \omega^2$$

$$\Rightarrow v_x^2 + v_y^2 = v^2 \quad \text{--- (b)}$$

Now squaring and adding (3) & (6) we get

$$a_x^2 + a_y^2 = R^2 \omega^4 \cos^2(\omega t + \phi) + R^2 \omega^4 \sin^2(\omega t + \phi)$$

$$\Rightarrow a_x^2 + a_y^2 = R^2 \omega^4(1) = R^2 \omega^4$$

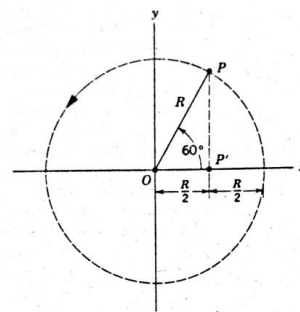
$$\Rightarrow a_x^2 + a_y^2 = (R\omega^2)^2$$

$$\Rightarrow a_x^2 + a_y^2 = a^2$$

So from the above discussion we conclude that

“The circular motion is combinations of two identical SHM acting perpendicular to each other.

Sample problem # 8: Consider a body executing a horizontal simple harmonic motion. The equation of that motion is $x = 0.35 \cos(8.3t)$ where x is in meters and t in seconds. This motion can also be represented as the projection of uniform circular motion along a horizontal diameter. (a) Give the properties of the corresponding uniform circular motion. (b) From the motion of reference point determine the time required for the body to come halfway in toward the center of motion from its initial position.



Sample Problem 8. The radius OP moves from $\phi = 0$ at $t = 0$ to $\omega t = 60^\circ$ at time t . The projection P' moves correspondingly from $x = R$ to $x = R/2$.

Given Data: Equation of motion of SHO: $x = 0.35 \cos(8.3t)$ — — — — (1)

To Determine: (a) Properties of the corresponding uniform motion =? i.e. $R = ?$, $\phi = ?$, $\omega = ?$

(b) Time required for the body to come half way $t = ?$

Calculations: (a) As we know that $x = R \cos(\omega t + \phi)$ — — — — (2)

Comparing (1) and (2), we have: $R = 0.35m$, $\phi = 0^\circ$ and $\omega = 8.3 \frac{\text{rad}}{\text{sec}}$

(b) As the body moves half way in, The angle covered $\theta = 60^\circ = \frac{\pi}{3} \text{ rad}$

Now as $\theta = \omega t \Rightarrow t = \frac{\theta}{\omega} = \frac{\pi}{3} \times \frac{1}{\omega} = \frac{\pi}{3} \times \frac{1}{8.3} = 0.13 \text{ s}$

4. Combination of Harmonic Motion (Lissajous Figures)

“When a particle is subjected to two SHMs at right angle to each other, the paths traced by the particle are called Lissajous figures.”

So the phenomenon of Lissajous figures arises due to the superposition of two S.H.Ms at right angle to each other. Let us consider two simple harmonic motion along x and y -axis having same frequencies but different amplitudes. They are given as

$$x = a \sin(\omega t + \phi) \text{ along } x\text{-axis} \text{ — — — — — (1)}$$

$$y = b \sin \omega t \text{ along } y\text{-axis} \text{ — — — — — (2)}$$

From (1): $\frac{x}{a} = \sin(\omega t + \phi) \Rightarrow \frac{x}{a} = \sin \omega t \cos \phi + \cos \omega t \sin \phi$ — — — — — (3)

From (2): $\frac{x}{a} = \sin(\omega t)$ — — — — — (4)

As $\cos \omega t = \sqrt{1 - \sin^2 \omega t} \Rightarrow \cos \omega t = \sqrt{1 - \frac{y^2}{b^2}}$

The equation (3) takes the form:

$$\begin{aligned} \frac{x}{a} &= \frac{y}{b} \cos \phi + \sqrt{1 - \frac{y^2}{b^2}} \sin \phi \\ \Rightarrow \frac{x}{a} - \frac{y}{b} \cos \phi &= \sqrt{1 - \frac{y^2}{b^2}} \sin \phi. \end{aligned}$$

Squaring both sides we get

$$\left(\frac{x}{a} - \frac{y}{b} \cos \phi\right)^2 = \left(1 - \frac{y^2}{b^2}\right) \sin^2 \phi.$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} \cos^2 \phi - \frac{2xy}{ab} \cos \phi = \sin^2 \phi - \frac{y^2}{b^2} \sin^2 \phi.$$

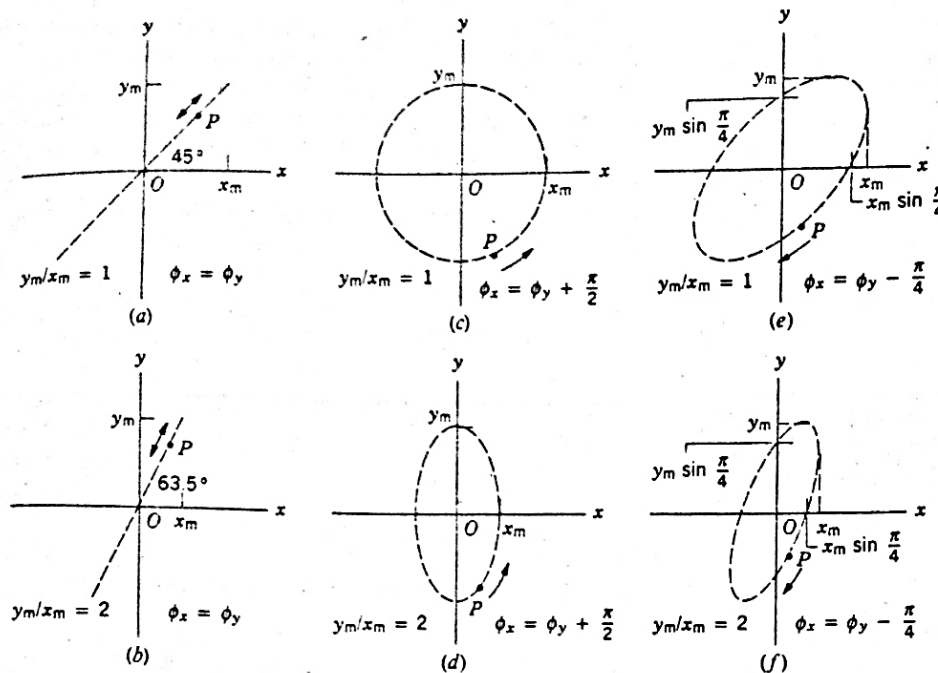
$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} \cos^2 \phi + \frac{y^2}{b^2} \sin^2 \phi - \frac{2xy}{ab} \cos \phi = \sin^2 \phi.$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} (\cos^2 \phi + \sin^2 \phi) - \frac{2xy}{ab} \cos \phi = \sin^2 \phi \quad \therefore \cos^2 \phi + \sin^2 \phi = 1$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \cos \phi = \sin^2 \phi \quad \text{----- (5)}$$

This is the general equation of Lissajous figure in xy-plane

Special Cases: Let us now consider some special cases:



Combinations of simple harmonic motions along two perpendicular directions. Each figure shows the motion of point P when the amplitudes and phases of the motions have the indicated relationships. The x and y motions have equal frequencies.

(i) When $\phi = 0$ Then equation (3) becomes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} = 0$$

$$\Rightarrow \left(\frac{x}{a} - \frac{y}{b}\right)^2 = 0 \Rightarrow \frac{x}{a} - \frac{y}{b} = 0$$

$$\Rightarrow \frac{x}{a} = \frac{y}{b}$$

This is the equation of a straight line AB of slope $\frac{b}{a}$ inclined to +ve x -axis at an angle $\theta = \tan^{-1} \left(\frac{b}{a}\right)$

(ii) When $\phi = \pi$, then equation (3) becomes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} (-1) = 0$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{2xy}{ab} = 0 \Rightarrow \left(\frac{x}{a} + \frac{y}{b}\right)^2 = 0 \Rightarrow \frac{x}{a} + \frac{y}{b} = 0$$

$$\Rightarrow \frac{x}{a} = -\frac{y}{b}$$

$$\text{or } y = -\frac{b}{a}x$$

This is the equation of a straight line of slope $-\frac{b}{a}$ inclined to $-ve$ x -axis at an angle $\theta = \tan^{-1} \frac{b}{a}$

(iii) When $\theta = \frac{\pi}{2}$, then equation (3) becomes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}(0) = 0$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

This is the equation of an ellipse having major axis = $2a$ (if $a > b$) and minor axis = $2b$.

- If $a = b$, then $x^2 + y^2 = a^2$ which is the equation of a circle of radius ' a ' in the xy -plane.

(iv) When $\phi = \frac{\pi}{4}$, then equation (3) becomes

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \frac{1}{\sqrt{2}} = \frac{1}{2}$$

This is the equation of an oblique ellipse.

So

“When a particle is subjected to two simple harmonic motion at right angle to each other the particle may move along different paths. Such paths are called Lissajous figures.”

Damped Harmonic Oscillator

If an oscillator moves in a resistive medium, its amplitude goes on decreasing. So energy of the oscillator is used in doing work against the resistance of the medium. The motion of the oscillator is then said to be damped. The damping force or the resistive force F_r is proportional to the velocity of the oscillator.

$$F_r \propto \dot{x}$$

$$\Rightarrow F_r = -\gamma \dot{x}$$

when γ is called the damping co-efficient. So for the damped oscillator, these are two forces acting on it i.e. (i) Elastic Restoring Force,

(ii) Damping force. Thus the equation of motion of damped oscillator is given as:

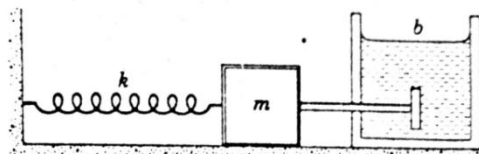
$$F = F_e + F_r \quad \text{where } F_e \text{ is the elastic restoring force}$$

$$\Rightarrow F = -kx - \gamma \dot{x}$$

$$\Rightarrow m\ddot{x} = -kx - \gamma \dot{x}$$

$$\Rightarrow \ddot{x} = -\frac{k}{m}x - \frac{\gamma}{m}\dot{x}$$

$$\Rightarrow \ddot{x} + \frac{k}{m}x - \frac{\gamma}{m}\dot{x} = 0$$



A representation of a damped harmonic oscillator.

We consider the oscillating body (of mass m) to be attached to a (massless) vane immersed in a fluid, in which it experiences a viscous damping force $-bv$. We do not consider sliding friction at the horizontal surface.

$$\text{Let } \frac{k}{m} = \omega_o^2 \text{ and } \frac{\gamma}{m} = 2\beta$$

$$\Rightarrow \ddot{x} + \omega_o^2 x + 2\beta \dot{x} = 0 \quad \text{--- (1)}$$

This is a linear second order homogeneous differential equations. To solve this equation, we put:

$$x = e^{\mu t} \quad \text{--- (a)}$$

$$\dot{x} = \mu e^{\mu t}.$$

$$\text{and } \ddot{x} = \mu^2 e^{\mu t}.$$

Equation (1) becomes

$$\mu^2 e^{\mu t} + 2\beta \mu e^{\mu t} + \omega_o^2 e^{\mu t} = 0$$

$$\Rightarrow e^{\mu t} (\mu^2 + 2\beta \mu + \omega_o^2) = 0$$

$$\Rightarrow \mu^2 + 2\beta \mu + \omega_o^2 = 0 \quad \therefore e^{\mu t} \neq 0$$

It is a quadratic equation in μ , therefore:

$$\mu = \frac{-2\beta \pm \sqrt{4\beta^2 - 4\omega_o^2}}{2}$$

$$\Rightarrow \mu = \frac{-2\beta \pm 2\sqrt{\beta^2 - \omega_o^2}}{2}$$

$$\Rightarrow \mu = 2 \left(\frac{-\beta \pm \sqrt{\beta^2 - \omega_o^2}}{2} \right)$$

$$\Rightarrow \mu = -\beta \pm \sqrt{\beta^2 - \omega_o^2}$$

$$\text{Let } \mu_1 = -\beta + \sqrt{\beta^2 - \omega_o^2} \text{ and } \mu_2 = -\beta - \sqrt{\beta^2 - \omega_o^2}$$

Therefore due to two values of ' μ ', we get two solutions from equation (a)

$$x_1 = Ae^{\mu_1 t} \text{ and } x_2 = Be^{\mu_2 t}$$

The complete solution of equation (1) is then sum of x_1 & x_2

$$x = x_1 + x_2$$

$$\Rightarrow x = Ae^{\mu_1 t} + Be^{\mu_2 t}$$

where A, B are constants to be determined. Putting the values of μ_1 and μ_2 , we get:

$$x = Ae^{(-\beta + \sqrt{\beta^2 - \omega_o^2})t} + Be^{(-\beta - \sqrt{\beta^2 - \omega_o^2})t}$$

This is the general solution of equation (1). It is clear that for oscillating motion, the damping force should be less than restoring i.e. $\omega_o^2 > \beta^2$. The above solution takes the form:

$$x = Ae^{(-\beta + \sqrt{(\omega_o^2 - \beta^2)})t} + Be^{(-\beta - \sqrt{(\omega_o^2 - \beta^2)})t}$$

$$\Rightarrow x = Ae^{(-\beta + i\sqrt{(\omega_o^2 - \beta^2)})t} + Be^{(-\beta - i\sqrt{(\omega_o^2 - \beta^2)})t}$$

$$\Rightarrow x = Ae^{-\beta t + i\omega t} + Be^{-\beta t - i\omega t} \quad \therefore \sqrt{(\omega_o^2 - \beta^2)} = \omega$$

$$\Rightarrow x = Ae^{-\beta t} \cdot e^{i\omega t} + Be^{-\beta t} \cdot e^{-i\omega t}$$

$$\Rightarrow x = e^{-\beta t} (Ae^{i\omega t} + Be^{-i\omega t})$$

As x is real, so R. H. S must be real. This is only possible when A and B are complex conjugates i.e.,

$A = a + ib$ & $B = a - ib$. So the above equation becomes:

$$x = e^{-\beta t} [(a + ib)e^{i\omega t} + (a - ib)e^{-i\omega t}].$$

$$\Rightarrow x = e^{-\beta t} [ae^{i\omega t} + ae^{-i\omega t} + ibe^{i\omega t} - ibe^{-i\omega t}].$$

$$\Rightarrow x = e^{-\beta t} [a(e^{i\omega t} + e^{-i\omega t}) + ib(e^{i\omega t} - e^{-i\omega t})].$$

$$\Rightarrow x = e^{-\beta t} \left[2a \left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} \right) + 2i^2 b \left(\frac{e^{i\omega t} - e^{-i\omega t}}{2i} \right) \right].$$

$$\Rightarrow x = e^{-\beta t} \left[2a \left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} \right) - 2b \left(\frac{e^{i\omega t} - e^{-i\omega t}}{2i} \right) \right].$$

Now $\left(\frac{e^{i\omega t} + e^{-i\omega t}}{2} \right) = \cos \omega t$ and $\left(\frac{e^{i\omega t} - e^{-i\omega t}}{2i} \right) = \sin \omega t$. Therefore,

$$x = e^{-\beta t} (2a \cos \omega t - 2b \sin \omega t)$$

Let $2a = x_0 \cos \phi$, $2b = x_0 \sin \phi$, where $x_0 = \sqrt{4a^2 + 4b^2} = \sqrt{4(a^2 + b^2)} = 2\sqrt{(a^2 + b^2)}$

and $\tan \phi = \frac{b}{a} \Rightarrow \phi = \tan^{-1} \frac{b}{a}$. The solution takes the form:

$$x = e^{-\beta t} (x_0 \cos \phi \cos \omega t - x_0 \sin \phi \sin \omega t)$$

$$\Rightarrow x = x_0 e^{-\beta t} (\cos \phi \cos \omega t - \sin \phi \sin \omega t)$$

$$\Rightarrow x = x_0 e^{-\beta t} \cos(\omega t + \phi)$$

Suppose $\beta = \frac{1}{2T}$ where T is called the relaxation time. It is the time during which amplitude

drops to $\frac{1}{e}$ of its initial value. Therefore,

$$x = x_0 e^{-\frac{t}{2T}} \cos(\omega t + \phi)$$

This is the solution of equation (1). It describes the motion of damped harmonic oscillator.

Parameters of Damped Harmonic Oscillator

Amplitude: The amplitude of the damped oscillator is given by:

$$A = x_0 e^{-\frac{t}{2T}}$$

We find that amplitude goes on decreasing exponentially

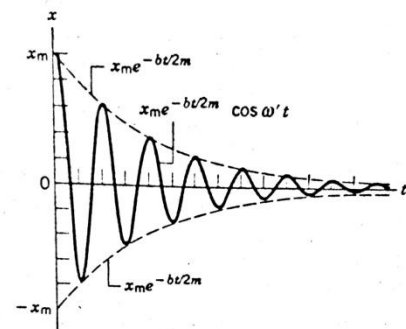
with time t due to the factor $e^{-\frac{t}{2T}}$. Figure shows that variation of amplitude with time.

Frequency

The frequency of the oscillator is given by,

$$f = \frac{\omega}{2\pi}$$

$$\Rightarrow f = \frac{\sqrt{\omega_0^2 - \beta^2}}{2\pi} \quad \therefore \omega = \sqrt{\omega_0^2 - \beta^2}$$



Damped harmonic motion. The displacement x is plotted against the time t with the phase constant ϕ taken to be 0. The motion is oscillatory, but the amplitude decreases exponentially with time.

This shows that the frequency of the damped oscillator is somewhat smaller than the undamped oscillator. For weak damping, the frequency is nearly the same as that of undamped.

Energy of Damped Harmonic Oscillator

Consider a damped oscillator whose vibrations are counted from the mean position ($\phi = 0$). The damped instantaneous displacement of damped oscillator is described as:

$$x = x_0 e^{-\frac{t}{2T}} \cos \omega t \quad \therefore \omega = \sqrt{\omega_0^2 - \beta^2}$$

For weak damping, $\omega = \omega_o$, when $\beta^2 = 0$. Therefore, $\sqrt{\omega_o^2 - \beta^2} = \omega_o$ because $\beta \ll$.

So the above expression becomes

$$x = x_o e^{-\left(\frac{t}{2T}\right)} \cos \omega_o t$$

Now average K. E over one period is given by:

$$\langle K \rangle = \frac{1}{4} m \omega_o^2 A^2 \quad \therefore A = x_o e^{-\left(\frac{t}{2T}\right)} \text{ is the amplitude}$$

$$\Rightarrow \langle K \rangle = \frac{1}{4} m \omega_o^2 x_o^2 e^{-\left(\frac{2t}{T}\right)}$$

$$\Rightarrow \langle K \rangle = \frac{1}{4} m \omega_o^2 x_o^2 e^{-\frac{t}{T}} \quad \text{--- (a)}$$

Now average P. E over one period is given by:

$$\langle U \rangle = \frac{1}{4} m \omega_o^2 A^2.$$

$$\Rightarrow \langle U \rangle = \frac{1}{4} m \omega_o^2 x_o^2 e^{-\frac{t}{T}} \quad \text{--- (b)}$$

Therefore average total energy over one period is

$$\langle E \rangle = \langle K \rangle + \langle U \rangle$$

$$\Rightarrow \langle E \rangle = \frac{1}{4} m \omega_o^2 x_o^2 e^{-\frac{t}{T}} + \frac{1}{4} m \omega_o^2 x_o^2 e^{-\frac{t}{T}}$$

$$\Rightarrow \langle E \rangle = \frac{1}{2} m \omega_o^2 x_o^2 e^{-\frac{t}{T}} \quad \text{--- (c)}$$

From (a), (b) and (c), we find that $\langle K \rangle$, $\langle U \rangle$ and $\langle E \rangle$ decreases exponentially with time. Now by putting $m \omega_o^2 = k$ (the spring constant) and $x_o = x_m$ (amplitude), the equation (c) takes the form:

$$\langle E \rangle = \frac{1}{2} k x_m^2 e^{-\frac{t}{T}}$$

$$\Rightarrow \langle E \rangle = \frac{1}{2} k x_m^2 e^{-\frac{t\gamma}{m}} \quad \therefore 2\beta = \frac{\gamma}{m} = \frac{1}{T}$$

$$\Rightarrow \langle E \rangle = \frac{1}{2} k x_m^2 e^{-\left(\frac{\gamma}{m}\right)t}$$

If $\gamma = b = \text{damping coefficient}$, then

$$\langle E \rangle = \frac{1}{2} k x_m^2 e^{-\left(\frac{bt}{m}\right)}$$

This is the expression for average energy of damped harmonic oscillator.

Sample problem # 9. In damped oscillator $m = 250 \text{ g}$, $k = 85 \text{ N/m}$ and $b = 0.0070 \text{ kg/s}$. In how many periods of oscillations, will the mechanical energy of the oscillator drop to half of its initial value.

Given Data: $m = 250 \text{ g} = 0.250 \text{ kg}$, $k = 85 \text{ N/m}$, $b = 0.0070 \text{ kg/s}$

To Determine: Periods of Oscillations in which the energy of oscillator become half $n = ?$

Calculations: Period of Oscillation $T = 2\pi \sqrt{\frac{m}{k}} = 2 \times 3.14 \sqrt{\frac{0.25}{85}} = 6.28 \times 0.054 = 0.34 \text{ s}$

The energy of damped oscillator: $E(t) = \frac{1}{2} k x_m^2 e^{-\left(\frac{bt}{m}\right)}$

The initial energy of oscillator is: $E(0) = \frac{1}{2} k x_m^2$

According to statement: $E(t) = \frac{1}{2}E(0) \Rightarrow \frac{1}{2}kx_m^2 e^{-\left(\frac{bt}{m}\right)} = \frac{1}{2}\left(\frac{1}{2}kx_m^2\right) \Rightarrow e^{-\left(\frac{bt}{m}\right)} = \frac{1}{2} \Rightarrow e^{\left(\frac{bt}{m}\right)} = 2$

$$\Rightarrow \frac{bt}{m} = \ln(2) \Rightarrow t = \ln(2) \frac{m}{b} = \frac{0.25}{0.070} \times 0.693 = 2.5 \text{ s}$$

Now the number of cycles are given by: $n = \frac{t}{T} = \frac{2.5}{0.34} = 7.4$

So after 7.4 cycles of oscillations, the mechanical energy of the system drop to one half of its initial value.

Forced Harmonic Oscillator

A damped harmonic oscillation under the action of a periodic force $F_p = F_m \sin \omega t$ is called forced harmonic oscillations. The examples of forced harmonic oscillations are:

- A bridge vibrating under the influence of marching soldiers
- The ear drum of a person vibrates due to periodic force of sound waves

Let us consider a damped harmonic oscillator under the action of periodic force $F_p = F_m \sin \omega t$, where ω is the frequency of applied force and F_m is the maximum value of external force. The total force on the oscillator is given as:

$$F = F_e + F_r + F_p$$

$$\text{where } F_e = -kx, F_r = -\gamma \dot{x}, F_p = F_m \sin \omega t$$

$$\Rightarrow m\ddot{x} = -kx - \gamma \dot{x} + F_m \sin \omega t$$

$$\Rightarrow \ddot{x} + \frac{k}{m}x + \frac{\gamma}{m}\dot{x} = \frac{F_m}{m} \sin \omega t$$

$$\Rightarrow \ddot{x} + \frac{\gamma}{m}\dot{x} + \frac{k}{m}x = \frac{F_m}{m} \sin \omega t$$

$$\text{Let } \frac{\gamma}{m} = 2\beta \text{ and } \frac{k}{m} = \omega_o^2$$

$$\Rightarrow \ddot{x} + 2\beta\dot{x} + \omega_o^2 x = \frac{F_m}{m} \sin \omega t \quad \text{----- (1)}$$

It is linear and non-homogeneous differential equation. To solve this equation, we change the variables x into a complex variable X such that x is the imaginary part of complex variable X . The above equation takes the form:

$$\ddot{X} + 2\beta\dot{X} + \omega_o^2 X = \frac{F_m}{m} \sin \omega t$$

$$\Rightarrow \ddot{X} + 2\beta\dot{X} + \omega_o^2 X = \frac{F_m}{m} e^{i\omega t} \quad \text{----- (2)}$$

$$\text{Let } X = x_m e^{i\omega t}$$

$$\dot{X} = i\omega x_m e^{i\omega t}$$

$$\ddot{X} = (i\omega)^2 x_m e^{i\omega t}$$

$$\ddot{X} = -\omega^2 x_m e^{i\omega t}$$

Equation (2) becomes:

$$-\omega^2 x_m e^{i\omega t} + 2\beta i\omega x_m e^{i\omega t} + \omega_o^2 x_m e^{i\omega t} = \frac{F_m}{m} e^{i\omega t}$$

$$\Rightarrow -\omega^2 x_m + 2\beta i \omega x_m + \omega_o^2 x_m = \frac{F_m}{m}$$

$$\Rightarrow x_m [(\omega_o^2 - \omega^2) + 2i\beta\omega] = \frac{F_m}{m}$$

$$\Rightarrow x_m = \frac{\frac{F_m}{m}}{(\omega_o^2 - \omega^2) + 2i\beta\omega}$$

Multiplying and dividing R. H. S by $(\omega_o^2 - \omega^2) - 2i\beta\omega$

$$x_m = \frac{\frac{F_m}{m}}{(\omega_o^2 - \omega^2) + 2i\beta\omega} \left[\frac{(\omega_o^2 - \omega^2) - 2i\beta\omega}{(\omega_o^2 - \omega^2) - 2i\beta\omega} \right]$$

$$\Rightarrow x_m = \left(\frac{F_m}{m} \right) \frac{(\omega_o^2 - \omega^2) - 2i\beta\omega}{(\omega_o^2 - \omega^2) + 4\beta^2\omega^2}$$

$$\Rightarrow x_m = \left(\frac{F_m}{m} \right) \frac{[(\omega_o^2 - \omega^2) - 2i\beta\omega]}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2} \times \sqrt{(\omega_o^2 - \omega^2) + 4\beta^2\omega^2}}$$

$$\Rightarrow x_m = \frac{\left(\frac{F_m}{m} \right)}{\sqrt{(\omega_o^2 - \omega^2) + 4\beta^2\omega^2}} \left[\frac{(\omega_o^2 - \omega^2) - 2i\beta\omega}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} \right]$$

$$\Rightarrow x_m = \frac{\left(\frac{F_m}{m} \right)}{\sqrt{(\omega_o^2 - \omega^2) + 4\beta^2\omega^2}} \left[\frac{(\omega_o^2 - \omega^2)}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} + \frac{i - 2\beta\omega}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} \right]$$

$$\text{Let } \frac{(\omega_o^2 - \omega^2)}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} = \cos \phi \quad \text{--- (a)}$$

$$\text{and } \frac{-2\beta\omega}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} = \sin \phi \quad \text{--- (b)}$$

The above expression becomes:

$$x_m = \frac{\left(\frac{F_m}{m} \right)}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} (\cos \phi + i \sin \phi)$$

$$\Rightarrow x_m = \frac{\left(\frac{F_m}{m} \right)}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} (e^{i\phi}) \quad \because \cos \phi + i \sin \phi = e^{i\phi}$$

Now as $X = x_m e^{i\omega t}$. Putting the value of x_m , we have:

$$X = \frac{\left(\frac{F_m}{m} \right)}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} (e^{i\phi}) (e^{i\omega t})$$

$$\Rightarrow X = \frac{\left(\frac{F_m}{m} \right)}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} e^{i(\omega t + \phi)}$$

This is the solution of complex equation (1). As x is imaginary part of X . So the solution of equation (2) is the imaginary part of solution of equation (2). Therefore,

$$x = \frac{\left(\frac{F_m}{m} \right)}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} \sin(\omega t + \phi)$$

$$\Rightarrow x = \frac{F_m}{m\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} \sin(\omega t + \phi)$$

$$\Rightarrow x = \frac{F_m}{\sqrt{m^2(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2 m^2}} \sin(\omega t + \phi)$$

Putting $\sqrt{m^2(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2 m^2} = G$

$$x = \frac{F_m}{G} \sin(\omega t + \phi) \quad \text{--- (3)}$$

This is solution of equation (1). The phase constant ϕ is given by:

$$\tan \phi = \left(\frac{\sin \phi}{\cos \phi} \right).$$

$$\Rightarrow \phi = \tan^{-1} \left(\frac{\sin \phi}{\cos \phi} \right)$$

Putting the values of $\sin \phi$ and $\cos \phi$ from equations (a) and (b), we get:

$$\Rightarrow \phi = \tan^{-1} \left(\frac{-2\beta\omega}{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}} \times \frac{\sqrt{(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2}}{(\omega_o^2 - \omega^2)} \right)$$

$$\Rightarrow \phi = \tan^{-1} \left(\frac{-2\beta\omega}{(\omega_o^2 - \omega^2)} \right).$$

where ω_o is the natural frequency and ω is the frequency of driving force.

Resonance Phenomenon

Resonance is the increase in the amplitude of a vibrating body under the action of a periodic force whose frequency is equal to the natural frequency of the body. So, for resonance $\omega = \omega_o$. So the amplitude in equation (3) takes the form:

$$\text{Amplitude} = \frac{F_m}{G} = \frac{F_m}{(m^2(\omega_o^2 - \omega^2)^2 + 4\beta^2\omega^2 m^2)^{\frac{1}{2}}}$$

$$\Rightarrow \text{Amplitude} = \frac{F_m}{\sqrt{4\beta^2\omega^2 m^2}} \quad \therefore \omega = \omega_o$$

$$\Rightarrow \text{Amplitude} = \frac{F_m}{2\beta\omega m}$$

$$\Rightarrow \text{Amplitude} = \frac{F_m}{\left(\frac{\gamma}{m}\right)\omega m} \quad \therefore 2\beta = \frac{\gamma}{m}$$

$$\Rightarrow \text{Amplitude} = \frac{F_m}{\gamma\omega m} \quad \therefore \omega = \omega_o$$

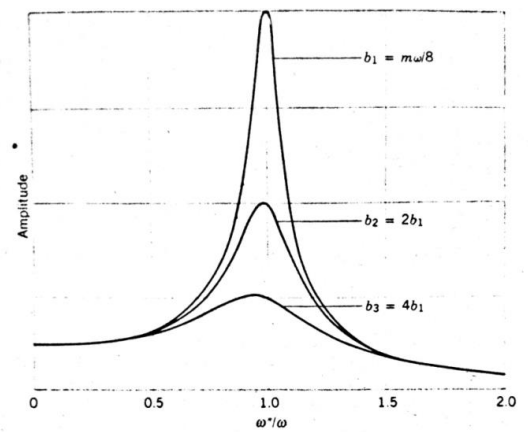
$$\Rightarrow \text{Amplitude} = \frac{F_m}{\gamma\omega_o}$$

This shows that the amplitude of the oscillator becomes infinity ∞ if there is no damping i.e., $\gamma = 0$.

But γ can't be zero. So due to damping, the amplitude never become ∞ . In the figure shown, we have three curves which shown the variation of amplitude with ratio of $\frac{\omega}{\omega_o}$. It is clear that

(i) When $\frac{\omega}{\omega_o} = 1$ and damping is smallest, then curve is sharp and amplitude is maximum.

(ii) When $\frac{\omega}{\omega_o} > 1$, and damping increases, the amplitude decreases and curve becomes wider.



The amplitude F_m/G of a forced oscillator as the angular frequency ω of the driving force is varied. The three curves correspond to different levels of damping, the smallest damping giving the sharpest resonance curve.

Wave Motion

Wave motion appears in almost every branch of Physics. Surface waves on bodies of water are commonly observed. Sound waves and light waves are essential to our perception of the environment, because we have evolved receptors (eyes and ears) capable of their detection. In the past century, we have learned how to produce and use radio waves. We can understand the structure of atoms and subatomic systems based on the wavelike properties of their constituent particles. The similarity of the physical and mathematical description of these different kinds of waves indicates that wave motion is one of the unifying themes of Physics.

19.1 Mechanical Waves

“The waves which require a material medium for their propagation are called mechanical waves”.

Mechanical waves are of two types.

- (i) Transverse waves
- (ii) Longitudinal waves

The waves in which particles of the medium vibrate perpendicular to the direction of propagation of waves are called *transverse waves*. When a string is tied to a hook from one end and the free end is displaced up and down then transverse waves are produced. The disturbance moves along the string but the particles vibrate perpendicularly to the direction of propagation of disturbance. Water waves are also transverse in nature.

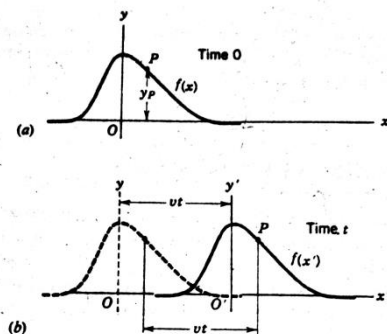
The waves in which particle of the medium vibrate along the direction of propagation of waves are called longitudinal waves. For example, when a spring under tension is made to vibrate back and forth at one end, a longitudinal wave is produced. The coils of spring vibrate back and forth parallel to the direction of propagation of disturbance. Sound waves in a gas are longitudinal waves or compressional waves.

19.2 Traveling Waves

The waves which transfer energy from one point to another moving away from source of disturbance are called traveling waves. They carry energy and momentum with them.

Consider a string whose one end is fixed to a hook and other end is held in hand. If we move the free end up and down at regular intervals, a series of identical waves travel along the string. Consider a transfer pulse in a long stretched string at $t = 0$, we suppose the shape of pulse remain unchanged as it travels the pulse lies in xy -plain.

Let the pulse be traveling in positive x direction with speed “ v ”. After some time “ t ” the pulse covers a



(a) A transverse pulse, shown as a snapshot at time $t = 0$. The point P represents a particular location on the phase of the pulse, *not* a particular point of the medium (the string, for instance). (b) At a time t later, the pulse has moved a distance vt in the positive x direction. The point P on the phase has also moved a distance vt . The peak of the pulse defines the origin of the x' coordinate.

distance “ vt ” as shown in figure (b) and its shape is the same as at $t = 0$. The y coordinate represent the transverse displacement of a point on the string. Which depends both on position x and time t i.e; y is a function of x and t .

The wave form of the pulse at $t = 0$ is given by:

$$y(x, 0) = f(x) \quad \text{--- -- -- -- --} \quad (1)$$

where the function ‘ f ’ gives the shape of the wave.

After time ‘ t ’ the shape of the wave remains unchanged. So after time ‘ t ’, the wave form is represented by the same function ‘ f ’. Relative to the origin O' of a reference frame that travels with the pulse, the shape is described by function $f(x')$ as shown in the figure b. The relationship between the x -coordinates in the two reference frames is described by formula:

$$x' = x - vt \quad \text{--- -- -- -- --} \quad (2) \quad \because x = x' + vt$$

where x' is the displacement of point ‘P’ along x -axis w.r.t O' and x is the displacement of point ‘P’ w.r.t O . So the wave form of the pulse at time ‘ t ’ is given by:

$$y(x, t) = f(x') \\ \Rightarrow y(x, t) = f(x - vt) \quad \text{--- -- -- -- --} \quad (3)$$

From equation (1) and (3) we see that we can change function of any shape into a wave traveling in the positive ‘ x ’ direction simply by replacing x by $x - vt$ in the function $f(x)$.

Let us follow the motion of a particular phase of the wave, such as that of location of P of the waveform of figure. If the wave is to keep its shape as it travels, then the y -coordinate y_P of point P must not change. That is, y_P remains same for all values of $x - vt$. This remains true for any location and for all times t . Thus for the motion of any particular phase of the wave, we must have:

$$x - vt = \text{constant} \quad \text{--- -- -- -- --} \quad (4)$$

We can verify that equation (4) characterizes the motion of the phase of the waveform by differentiating with respect to time, which gives:

$$\frac{dx}{dt} - v = 0 \\ \Rightarrow \frac{dx}{dt} = v$$

The velocity $\frac{dx}{dt}$ describe the motion of the phase of the wave, and so it is known as the phase velocity.

If the wave moves in the negative x direction, all we need do is replace v by $-v$. In this case we would obtain:

$$y(x, t) = f(x + vt)$$

The motion of any phase of the wave would then be characterized by the requirement that

$$x + vt = \text{constant}$$

The phase velocity in this case would be:

$$\frac{dx}{dt} = -v$$

Important Note

The function $y(x, t)$ contains complete description of the shape of the wave and its motion. At any particular time, say t_1 , the function $y(x, t_1)$ gives y as function of x , which defines a curve; this curve represents the actual shape of the string at that time t_1 and can be regarded as a 'snapshot' of the wave.

On the other hand, we can consider the motion of a particular point on the string, say at the fixed coordinate x_1 . The function $y(x_1, t)$ tells the y coordinate of that point as a function of the time. For a transverse wave in string, the point is not moving at all at time $t = 0$. It then begin to move gradually, as the leading edge of the pulse arrives. After the peak of the wave passes, the displacement of the point drops rapidly back to zero as the trailing edge passes.

Problem: At $t = 0$ transverse wave pulse in the wire is described by formula $y(x, 0) = \frac{6}{x^2+3}$.

Describe the function $y(x, t)$ that describe this wave if it is travelling in the positive x-direction with speed of 4.5 m/s.

Given Speed $v = 4.5 \frac{m}{s}$, Wave Pulse at $t = 0$: $y(x, 0) = \frac{6}{x^2+3}$

The required function will be:

$$y(x, t) = \frac{6}{(x - vt)^2 + 3}$$

$$\Rightarrow y(x, t) = \frac{6}{(x - 4.5t)^2 + 3}$$

Problem: A travelling wave is represented by wave function $y(x, t) = \frac{2}{(x-3t)^2+1}$. (a) Find $v = ?$

(b) Find vertical displacement at $x = 0.5 \text{ m}$, $t = 1, 2, 1.5 \text{ s}$.

(a) In given equation $x - vt = x - 3t \Rightarrow v = 3 \text{ ms}^{-1}$

(b) At $t = 1 \text{ s}$, $x = 0.5 \text{ m}$

$$y(0.5, 1) = \frac{2}{(0.5 - 3 \times 1)^2 + 1} = \frac{2}{7.25} = 0.2758 \text{ m}$$

At $t = 2 \text{ s}$, $x = 0.5 \text{ m}$

$$y(0.5, 2) = \frac{2}{(0.5 - 3 \times 2)^2 + 1} = \frac{2}{31.25} = 0.064 \text{ m}$$

At $t = 1.5 \text{ s}$, $x = 0.5 \text{ m}$

$$y(0.5, 1.5) = \frac{2}{(0.5 - 3 \times 1.5)^2 + 1} = \frac{2}{17} = 0.118 \text{ m}$$

19.3 Sinusoidal Waves

Consider a transverse wave of sinusoidal shape. Suppose at time = 0, the wave train is moving along x-axis and is given by:

$$y(x, 0) = y_m \sin kx$$

Where $k = \frac{2\pi}{\lambda}$ is the wave number having unit rad/m.

$$y(x, 0) = y_m \sin\left(\frac{2\pi}{\lambda}\right)x \quad \text{--- (1)}$$

Here y_m is the maximum displacement and is called the amplitude of the wave and λ is the wavelength.

Thus a wave travelling in the x-direction with phase velocity v at any time t is given by:

$$y(x, t) = y_m \sin\left(\frac{2\pi}{\lambda}\right)(x - vt) \quad \text{--- (2)}$$

Now time period T of wave is defined as the time during which the wave covers a distance ' λ ' with phase velocity v .

$$\lambda = vT$$

$$\text{or } v = \frac{\lambda}{T}$$

Hence Equation (2) becomes

$$y(x, t) = y_m \sin\left(\frac{2\pi}{\lambda}\right)\left(x - \frac{\lambda t}{T}\right)$$

$$\Rightarrow y(x, t) = y_m \sin 2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right) \quad \text{--- (3)}$$

$$\Rightarrow y(x, t) = y_m \sin\left(\frac{2\pi x}{\lambda} - \frac{2\pi t}{T}\right)$$

$$\Rightarrow y(x, t) = y_m \sin(kx - \omega t) \quad \text{--- (4)}$$

Transverse Displacement of Sinusoidal Travelling is Periodic as the Function of Position

The transverse displacement ' y ' at any time ' t ' has the same value at $x, x + \lambda, x + 2\lambda$ and so on. Consider

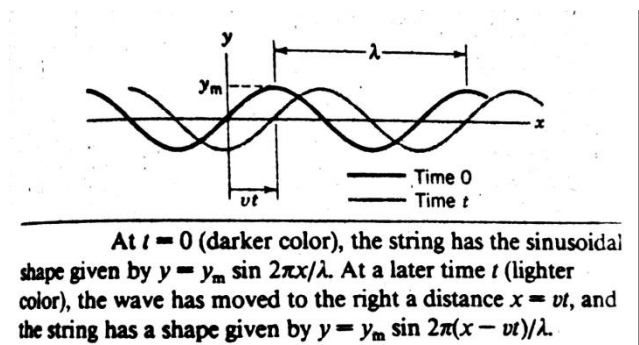
$$y(x + \lambda, t) = y_m \sin 2\pi\left(\frac{x + \lambda}{\lambda} - \frac{t}{T}\right)$$

$$\Rightarrow y(x + \lambda, t) = y_m \sin 2\pi\left(\frac{x}{\lambda} + 1 - \frac{t}{T}\right)$$

$$\Rightarrow y(x + \lambda, t) = y_m \sin\left(\frac{2\pi x}{\lambda} - \frac{2\pi t}{T} + 2\pi\right)$$

$$\Rightarrow y(x + \lambda, t) = y_m \sin\left(\frac{2\pi x}{\lambda} - \frac{2\pi t}{T}\right)$$

$$\Rightarrow y(x + \lambda, t) = y_m \sin 2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)$$



$$\Rightarrow y(x + \lambda, t) = y(x, t)$$

Transverse Displacement of Sinusoidal Travelling is Periodic as the Function of Time

Transverse displacement 'y' at any given position has the same value at time $t, t + T, t + 2T$ and so on. Consider

$$y(x, t + T) = y_m \sin 2\pi \left(\frac{x}{\lambda} - \frac{t + T}{T} \right)$$

$$\Rightarrow y(x, t + T) = y_m \sin 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} - 1 \right)$$

$$\Rightarrow y(x, t + T) = y_m \sin \left(\frac{2\pi x}{\lambda} - \frac{2\pi t}{T} - 2\pi \right)$$

$$\Rightarrow y(x, t + T) = y_m \sin \left(\frac{2\pi x}{\lambda} - 2\pi - \frac{2\pi t}{T} \right)$$

$$\Rightarrow y(x, t + T) = y_m \sin \left(\frac{2\pi x}{\lambda} - \frac{2\pi t}{T} \right)$$

$$\Rightarrow y(x, t + T) = y_m \sin 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right)$$

$$\Rightarrow y(x, t + T) = y(x, t)$$

19.4 Relation between Wave Number and Angular Frequency

Wave number is given by

$$k = \frac{2\pi}{\lambda} \quad \text{--- (1)}$$

Angular frequency is given by

$$\omega = \frac{2\pi}{T} \quad \text{But } \frac{1}{T} = f$$

$$\omega = 2\pi f \quad \text{--- (2)}$$

$$\text{As } v = f\lambda \Rightarrow f = \frac{v}{\lambda}$$

Relation (2) becomes:

$$\omega = 2\pi \frac{v}{\lambda}$$

$$\Rightarrow \omega = \frac{2\pi}{\lambda} v$$

$$\Rightarrow \omega = kv$$

$$\Rightarrow v = \frac{\omega}{k}$$

Problem: In an electric shaver, the blade moves back and forth over a distance of 2 mm. the motion is simple harmonic with frequency 120 Hz. Find (i) amplitude, (ii) maximum blade speed and (iii) maximum blade acceleration.

Given Data: Length $l = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$, Frequency $f = 120 \text{ Hz}$

- (i) Amplitude $x_m = \frac{\text{Length}}{2} = \frac{2 \times 10^{-3} \text{ m}}{2} = 1 \times 10^{-3} \text{ m} = 1 \text{ mm}$
- (ii) Maximum Speed $v_m = \omega x_m = 2\pi f x_m = 2 \times 3.14 \times 120 \times 1 \times 10^{-3} = 0.754 \frac{\text{m}}{\text{s}}$
- (iii) Maximum Acceleration $a_m = \omega^2 x_m = (2\pi f)^2 x_m = 4\pi^2 f^2 x_m$
 $= 4 \times (3.14)^2 \times (120)^2 \times 1 \times 10^{-3} = 0.568 \frac{\text{m}}{\text{s}^2}$

19.5 Group Speed and Dispersion

The speed of wave in any medium is called phase speed. But the term phase speed is used for those waves which keep their shape unchanged while traveling like sine wave. But in case of other types of waves e.g., square wave, saw tooth wave, we use the term group speed instead of phase speed.

“Group speed is the speed of group of waves in complex wave.”

This is the speed at which energy travels in a real wave.

Dispersive and Non-Dispersive Medium

Now the medium in which velocity of wave depends on wavelength is called dispersive medium. The wave, whose velocity changes with change in wavelength, is called dispersive wave. Hence, in a dispersive medium, different component of waves have different phase speeds. In a dispersive medium like glass, the group speed and phase speed of light waves become different.

In a non-dispersive medium, all component waves travel with the same speed e.g; sound waves in air and light waves in air.

19.6 Wave Speed

The wave speed means the phase speed of a single wave or group speed of a wave in non-dispersive medium. In non-dispersive medium the speed is independent of frequency or wave length. Here we will find the speed of a transverse wave in a stretched string. The wave speed can be calculated by two ways.

- (i) By Dimensional Analysis
- (ii) By Mechanical Analysis

19.7 Dimensional Analysis

The speed of waves on the string depends on the mass of the element of the string and on the force between the neighboring elements, which is the tension F under which the string is stretched.

We characterize the mass of the element of string in terms of the linear mass density μ , the mass per unit length of the string. Assuming that the wave speed v depends on only on F and μ . Thus we can describe the wave speed v as:

$$v \propto F^a \mu^b$$

$$\Rightarrow v = C F^a \mu^b \quad \text{--- (1)}$$

where C is the constant of proportionality and a and b are the exponents to be determined from the dimensional analysis. Therefore,

$$\begin{aligned}[v] &= C [F]^a [\mu]^b \\ \Rightarrow [LT^{-1}] &\propto [MLT^{-2}]^a [ML^{-1}]^b \\ \Rightarrow [LT^{-1}] &\propto [M^a L^a T^{-2a}] [M^b L^{-b}] \\ \Rightarrow [M^0 L T^{-1}] &\propto [M^{a+b} L^{a-b} T^{-2a}]\end{aligned}$$

Comparing the powers of the both sides, we have:

$$M: \quad a + b = 0 \Rightarrow a = -b \quad \text{--- (2)}$$

$$L: \quad a - b = 1 \Rightarrow -b - b = 1 \Rightarrow -2b = 1 \Rightarrow b = -\frac{1}{2}$$

Equation (2) implies:

$$a = -\left(-\frac{1}{2}\right) \Rightarrow a = \frac{1}{2}$$

Equation (1) becomes:

$$\begin{aligned}v &= C (F)^{\frac{1}{2}} (\mu)^{-\frac{1}{2}} \\ \Rightarrow v &= C \sqrt{\frac{F}{\mu}}\end{aligned}$$

The most we can say from this analysis is that the wave speed is equal to the dimensionless constant times $\sqrt{\frac{F}{\mu}}$. The value of the constant can be obtained from a mechanical analysis or from experiment.

19.8 Mechanical Analysis

Consider a stretched string in which transverse wave is produced which is moving with speed 'v'. Let us take a small element of the string of length δl . This element forms an arc of circle. Let R be the radius of circle. If μ is linear mass density of line string then mass of element of string is:

$$\delta m = \mu \delta l$$

Let F be the tension at the two ends of the element. This tension is along the tangent at each end of the string elements. The horizontal components of F cancel each other and the vertical components are added up.

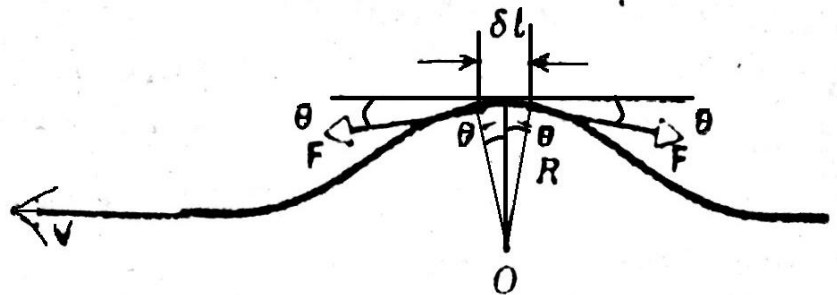
Total vertical force $F_{\perp}$ is

$$\begin{aligned}F_{\perp} &= F \sin \theta + F \sin \theta \\ \Rightarrow F_{\perp} &= 2F \sin \theta\end{aligned}$$

For small θ : $\sin \theta \approx \theta$. Therefore,

$$\begin{aligned}F_{\perp} &= 2F\theta \\ F_{\perp} &= F(2\theta) \quad \text{--- (1)}\end{aligned}$$

By using the relation $s = r\theta$, we have $\delta l = R(2\theta)$.



$$2\theta = \frac{\delta l}{R} \quad \text{--- -- -- -- --} \quad (2)$$

Putting (2) in (1), we get:

$$F_{\perp} = F \left(\frac{\delta l}{R} \right) \quad \text{--- -- -- -- --} \quad (3)$$

But $F_{\perp}$ is directed towards center of circular arc. So this becomes the centripetal force.

$$F_{\perp} = \frac{\delta m v^2}{R} \quad \text{--- -- -- -- --} \quad (4)$$

Comparing (3) and (4) we get,

$$\frac{\delta m v^2}{R} = F \left(\frac{\delta l}{R} \right)$$

$$\delta m v^2 = F(\delta l)$$

$$v^2 = \frac{F \delta l}{\delta m}$$

But $\frac{\delta m}{\delta l} = \mu \Rightarrow \frac{\delta l}{\delta m} = \frac{1}{\mu}$. Thus,

$$v^2 = \frac{F}{\mu}$$

$$\Rightarrow v = \sqrt{\frac{F}{\mu}}$$

This is the expression for wave speed for small transverse displacement.

Problem: When a particular wire is vibrating with frequency of 4 Hz. A transverse wave of wavelength 60 cm is produced. Determine the speed of wave pulse along wire

Given Data: Frequency $f = 4 \text{ Hz}$, Wavelength $\lambda = 60 \text{ cm} = 0.6 \text{ m}$

To Determine: Wave Speed $v = ?$

Calculations: As $v = f \lambda = 4 \times 0.6 = 2.4 \frac{\text{m}}{\text{s}}$

Problem. A sinusoidal travelling wave in x-direction has amplitude of 15 cm, wavelength of 40 cm and the frequency of 8 Hz. The vertical displacement at $x = 0$ and $t = 0$ is 15 cm. Find (a) Wave Number (b) Time Period, (c) Wave Speed and (d) Angular Frequency of wave.

Given Amplitude $A = 15 \text{ cm} = 15 \times 10^{-2} \text{ m}$, Wavelength $\lambda = 40 \text{ cm} = 40 \times 10^{-2} \text{ m}$,

Frequency $f = 8 \text{ Hz}$

(a) As Wave Number $k = \frac{2\pi}{\lambda} = \frac{2\pi}{40 \times 10^{-2}} = 15.7 \frac{\text{rad}}{\text{m}}$

(b) Time Period $T = \frac{1}{f} = \frac{1}{8} = 0.125 \text{ s}$

(c) As $S = vt \Rightarrow \lambda = vT \Rightarrow v = \frac{\lambda}{T} = \frac{40 \times 10^{-2}}{0.125} = 3.2 \frac{\text{m}}{\text{s}}$

(d) $\omega = 2\pi f = 6.28 \times 8 = 50.24 \frac{\text{rad}}{\text{s}}$

Problem: A uniform cord has a mass of 0.3 kg and length of 6 m. the cord passes over pulley and supports a 2 kg object. (a) What will be the speed of wave? (b) What will be the time for wave to travel the distance of 6 m?

(a) Given Mass of String $m_1 = 0.3 \text{ kg}$, Length of String $l = 6 \text{ m}$, Mass suspended $m_2 = 2 \text{ kg}$

$$\text{As Linear Mass Density } \mu = \frac{m}{l} = \frac{0.3}{6} = 0.05 \frac{\text{kg}}{\text{m}}$$

$$\text{Tension in String } F = mg = 2 \times 9.8 = 19.6 \text{ N}$$

$$\text{Now } v = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{19.6}{0.05}} = 19.77 \text{ ms}^{-1}$$

(b) Distance Covered by Wave $S = 6 \text{ m}$

Time $t = ?$

$$\text{As } S = vt \Rightarrow t = \frac{S}{v} = \frac{6}{19.77} = 0.304 \text{ s}$$

Problem: Transverse wave with speed of 50 m/s are to be produced in taut string. A 5 m length of the string with a mass of 0.060 kg is used. Find out the tension in the string.

Given Speed of Wave $v = 50 \frac{\text{m}}{\text{s}}$, Length of String $l = 5 \text{ m}$, Mass of string $m = 0.05 \text{ kg}$

Tension in String $F = ?$

$$\text{As Linear Mass Density } \mu = \frac{m}{l} = \frac{0.06}{5} = 0.012 \frac{\text{kg}}{\text{m}}$$

$$\text{Now } v = \sqrt{\frac{F}{\mu}} \Rightarrow v^2 = \frac{F}{\mu} \Rightarrow F = \mu v^2 = 0.012 \times 50^2 = 30 \text{ N}$$

Problem: Transverse wave travel with speed of 20 m/s in a string under the tension of 6 N. What tension is required to produce a wave travelling with speed 30 m/s in the same string?

Case 1: Given Speed of wave $v_1 = 20 \frac{\text{m}}{\text{s}}$, Tension $F_1 = 6 \text{ N}$

Linear Mass Density $\mu = ?$

$$\text{Now } v_1 = \sqrt{\frac{F_1}{\mu}} \Rightarrow (v_1)^2 = \frac{F_1}{\mu} \Rightarrow \mu = \frac{F_1}{(v_1)^2} = \frac{6}{(20)^2} = 0.015 \frac{\text{kg}}{\text{m}}$$

Case 2: Given Speed of Wave $v_2 = 30 \frac{\text{m}}{\text{s}}$

To Determine Tension in String $F_2 = ?$

$$\text{For present case: } v_2 = \sqrt{\frac{F_2}{\mu}} \Rightarrow (v_2)^2 = \frac{F_2}{\mu} \Rightarrow F_2 = \mu (v_2)^2 = 0.015 \times (30)^2 = 13.5 \text{ N}$$

19.9 The Wave Equation

In case of SHM of a mass-spring mass system, the equation of motion is:

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

The solution of the equation is

$$x = x_m \cos(\omega t + \varphi)$$

Consider a string lying along x-axis. Let the given string be stretched between two fixed supports along x-axis in the equilibrium position. Let F_c be the tension in the string in equilibrium position.

Consider a small element of string. Let ∂x and ∂m are the arc length and mass of this string, then

linear mass density $\mu = \frac{\partial m}{\partial x}$. When the string is plucked (displaced) along y-axis then it does not remain straight but become slight curved as shown. Let F_1 and F_2 are the tension at the ends in the displaced position. The angle θ_1 and θ_2 be not equal. Now we resolve the tangent into its components. The horizontal components $F_1 \cos \theta_1$ and $F_2 \cos \theta_2$ being equal and opposite will cancel each other. The net force acting on the element in the upward direction is given as:

$$F_y = F_2 \sin \theta_2 - F_1 \sin \theta_1$$

$$\Rightarrow F_y = F_2 \cos \theta_2 \left(\frac{\sin \theta_2}{\cos \theta_2} \right) - F_1 \cos \theta_1 \left(\frac{\sin \theta_1}{\cos \theta_1} \right)$$

$$\Rightarrow F_y = F_2 \cos \theta_2 \tan \theta_2 - F_1 \cos \theta_1 \tan \theta_1$$

Let $F_2 \cos \theta_2 = F_1 \cos \theta_1 = F$ (say)

$$\Rightarrow F_y = F \tan \theta_2 - F \tan \theta_1$$

$$\Rightarrow F_y = F \partial(\tan \theta) \quad \text{--- (1) where } \partial(\tan \theta) = \tan \theta_2 - \tan \theta_1$$

But this net force on the element of string is given by Newton 2nd law of motion

$$F_y = \partial m a_y = \mu \partial x a_y \quad \text{--- (2) } \because \frac{\partial m}{\partial x} = \mu$$

From (1) and (2) we get:

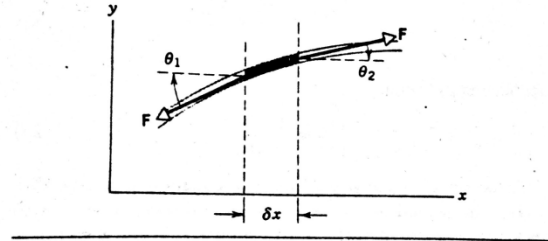
$$F \partial(\tan \theta) = \mu \partial x a_y$$

$$\frac{\partial(\tan \theta)}{\partial x} = \frac{\mu a_y}{F} \quad \text{--- (3)}$$

Now $\tan \theta = \frac{\partial y}{\partial x}$ and $a_y = \frac{\partial^2 y}{\partial t^2}$

Eq (3) become

$$\frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x} \right) = \frac{\mu}{F} \left(\frac{\partial^2 y}{\partial t^2} \right)$$



A small element of length δx of a long string under tension F . The figure represents a snapshot of the element at a particular time during the passage of a wave.

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{F} \frac{\partial^2 y}{\partial t^2} \quad \text{--- (4)}$$

$$\text{But } v = \sqrt{\frac{F}{\mu}} \Rightarrow \frac{1}{v} = \sqrt{\frac{\mu}{F}} \Rightarrow \frac{\mu}{F} = \frac{1}{v^2}$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

This is equation of motion of transverse oscillation of the string. So it is also called wave equation.

19.10 Solution of Wave Equation

As the wave eq is given by:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \left(\frac{\partial^2 y}{\partial t^2} \right) \quad \text{--- (1)}$$

We want to find out the solution of this equation.

We know that general formula for traveling wave is

$$y(x, t) = f(x \pm vt) \quad \text{--- (2)}$$

Let us suppose that equation (2) is the solution of (1).

Suppose $x \pm vt = z$ --- (A)

Therefore, equation (2) can be written as

$$y(x, t) = f(z) \quad \text{--- (B)}$$

Differentiating (B) w.r.t x

$$\begin{aligned} \frac{\partial y}{\partial x} &= \frac{\partial f(z)}{\partial x} \\ \Rightarrow \frac{\partial y}{\partial x} &= \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} \\ \Rightarrow \frac{\partial y}{\partial x} &= \frac{\partial f}{\partial z} \quad \because \frac{\partial z}{\partial x} = \frac{\partial(x \pm vt)}{\partial x} = 1 \end{aligned}$$

Again differentiating with respect to x we get:

$$\begin{aligned} \frac{\partial^2 y}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial z} \right) \\ \Rightarrow \frac{\partial^2 y}{\partial x^2} &= \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial x} \right) \\ \Rightarrow \frac{\partial^2 y}{\partial x^2} &= \frac{\partial}{\partial z} \cdot \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} \quad \because \frac{\partial z}{\partial x} = \frac{\partial(x \pm vt)}{\partial x} = 1 \\ \Rightarrow \frac{\partial^2 y}{\partial x^2} &= \frac{\partial^2 f}{\partial z^2} \quad \text{--- (3)} \end{aligned}$$

Similarly differentiating (B) with respect to t , we have:

$$\frac{\partial y}{\partial t} = \frac{\partial f(z)}{\partial t} = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial t}$$

$$\Rightarrow \frac{\partial y}{\partial t} = \frac{\partial f}{\partial z}(\pm v) \quad \because \frac{\partial z}{\partial t} = \frac{\partial(x \pm vt)}{\partial t} = \pm v$$

Again Differentiating with respect to t:

$$\begin{aligned} \frac{\partial^2 y}{\partial t^2} &= \frac{\partial}{\partial t} \left(\pm v \frac{\partial f}{\partial z} \right) \\ \Rightarrow \frac{\partial^2 y}{\partial t^2} &= (\pm v) \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial t} \right) = (\pm v) \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial t} \right) = (\pm v) \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial t} \right) \\ \Rightarrow \frac{\partial^2 y}{\partial t^2} &= (\pm v) \frac{\partial}{\partial z} \left[\frac{\partial f}{\partial z} (\pm v) \right] = (\pm v)^2 \frac{\partial}{\partial z} \left[\frac{\partial f}{\partial z} \right] \\ \Rightarrow \frac{\partial^2 y}{\partial t^2} &= v^2 \frac{\partial^2 f}{\partial x^2} \\ \Rightarrow \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} &= \frac{\partial^2 f}{\partial x^2} \quad \text{----- (4)} \end{aligned}$$

From (3) and (4), we have:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

So $y(x, t) = f(x \pm vt)$ is the solution of wave equation.

Problem: Show that the wave-function $y = \ln[b(x - vt)]$ is the solution of wave equation, where b is constant.

The wave equation is: $\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$

We have to show that $y = \ln[b(x - vt)]$ is the solution of wave equation.

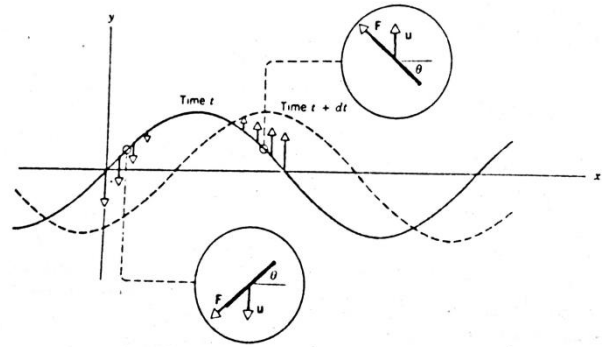
$$\begin{aligned} L.H.S &= \frac{\partial^2 y}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \ln[b(x - vt)] \right) \\ &= \frac{\partial}{\partial x} \left(\frac{1}{b(x - vt)} \frac{\partial}{\partial x} [b(x - vt)] \right) = \frac{\partial}{\partial x} \left(\frac{b}{b(x - vt)} \right) = \frac{\partial}{\partial x} (x - vt)^{-1} \\ &= (-1)(x - vt)^{-2} \frac{\partial}{\partial x} (x - vt) = \frac{-1}{(x - vt)^2} \\ R.H.S &= \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} = \frac{1}{v^2} \frac{\partial}{\partial t} \left(\frac{\partial y}{\partial t} \right) = \frac{1}{v^2} \frac{\partial}{\partial t} \left(\frac{\partial}{\partial t} \ln[b(x - vt)] \right) \\ &= \frac{1}{v^2} \frac{\partial}{\partial t} \left(\frac{1}{b(x - vt)} \frac{\partial}{\partial t} [b(x - vt)] \right) = \frac{1}{v^2} \frac{\partial}{\partial t} \left(\frac{-bv}{b(x - vt)} \right) = -\frac{1}{v} \frac{\partial}{\partial t} (x - vt)^{-1} \\ &= -\frac{1}{v} (-1)(x - vt)^{-2} \frac{\partial}{\partial t} (x - vt) = -\frac{1}{v} (-1) \frac{1}{(x - vt)^2} (-v) = \frac{-1}{(x - vt)^2} \end{aligned}$$

As L.H.S = R.H.S, so $y = \ln[b(x - vt)]$ is the solution of wave equation.

19.11 Power and Intensity in Wave Motion

When move the end of string up and down we give energy to string .This energy is transmitted through the string to the other end. Figure shows the snapshot of the waves at time t and $t + dt$.

When wave passes along the string then each particle perform work to the other particle to its right and work is done on that particle by the particle to its left. In this way energy is transferred from particle to particle.



When the string is moved, the particles of string have transverse velocity u given as follow:

$$u(x, t) = \frac{\partial y}{\partial t} = \frac{\partial}{\partial t} [y_m \sin(kx - \omega t)] \quad \because y = y_m \sin(kx - \omega t)$$

$$\Rightarrow u(x, t) = -\omega y_m \cos(kx - \omega t) \quad \text{----- (a)}$$

The force F exerted on an element of the string by the element to its left is shown in the figure. Now the power transmitted is given as:

$$P = uF_y \quad \text{----- (1)}$$

Note that y-component of force F is parallel to u , no matter whether that element of the string happens to be moving up or down. Thus the power transmitted is never negative during the cycle of oscillation. Now as the vertical force on the element of string is:

$$F_y = -F \sin \theta$$

For small angle $\sin \theta \approx \tan \theta$

$$F_y = -F \tan \theta$$

$$\Rightarrow F_y = -F \left(\frac{\partial y}{\partial x} \right) \quad \text{----- (b)}$$

Putting values of equation (a) and (b) in equation (1), we get:

$$P = [-\omega y_m \cos(kx - \omega t)] \left(-F \frac{\partial y}{\partial x} \right)$$

$$\Rightarrow P = [-\omega y_m \cos(kx - \omega t)] (-F) \frac{\partial}{\partial x} [y_m \sin(kx - \omega t)]$$

$$\Rightarrow P = [-\omega y_m \cos(kx - \omega t)] (-F) [y_m k \cos(kx - \omega t)]$$

$$\Rightarrow P = y_m^2 \omega k F \cos^2(kx - \omega t)$$

$$\Rightarrow P = y_m^2 \omega k \mu v^2 \cos^2(kx - \omega t) \quad \because v = \sqrt{\frac{F}{\mu}} \Rightarrow v^2 = \frac{F}{\mu} \Rightarrow F = \mu v^2$$

$$\Rightarrow P = y_m^2 \omega (kv) \mu v \cos^2(kx - \omega t)$$

$$\Rightarrow P = y_m^2 \omega(\omega) \mu v \cos^2(kx - \omega t) \quad \because kv = \omega$$

$$\Rightarrow P = y_m^2 \omega^2 \mu v \cos^2(kx - \omega t)$$

Average Power

The power or rate of flow of energy is not constant. This is because the input power oscillates: the work done by the hand that is moving the end of the string varies with the transverse displacement of that point. The power input to the string is often taken to be the average over one period of motion. The average power delivered is:

$$\bar{P} = y_m^2 \mu v \omega^2 < \cos^2(kx - \omega t) >$$

$$\Rightarrow \bar{P} = \frac{1}{2} y_m^2 \mu v \omega^2 \quad \because < \cos^2(kx - \omega t) > = \frac{1}{2}$$

This shows that average power does not depend on x and t but it depends on square of amplitude and square of frequency.

Problem: A taut string for which $\mu = 5 \times 10^{-2} \frac{kg}{m}$ is under tension of 80 N. How much power must be supplied to the string to generate sinusoidal waves at frequency of 60 Hz and an amplitude of 6 m.

Given Linear Mass Density $\mu = 5 \times 10^{-2} \frac{kg}{m}$, Tension $F = 80 \text{ N}$, Frequency $f = 60 \text{ Hz}$,

Amplitude $y_m = 6 \text{ m}$

Average Power $\langle P \rangle = \frac{1}{2} y_m^2 \mu v \omega^2 = ?$

As $\omega = 2\pi f = 2 \times 3.14 \times 60 = 376.8 \frac{rad}{s}$

$$v = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{80}{5 \times 10^{-2}}} = \sqrt{1600} = 40 \frac{m}{s}$$

$$\text{Therefore } \langle P \rangle = \frac{1}{2} y_m^2 \mu v \omega^2 = \frac{1}{2} \times (6)^2 \times 5 \times 10^{-2} \times 40 \times (376.8)^2 = 5111 \text{ W}$$

Problem: Transverse waves are being generated on a rope under constant tension. By what factor is the required power increased or decreased, if: (a) Length of rope is doubled and angular frequency remains constant. (b) The amplitude is doubled and angular frequency is halved. (c) Both the wavelength and the amplitude are doubled. (d) Both the wavelength and length of rope are halved.

$$\text{Average Power } \langle P \rangle = \frac{1}{2} y_m^2 \mu v \omega^2 = \frac{1}{2} y_m^2 \left(\frac{m}{l} \right) v \omega^2 \quad \because \mu = \frac{m}{l}$$

(i) Length of rope is doubled and angular frequency remains constant. Replace l by $2l$

$$\langle P_1 \rangle = \frac{1}{2} y_m^2 \left(\frac{m}{2l} \right) v \omega^2 = \frac{1}{2} \left[\frac{1}{2} y_m^2 \left(\frac{m}{l} \right) v \omega^2 \right] = \frac{1}{2} \left[\frac{1}{2} y_m^2 \mu v \omega^2 \right] = \frac{1}{2} \langle P \rangle$$

The average power delivered will be decreased of 50% of its initial value.

(ii) The amplitude is doubled and angular frequency is halved. Replace y_m by $2y_m$ and ω by $\frac{\omega}{2}$

$$\langle P_2 \rangle = \frac{1}{2} (2y_m)^2 \mu v \left(\frac{\omega}{2} \right)^2 = \frac{1}{2} (4y_m^2) \mu v \omega^2 \mu v \frac{\omega^2}{4} = \frac{1}{2} y_m^2 \mu v \omega^2 = \langle P \rangle$$

The average power delivered will remain unchanged.

(iii) Both the wavelength and the amplitude are doubled.

$$\text{Consider } \langle P \rangle = \frac{1}{2} y_m^2 \mu v \omega^2 = \frac{1}{2} y_m^2 \mu f \lambda \omega^2$$

Replace y_m by $2y_m$ and λ by 2λ :

$$\langle P_3 \rangle = \frac{1}{2} (2y_m)^2 \mu f (2\lambda) \omega^2 = \frac{1}{2} 2 (4y_m^2) \mu f \lambda \omega^2 = 8 \left[\frac{1}{2} y_m^2 \mu v \omega^2 \right] = 8 \langle P \rangle$$

The average power delivered will increase 8 times.

(iv) Both the wavelength and the length of rope are halved.

$$\text{Consider } \langle P \rangle = \frac{1}{2} y_m^2 \mu v \omega^2 = \frac{1}{2} y_m^2 \left(\frac{m}{l} \right) f \lambda \omega^2$$

Replace l by $\frac{l}{2}$ and λ by $\frac{\lambda}{2}$:

$$\langle P_4 \rangle = y_m^2 \left(\frac{m}{2l} \right) f (2\lambda) \omega^2 = \langle P \rangle$$

The average power delivered will remain unchanged.

19.12 Intensity of Wave of a 3D Wave

In a three dimensional wave, such as a light wave or a sound wave from a point source, it is often more useful to specify the intensity of the wave. The intensity I is defined as:

“ The average power per unit area transmitted across an area normal to the direction in which the wave is travelling ”

$$I = \frac{\bar{P}}{A}$$

The intensity of a wave travelling along a string is proportional to the square of its amplitude but it is not true for a circular or spherical wave because amplitude of spherical wave is not constant and is a function of distance from the point source. The intensity of a spherical wave from a point source located in an isotropic medium is given by,

$$I = \frac{P}{A} = \frac{P}{4\pi r^2} = \frac{P}{4\pi} \left(\frac{1}{r^2} \right)$$

$$\Rightarrow I = \text{constant} \times \frac{1}{r^2}$$

Since $I \propto (\text{Amplitude})^2$. Therefore, for present case:

$$(\text{Amplitude})^2 \propto \frac{1}{r^2}$$

$$\Rightarrow (\text{Amplitude}) \propto \frac{1}{r}$$

So we find that amplitude of spherical wave is inversely proportional to distance from the point source i.e if the distance from the source is doubled, the amplitude reduces to half and intensity decreases four times.

Problem: A sound wave of intensity $1.6 \frac{\mu W}{cm^2}$ passes through a surface of area $4.70 cm^2$. How much energy passes through the surface in one hour?

Given Data: Intensity $I = 1.6 \frac{\mu W}{cm^2} = 1.6 \times \frac{10^{-6} W}{10^{-4} m^2} = 1.6 \times 10^{-2} \frac{W}{m^2}$,

Area $A = 4.70 cm^2 = 4.70 \times 10^{-4} m^2$, Time $t = 60 s$

To Determine: Energy $E = ?$

Calculation: $I = \frac{P}{A} \Rightarrow I = \frac{E}{A \times t} \Rightarrow E = A \times t \times I = 4.70 \times 10^{-4} \times 60 \times 1.6 \times 10^{-2}$

$E = 4.512 \times 10^{-4} J$

19.13 The Principal of Superposition

When two or more than waves travelling through the same medium combine at a point then the resultant displacement of any particle at any given time is equal to the vector sum of displacement of individual waves given to the particle. For example, if two waves travel simultaneously along a stretched string and $y_1(x, t)$, $y_2(x, t)$ are the displacements due to the two waves separately then the resultant displacement of the particle is

$$y(x, t) = y_1(x, t) + y_2(x, t)$$

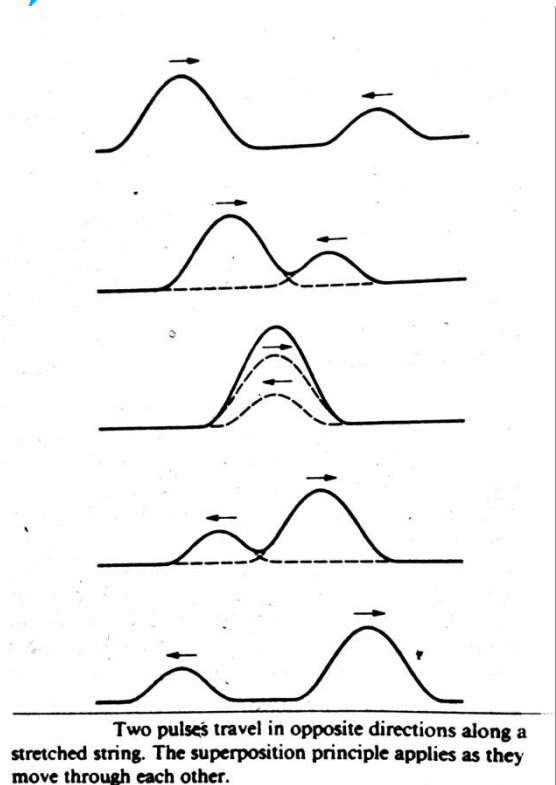
This is called superposition principle. There are three application of principle of superposition.

(i) **Interference:** Superposition of two waves having same frequency and are moving in same direction will result in the phenomenon of interference.

(ii) **Beats:** Periodic rise and fall in the amplitude of resultant wave due to superposition of two waves having slightly different frequency but moving in a same direction are called beats.

(iii) **Stationary Waves:** Superposition of two waves

having same frequency but moving in opposite direction will result in phenomenon of stationary waves.



19.14 Interference of Waves

Superposition of two waves having same frequency and are moving in same direction will result in the phenomenon of interference.

Consider two sinusoidal waves of same amplitude and frequency travelling along x-axis with the same speed. These waves are represented as

$$y_1(x, t) = y_m \sin(kx - \omega t - \varphi_1)$$

$$y_2(x, t) = y_m \sin(kx - \omega t - \varphi_2)$$

Where φ_1 and φ_2 are phase constants for two waves.

The resultant wave is given by the principle of superposition as

$$y = y_1(x, t) + y_2(x, t)$$

$$\Rightarrow y = y_m \sin(kx - \omega t - \varphi_1) + y_m \sin(kx - \omega t - \varphi_2)$$

$$\Rightarrow y = y_m [\sin(kx - \omega t - \varphi_1) + \sin(kx - \omega t - \varphi_2)]$$

By trigonometry $\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

$$y = y_m \left[2 \sin\left(\frac{kx - \omega t - \varphi_1 + kx - \omega t - \varphi_2}{2}\right) \cos\left(\frac{kx - \omega t - \varphi_1 - kx + \omega t + \varphi_2}{2}\right) \right]$$

$$\Rightarrow y = y_m \left[2 \sin\left(\frac{2kx - 2\omega t - \varphi_1 - \varphi_2}{2}\right) \cos\left(\frac{\varphi_2 - \varphi_1}{2}\right) \right]$$

$$\Rightarrow y = y_m \left[2 \sin\left(\frac{2kx - 2\omega t}{2} - \frac{\varphi_1 + \varphi_2}{2}\right) \cos\left(\frac{\varphi_2 - \varphi_1}{2}\right) \right]$$

$$\Rightarrow y = y_m \left[2 \sin\left((kx - \omega t) - \frac{\varphi_1 + \varphi_2}{2}\right) \cos\left(\frac{\varphi_2 - \varphi_1}{2}\right) \right]$$

$$\Rightarrow y = 2y_m \cos\left(\frac{\varphi_2 - \varphi_1}{2}\right) \sin\left((kx - \omega t) - \frac{\varphi_1 + \varphi_2}{2}\right)$$

Putting $\frac{\varphi_1 + \varphi_2}{2} = \varphi'$ and $\varphi_2 - \varphi_1 = \Delta\varphi$

$$y = 2y_m \cos\left(\frac{\Delta\varphi}{2}\right) \sin((kx - \omega t) - \varphi') \text{ --- (1)}$$

From equation (1) we find that the resultant waves has same frequency but its amplitude is $2y_m \cos\left(\frac{\Delta\varphi}{2}\right)$.

Constructive Interference

For the case of constructive interference, crest of one wave falls on crest of other wave and trough of one wave falls on trough of other wave. i.e., $\Delta\varphi = 0$. In this case,

$$\text{Amplitude of resultant wave} = 2y_m \cos\left(\frac{\Delta\varphi}{2}\right) = 2y_m \cos 0 = 2y_m$$

i.e., the amplitude of the resultant wave is twice the magnitude of either wave.

Condition for Constructive Interference

Now for condition of constructive interference, we have:

$$\cos\left(\frac{\Delta\phi}{2}\right) = \pm 1$$

$$\Rightarrow \frac{\Delta\phi}{2} = \cos^{-1}(\pm 1)$$

$$\Rightarrow \frac{\Delta\phi}{2} = 0, \pi, 2\pi, \dots$$

$$\Rightarrow \frac{\Delta\phi}{2} = n\pi$$

$$\Rightarrow \Delta\phi = 2n\pi$$

Path difference Δx and phase difference $\Delta\phi$ is related as by the expression

$$\frac{\Delta x}{\lambda} = \frac{\Delta\phi}{2\pi}$$

$$\Rightarrow \Delta x = \frac{2n\pi}{2\pi} \cdot \lambda$$

$$\Rightarrow \Delta x = n\lambda$$

Destructive Interference

For the case of destructive interference, crest of one wave falls on trough of other wave i.e., $\Delta\phi = 180^\circ$. In this case,

$$\text{Amplitude of resultant wave} = 2y_m \cos\left(\frac{\Delta\phi}{2}\right) = 2y_m \cos\left(\frac{180^\circ}{2}\right) = 2y_m \cos 90^\circ = 0$$

i.e., the two waves will cancel the effect of each other and the amplitude of the resultant wave is zero.

Condition for Destructive Interference

For condition of destructive interference, we have:

$$\cos\left(\frac{\Delta\phi}{2}\right) = 0$$

$$\Rightarrow \frac{\Delta\phi}{2} = \cos^{-1}(0)$$

$$\Rightarrow \frac{\Delta\phi}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\Rightarrow \Delta\phi = \pi, 3\pi, 5\pi, \dots$$

$$\Rightarrow \Delta\phi = (2n + 1)\pi$$

Path difference and phase difference is related by the expression:

$$\frac{\Delta x}{\lambda} = \frac{\Delta\phi}{2\pi}$$

$$\Rightarrow \Delta x = \frac{(2n + 1)\pi}{2\pi} \cdot \lambda$$

$$\Rightarrow \Delta x = (2n + 1) \frac{\lambda}{2}$$

Sample Problem 4: Two waves travel in the same direction along a string and interfere. The waves have the same amplitude and the same wavelength and travel with the same speed. The amplitude of the waves is 9.7mm and there is a phase difference of 110° between them. (a) What is the amplitude of the combined waves? To what value should the phase difference be changed so that the combined wave will have amplitude equal to that one of the original waves?

Solution:

Amplitude of one wave $y_m = 9.7\text{mm}$, Phase Difference $\Delta\phi = 110^\circ$

(a) Amplitude of Resultant Wave =?

$$\text{Amplitude of Resultant Wave} = 2y_m \cos\left(\frac{\Delta\phi}{2}\right)$$

$$= 2 \times 9.7 \times \cos \frac{110^\circ}{2} = 19.4 \times \cos 55^\circ = 19.4 \times (0.573) = 11.1 \text{ mm}$$

(b) $\Delta\phi = ?$ if Amplitude of Resultant = y_m

$$\text{According to the given condition } 2y_m \cos\left(\frac{\Delta\phi}{2}\right) = y_m$$

$$\Rightarrow 2\cos\left(\frac{\Delta\phi}{2}\right) = 1 \Rightarrow \frac{\Delta\phi}{2} = \cos^{-1}\left(\frac{1}{2}\right) \Rightarrow \frac{\Delta\phi}{2} = 60^\circ \Rightarrow \Delta\phi = 120^\circ$$

Sample Problem 5: A listener is seated at a point distant 1.2m directly in front of one speaker. The two speakers which are separated by a distance of $D = 2.3 \text{ m}$ emit pure tones of wavelength λ . The waves are in phase when they leave the speakers. For what wavelength will the listener hear a minimum in the sound intensity?

Solution: Given $x = 1.2\text{m}$, $D = 2.3\text{m}$,

We have to Determine $\lambda = ?$ for minimum sound.

As for minimum sound the two waves should interfere destructively

$$\text{Path difference} = \left(n + \frac{1}{2}\right)\lambda$$

$$\text{or } \Delta x = \left(n + \frac{1}{2}\right)\lambda \Rightarrow x_1 - x_2 = \left(n + \frac{1}{2}\right)\lambda \quad \text{--- (1)}$$

If the listener is seated in front of speaker 2 then

$$x_2 = 1.2\text{m} \quad \therefore x_1^2 = x_2^2 + y^2 \Rightarrow x_1 = \sqrt{(1.2)^2 + (2.3)^2}$$

$$\Rightarrow x_1 = \sqrt{1.44 + 5.29} = \sqrt{6.73} = 2.59 \Rightarrow x_1 = 2.6\text{m}$$

Putting the values in equation (1) we get

$$2.6 - 1.2 = \left(n + \frac{1}{2}\right)\lambda \Rightarrow 1.4 = \left(n + \frac{1}{2}\right)\lambda \Rightarrow \lambda = \frac{1.4}{\left(n + \frac{1}{2}\right)}$$

$$\lambda = \frac{1.4}{\left(\frac{1}{2}\right)}, \frac{1.4}{\left(\frac{3}{2}\right)}, \frac{1.4}{\left(\frac{5}{2}\right)}, \dots \dots \dots \Rightarrow \lambda = 2(1.4), \frac{2}{3}(1.4), \frac{2}{5}(1.4), \dots$$

$$\lambda = 2.8\text{m}, 0.93\text{m}, 0.56\text{m}, \dots \dots \dots$$

19.15 Standing Waves

Superposition of two exactly similar waves moving in a medium along the same straight line in opposite directions give rise to the phenomenon of standing waves

These waves are not actually stationary but only appear to be so. These waves are also called standing waves because no moving particle is visible on the string. The pattern formed seems to be stationary.

- In standing waves, there are certain points on the string which do not vibrate i.e., their amplitude of vibration is zero. Such points are called Nodes.
- Between two nodes there are points where amplitude of vibration is maximum. Such points are called Anti-Nodes. So standing waves consists of nodes and antinodes. The distance between two consecutive nodes or antinodes is equal to $\frac{\lambda}{2}$.

Mathematical Analysis of Standing Waves:

Consider two identical waves moving along a string in opposite direction. These waves are represented as

$$y_1(x, t) = y_m \sin(kx - \omega t) \quad \text{towards right}$$

$$y_2(x, t) = y_m \sin(kx + \omega t) \quad \text{towards left}$$

By the superposition principle the resultant wave is

$$y(x, t) = y_1(x, t) + y_2(x, t)$$

$$\Rightarrow y(x, t) = y_m \sin(kx - \omega t) + y_m \sin(kx + \omega t)$$

$$\Rightarrow y(x, t) = y_m [\sin(kx - \omega t) + \sin(kx + \omega t)]$$

By trigonometry $\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

$$y(x, t) = 2y_m \left[\sin\left(\frac{kx - \omega t + kx + \omega t}{2}\right) \cos\left(\frac{kx - \omega t - kx - \omega t}{2}\right) \right]$$

$$\Rightarrow y(x, t) = 2y_m [\sin(kx) \cos(-\omega t)] \quad \because \cos(-\theta) = \cos \theta$$

$$\Rightarrow y(x, t) = 2y_m [\sin(kx) \cos(\omega t)] \quad \text{--- (1)}$$

This is the equation of a standing wave. This equation shows that each particle performs S.H.M with same angular frequency ω but amplitude of different particles are different because amplitude varies with x .

Condition for Nodes (Points of Maximum Amplitude):

In equation (1) amplitude is given by

$$A = 2y_m \sin kx \quad \text{--- (2)}$$

We find that A is maximum ($= 2y_m$), when

$$\sin kx = 1$$

$$\Rightarrow kx = \sin^{-1}(1)$$

$$\Rightarrow kx = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\Rightarrow kx = (2n + 1) \frac{\pi}{2} \quad \because n \text{ is integer}$$

$$\Rightarrow \frac{2\pi}{\lambda} x = (2n + 1) \frac{\pi}{2} \quad \because k = \frac{2\pi}{\lambda}$$

$$\Rightarrow x = (2n + 1) \frac{\lambda}{4}$$

These points of maximum amplitude are called antinodes

Condition for Anti-Nodes (Points of Minimum Amplitude):

From the expression of amplitude of stationary wave $A = 2y_m \sin kx$, we find that the amplitude is zero at all points for which:

$$kx = 0, \pi, 2\pi, 3\pi, \dots$$

$$\Rightarrow kx = n\pi \quad \because n \text{ is integer}$$

$$\Rightarrow \frac{2\pi}{\lambda} x = n\pi \quad \because k = \frac{2\pi}{\lambda}$$

$$\Rightarrow x = n \left(\frac{\lambda}{2} \right)$$

19.16 Phase Change on Reflection

Reflection of Wave from Denser Medium

Consider a coil of spring with two ends connected to two hooks as shown in the figure. Suppose two hooks are denser than the spring. When an upward displacement is produced in the spring at right end, this disturbance travels to the left end of spring.

On reaching the left end, the upward displacement tries to raise the hook also. But the hook being denser is not raised. Due to the reaction of hook, the upward displacement becomes downward displacement. Thus we have generated a rule,

When a transverse wave is reflected from a denser medium, it is reflected with opposite phase i.e., upward displacement is reflected as a downward displacement etc. The reflected wave has a phase difference of 180° or path difference of $\frac{\lambda}{2}$.

Reflection of Wave from Rare Medium

Now suppose the left end of spring is tied to a medium rarer than the spring (e.g., nylon thread). Now upward displacement produced at right end can raise the nylon thread reaching it. A part of the pulse travels into the nylon thread and a part goes back as upward displacement. Thus we find that:

“When a transverse wave is reflected at the boundary of a rare medium, it will be reflected without any phase change i.e., upward displacement is reflected as upward displacement and vice versa”.

19.17 Natural Frequencies

When a system (simple pendulum) is made to oscillate, it vibrates with a certain time period and certain frequency. Whenever this pendulum is vibrating, it will always vibrate with the same period and frequency. This natural period and natural frequency depend upon the length of pendulum.

Consider a string of length 'L' stretched with the one fix hook and the other end held in hand. When we move the held in hand up and down, we send transverse waves to the hook. These waves reach the hook and are reflected back. Due to superposition between incident and reflected waves, stationary waves are formed. If we wiggle the string with different frequencies, the string vibrates in different modes. Figure shows different modes of oscillation of the string. These modes depend upon frequency. The end fixed to hook and the free end in hand acts as nodes.

To produce standing waves in the string of length L, the following condition should be satisfied.

$$L = \frac{n\lambda_n}{2} \quad n = 1, 2, 3, 4 \dots \dots \dots$$

where λ_n is the wavelength of nth mode. For nth mode, the wavelength to given by:

$$\lambda_n = \frac{2L}{n} \quad n = 1, 2, 3, 4 \dots \dots \dots$$

The condition for stationary waves can be described in terms of frequency by using relation

$$\begin{aligned} v &= f_n \lambda_n \\ \Rightarrow v &= \frac{2L}{n} f_n \\ \Rightarrow f_n &= n \left(\frac{v}{2L} \right) \quad n = 1, 2, 3, 4 \dots \dots \dots \end{aligned}$$

Thus the frequencies of stationary waves are quantized. It means that the string should be shaken at these particular frequencies in order to produce stationary waves. These frequency are called natural frequencies.

Problem: A 300 g wire is stretched between 2.7 cm apart. If the tension in the wire is 600 N, find the wire's first, second and third harmonic.

Given Mass of wire $m = 0.300 \text{ g} = 0.0003 \text{ kg}$, Length of wire $L = 2.7 \text{ cm} = 2.7 \times 10^{-2} \text{ m}$

Tension in String $F = 600 \text{ N}$

First Harmonic: $f_1 = \frac{v}{2L} = \frac{1}{2L} \sqrt{\frac{F}{\mu}} = \frac{1}{2L} \sqrt{\frac{F}{(m/L)}} = \frac{1}{2L} \sqrt{\frac{F \times L}{m}} = \frac{1}{2 \times 2.7 \times 10^{-2}} \sqrt{\frac{600 \times 2.7 \times 10^{-2}}{0.0003}} = 2618 \text{ Hz}$

Second Harmonic: $f_2 = 2f_1 = 2(2618) = 5236 \text{ Hz}$

Third Harmonic: $f_3 = 3f_1 = 3(2618) = 7854 \text{ Hz}$

Problem: What are three lowest frequency for stationary wave on a wire 9.88 m long having mass 0.107 kg which is stretched under tension of 236 N.

Given Mass of wire $m = 0.107 \text{ kg}$, Length of wire $L = 9.88 \text{ m}$

Tension in String $F = 236 \text{ N}$

First Harmonic: $f_1 = \frac{v}{2L} = \frac{1}{2L} \sqrt{\frac{F}{\mu}} = \frac{1}{2L} \sqrt{\frac{F}{(m/L)}} = \frac{1}{2L} \sqrt{\frac{F \times L}{m}} = \frac{1}{2 \times 9.88} \sqrt{\frac{236 \times 9.88}{0.107}} = 14.94 \text{ Hz}$

Second Harmonic: $f_2 = 2f_1 = 2(14.94) = 29.88 \text{ Hz}$

Third Harmonic: $f_3 = 3f_1 = 3(14.94) = 89.54 \text{ Hz}$

Problem: In an arrangement, vibrator sets the string into motion at a frequency of 120 Hz. The string has a length $L = 1.2 \text{ m}$ and its linear mass density is $1.6 \frac{\text{g}}{\text{m}}$. To what value must the tension be adjusted to obtain the pattern of motion having four loops?

Given Data: Length of string $L = 1.2 \text{ m}$, Frequency $f = 120 \text{ Hz}$,

Linear Mass Density $\mu = 1.6 \frac{\text{g}}{\text{m}} = 1.6 \times 10^{-3} \frac{\text{kg}}{\text{m}}$

For case of stationary waves: $f_4 = 4f_1 \Rightarrow 120 = 4 \left(\frac{v}{2L} \right) \Rightarrow 120 = 2 \left(\frac{v}{L} \right) \Rightarrow v = 60 L$

$$\Rightarrow v = 60 \times 1.2 \Rightarrow v = 72 \Rightarrow \sqrt{\frac{F}{\mu}} = 72 \Rightarrow F = (72)^2 \times \mu \Rightarrow F = (72)^2 \times 1.6 = 133 \text{ N}$$

Problem: A string can vibrate with frequency of 440 Hz and has a length 0.34 m. What are the three longest wavelengths of the resonance of the string?

Given Data: Frequency $f = 440 \text{ Hz}$, Length of the string $L = 0.34 \text{ m}$

To Determine: Three Longest Wavelengths $\lambda_1 = ?$, $\lambda_2 = ?$, $\lambda_3 = ?$

Calculations: We know that for stationary waves: $\lambda_n = \frac{2L}{n}$ — — — — (1)

First Harmonic Put $n = 1$ in equation (1): $\lambda_1 = \frac{2L}{1} = 2 \times 0.34 = 0.68 \text{ m}$

Second Harmonic Put $n = 2$ in equation (1): $\lambda_2 = \frac{2L}{2} = 0.34 \text{ m}$

Third Harmonic Put $n = 3$ in equation (1): $\lambda_3 = \frac{2L}{3} = \frac{2}{3} \times 0.34 = 0.23 \text{ m}$

A string is fixed at both ends is 8.36 m long, and has mass of 122 g. It is subjected to tension of 96.7 N and set vibrating. Find (a) What is speed of wave in string? (b) What is the wavelength of longest possible standing wave?

Given Mass of string $m = 122 \text{ g} = 0.122 \text{ kg}$, Length of string $L = 8.36 \text{ m}$, Tension $F = 96.7 \text{ N}$

(a) Speed of wave $v = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{F}{\left(\frac{m}{L}\right)}} = \sqrt{\frac{F \times L}{m}} = \sqrt{\frac{96.7 \times 8.36}{0.122}} = 81.66 \text{ ms}^{-1}$

(b) As Wavelength of standing wave produced in stretched string $\lambda_n = \frac{2L}{n}$

For longest wavelength, $n = 1$. Therefore,

Longest wavelength $\lambda_1 = \frac{2L}{1} = 2L = 2 \times 8.36 \text{ m} = 16.72 \text{ m}$

19.18 Resonance

“An increase the amplitude of a vibrating body under the action of periodic force whose frequency is equal to natural frequency if vibrating body, is called Resonance”.

Consider a string whose one end is fixed to a hook and other end is held in hand. If we move the free end up and down at regular intervals, a series of identical waves travel along the string. When the frequency of the applied produced force (to produce waves in string) becomes equal to any of natural frequencies, the string begins to vibrate with greater amplitude. This increase in amplitude is called resonance.

At resonance the string absorb as much energy as it can form the source. This is true for any vibrating system. e.g. in tuning a radio , the natural frequency of electronic circuit is change until it becomes equal to the particular frequency of radio waves i.e., broadcast from the station. At this point resonance produces and absorbs maximum energy from the signal. Similarly resonance can take place in sound, electromagnetisms, optics and atomic and nuclear physics.

Sample problem: In the arrangement vibrator meats the string in motion at a frequency of 120 Hz .the string has a length $L = 1.2\text{ m}$ and its linear mass density is 1.6 g/cm . To what value must tension be adjusted by increasing the hanging weight to obtain the pattern of motion with 4 loops.

Given Frequency $f_4 = 120\text{ Hz}$, Length of string $L = 1.2\text{ m}$,

Linear Mass Density $\mu = 1.6\frac{\text{g}}{\text{cm}} = 0.16\frac{\text{kg}}{\text{m}}$

As for stationary waves $f_n = n\left(\frac{v}{2L}\right) \Rightarrow f_4 = 4\left(\frac{v}{2L}\right) = \frac{2}{L}\sqrt{\frac{F}{\mu}} \Rightarrow (f_4)^2 = \frac{4}{L^2}\left(\frac{F}{\mu}\right)$

$$\Rightarrow (f_4)^2 = \frac{4}{L^2}\left(\frac{F}{\mu}\right) \Rightarrow F = \frac{\mu L^2 (f_4)^2}{4} = \frac{0.16 \times (1.2)^2 (120)^2}{4} = 829\text{ N}$$

Sound

20.1 Beats

The periodic alterations in the intensity of sound between minimum and maximum loudness are called beats.

The phenomenon of beats arises due to superposition of two sound waves of same amplitude but of slightly different frequencies.

When two sound waves of same amplitude and having slightly different frequency superpose each other, they produce a resultant wave whose amplitude varies with time i.e., the amplitude rises and falls periodically. These periodic changes in amplitudes produces rises and falls in the intensity of sound which we call beats.

Derivation:

Let us consider a point in space through which two sine waves are passing. Their frequencies are slightly different. Due to superposition, we get resultant waveform. The resultant pressure at a given point is equal to the sum of individual pressures as a function of time.

From the figure, we see that the amplitude of the resultant wave is not constant but varies with time. In case of sound waves, this variation in loudness called beats.

Let the variation in pressure with time produced by one wave at a given point is given by:

$$\Delta P_1(t) = \Delta P_m \sin \omega_1 t \quad \text{--- (1)}$$

The variation in pressure produced by other wave of equal amplitude at the same point is given by:

$$\Delta P_2(t) = \Delta P_m \sin \omega_2 t \quad \text{--- (2)}$$

For simplicity, we have taken $\phi_1 = \phi_2 = 0$

By the superposition principle, the resultant pressure is given by:

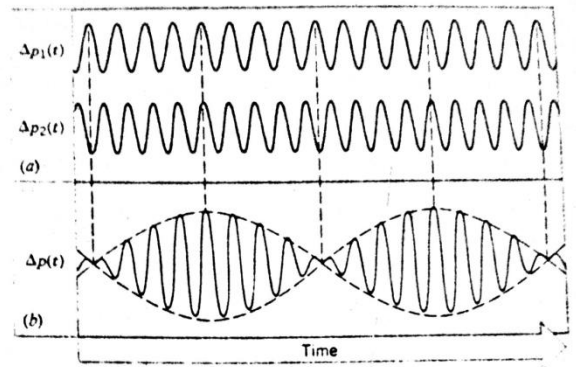
$$\begin{aligned} \Delta P &= \Delta P_1(t) + \Delta P_2(t) \\ \Rightarrow \Delta P &= \Delta P_m \sin \omega_1 t + \Delta P_m \sin \omega_2 t \\ \Rightarrow \Delta P &= \Delta P_m (\sin \omega_1 t + \sin \omega_2 t) \\ \Rightarrow \Delta P &= \Delta P_m \left[2 \sin \left(\frac{\omega_1 t + \omega_2 t}{2} \right) \cos \left(\frac{\omega_1 t - \omega_2 t}{2} \right) \right] \\ \Rightarrow \Delta P &= 2\Delta P_m \cos \left(\frac{\omega_1 - \omega_2}{2} t \right) \cdot \sin \left(\frac{\omega_1 + \omega_2}{2} t \right) \end{aligned}$$

Putting $\frac{\omega_1 - \omega_2}{2} = \omega_{amp}$ and $\frac{\omega_1 + \omega_2}{2} = \omega_{av}$, we have:

$$\Delta P = 2\Delta P_m \cos \omega_{amp} t \cdot \sin \omega_{av} t \quad \text{--- (2)}$$

The equation (2) represents the resultant wave having frequency $\frac{\omega_1 + \omega_2}{2}$ and amplitude

$$2\Delta P_m \cos \left(\frac{\omega_1 - \omega_2}{2} t \right)$$



(a) Two sinusoidal waveforms of nearly equal frequencies. (b) The superposition of the two waveforms. Note that the two waves in part (a) go from being in phase, giving a resultant of large amplitude, to being out of phase, giving a resultant of zero amplitude. The dashed curves show the sinusoidal variation of the modulating envelope with angular frequency ω_{amp} .

We find that the amplitude depends on time i.e., the amplitude varies between maximum value $2\Delta P_m$ and minimum value $-2\Delta P_m$ with frequency $\frac{\omega_1 - \omega_2}{2}$.

Since intensity is proportional to square of amplitude, so the intensity is maximum both when amplitude is maximum or minimum. So intensity attains maximum value twice in one cycle.

So we conclude that number of beats is twice the number of cycles per second. Hence beats frequency is twice the amplitude frequency i.e.,

$$\omega_{beat} = 2 \omega_{amp}$$

$$\Rightarrow \omega_{beat} = 2 \left(\frac{\omega_1 - \omega_2}{2} \right)$$

$$\Rightarrow \omega_{beat} = \omega_1 - \omega_2$$

$$\Rightarrow 2\pi f_{beat} = 2\pi f_1 - 2\pi f_2 \quad \because \text{As } \omega = 2\pi f$$

$$\Rightarrow 2\pi f_{beat} = 2\pi(f_1 - f_2)$$

$$\Rightarrow f_{beat} = f_1 - f_2$$

It shows that the number of beats produced per second is equal to the difference of the frequencies of the two waves.

20.2 The Doppler Effect

The apparent change in the pitch of sound due to relative motion between source and observer is called Doppler Effect.

Consider a source of sound and an observer (listener). It is found that as the source approaches the listener, the pitch of sound is found to be greater than the actual pitch of sound. Similarly if the source moves away from the listener, the pitch is found to decrease. The same happens when source remains at rest and the observer moves towards or away from the source.

This apparent change in the pitch of sound due to motion of source, observer or both is called Doppler Effect.

Let f be the actual frequency of source i.e., it emits f waves in one second at rest. Let v be the speed of sound of sound waves i.e., v is the distance which the sound waves cover in one second. If both the source and observer are stationary, then the frequency of sound waves is described as:

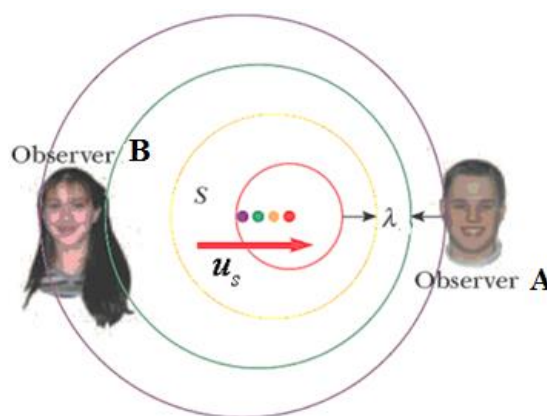
$$f = \frac{v}{\lambda}$$

where λ is the wavelength of sound waves emitted by source at rest.

We want to find out the expression of apparent change in the frequency of sound due to the relative motion between source and observer.

Case 1. When the Source moves towards the Stationary Observer

If the source is moving away towards the observer A with speed u_s , then the waves are compressed per second by an amount



$\Delta\lambda$, known as Doppler Shift:

$$\Delta\lambda = \frac{u_s}{f}$$

The compression of waves is due to the fact that same numbers of waves are contained in a shorter space depending upon the velocity of source. The wavelength for observer C is:

$$\lambda_A = \lambda - \Delta\lambda$$

$$\Rightarrow \lambda_A = \frac{v}{f} - \frac{u_s}{f} = \frac{v - u_s}{f}$$

The modified frequency for observer C is:

$$f_A = \frac{v}{\lambda_A}$$

$$\Rightarrow f_A = \frac{v}{\left(\frac{v - u_s}{f}\right)}$$

$$\Rightarrow f_A = \left(\frac{v}{v - u_s}\right)f \quad \text{----- (i)}$$

Since $\frac{v}{v - u_s} > 1$, therefore $f_A > f$

Result: The apparent frequency of the sound increases, when the source moves towards the stationary observer.

Case 2. When the Source moves away from the Stationary Observer

If the source is moving away towards the observer B with speed u_s , the wavelength of increases by an amount $\Delta\lambda$, known as Doppler Shift:

$$\Delta\lambda = \frac{u_s}{f}$$

The expansion of wavelength is due to the fact that same number of waves are contained in a larger space depending upon the velocity of source.

The wavelength for observer D is:

$$\lambda_B = \lambda + \Delta\lambda$$

$$\Rightarrow \lambda_B = \frac{v}{f} + \frac{u_s}{f} = \frac{v + u_s}{f}$$

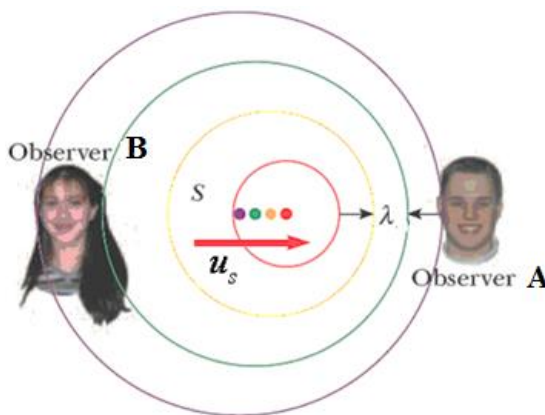
The modified frequency for observer D is:

$$f_B = \frac{v}{\lambda_B}$$

$$\Rightarrow f_B = \frac{v}{\left(\frac{v + u_s}{f}\right)} \Rightarrow f_B = \left(\frac{v}{v + u_s}\right)f \quad \text{----- (ii)}$$

Since $\frac{v}{v + u_s} < 1$, therefore $f_B < f$

Result: The apparent frequency of the sound decrease, when the source moves away from the stationary observer.



Generalization of Case-1 and Case-2

Equation (i) and (ii) are combined into a single relation as:

$$f' = \left(\frac{v}{v \pm u_s} \right) f \quad \text{--- (iii)}$$

where $-ve$ sign holds for motion towards the observer and the $+ve$ sign holds for motion away from the observer.

It should be noted that apparent change in pitch is due to increase or decrease of wavelength due to motion of source.

Case 3. When the observer moves towards the Stationary Source

If the observer C moves towards the stationary source with velocity u_o , then the relative velocity of the waves and the observer is increased to $(v + u_o)$. Then the frequency of sound received by the observer C is:

$$\begin{aligned} f_c &= \frac{v + u_o}{\lambda} \\ \Rightarrow f_c &= \frac{v + u_o}{\left(\frac{v}{f}\right)} \quad \because \text{As } \lambda = \frac{v}{f} \\ \Rightarrow f_c &= \left(\frac{v + u_o}{v}\right) f \quad \text{--- (iv)} \end{aligned}$$

Since $\frac{v+u_o}{v} > 1$, therefore $f_c > f$

Result: The apparent frequency of the sound increases, when the observer moves towards the stationary source of sound.

Case 4. When the observer moves away from Stationary Source

If the observer D moves away from stationary source with velocity u_o , then the relative velocity of the waves and the observer is increased to $(v - u_o)$. Then the frequency of sound received by the observer D is:

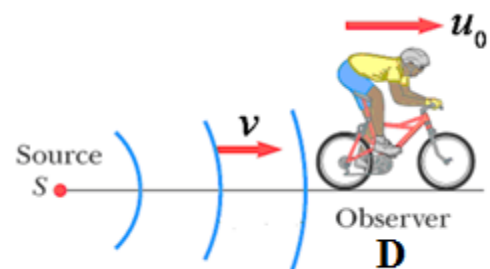
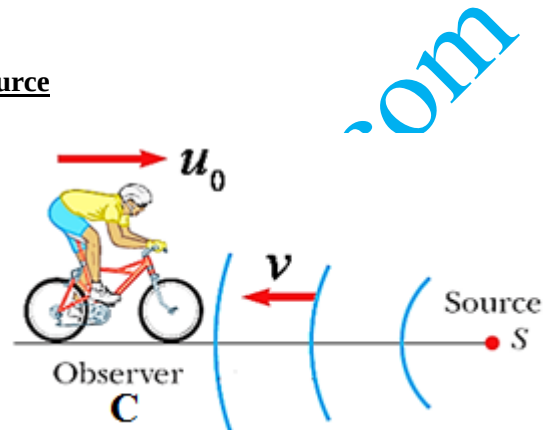
$$\begin{aligned} f_D &= \frac{v - u_o}{\lambda} \\ \Rightarrow f_D &= \frac{v - u_o}{\left(\frac{v}{f}\right)} \quad \because \text{As } \lambda = \frac{v}{f} \\ \Rightarrow f_D &= \left(\frac{v - u_o}{v}\right) f \quad \text{--- (v)} \end{aligned}$$

Since $\frac{v-u_o}{v} < 1$, therefore $f_D < f$

Result: The apparent frequency of the sound decreases, when the observer moves away from the stationary source of sound.

Generalization of Case-3 and Case-4

Equation (iv) and (v) can be combined into a single relation as:



$$f'' = \left(\frac{v \pm u_o}{v} \right) f \quad \text{--- -- -- -- --} \quad (vi)$$

where $+ve$ sign holds for motion towards the source and $-ve$ sign holds for motion away from source.

It should be noted that apparent change in pitch takes place because the observer receive more or less waves per second due to its motion.

Case-5: Both Source and Observer are in Motion

Let us consider the case when both source and observer move towards each other.

As we know that when source moves towards or away from observer **at rest** then frequency is given by equation (iii) as:

$$f' = \left(\frac{v}{v \pm u_s} \right) f$$

If observer also moves towards or away from source **in motion**, then apparent frequency f''' will be:

$$f''' = \left(\frac{v \pm u_o}{v} \right) f'$$

Putting values of f' from the relation get:

$$f''' = \left(\frac{v \pm u_o}{v} \right) \left(\frac{v}{v \pm u_s} \right) f$$

$$\Rightarrow f''' = \left(\frac{v \pm u_o}{v \pm u_s} \right) f$$

where

$f''' = \left(\frac{v+u_o}{v-u_s} \right) f$ represents the case **when both source and observer are moving towards each other**

and

$f''' = \left(\frac{v-u_o}{v+u_s} \right) f$ represents the case **when both source and the observer are moving away from each other.**

Interference of Light Waves

Interference

Superposition of two waves having same frequency and moving in the same direction will result in the phenomenon of interference. The phenomenon of interference is a wave phenomenon. Interference is of two types:

Constructive interference

If the two waves reach a point in phase i.e., crest of one wave falls on the crest of the other wave and trough of one wave falls on the trough of the other wave then the two waves reinforce each other and the net wave effect increases i.e., the net intensity is greater than the intensity of individual wave. Such interference is called constructive interference.

Destructive Interference

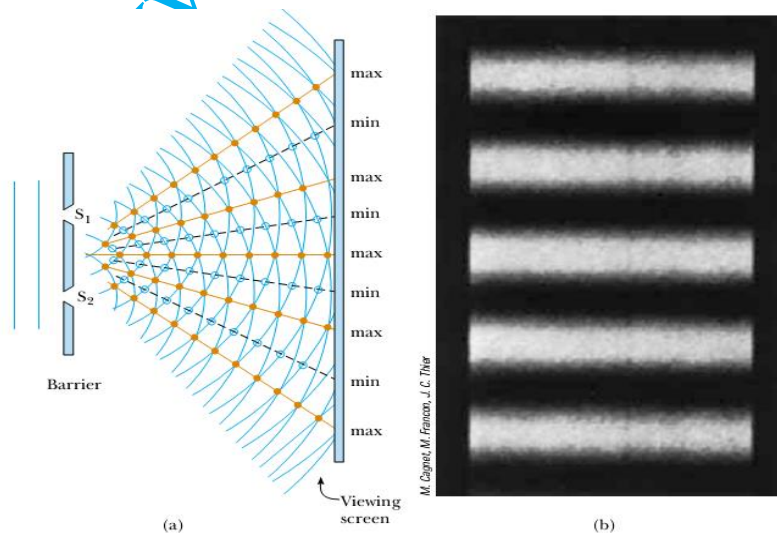
If the two waves reach a point out of phase i.e., crest of one wave falls on the trough of the other, then the two waves cancel each other's effect and the net intensity of individual wave. This type of interference is called destructive interference.

Coherent sources

The phenomenon of interference is a wave phenomenon. It takes place due to superposition of two exactly similar waves. It should be noted that the two waves interfere only if they have phase coherence i.e., they come from two coherent sources.

Two sources are said to be coherent sources if they produce waves of same amplitude, same frequency having either no initial phase difference or a constant phase difference.

(a) A train of plane light waves (for example, from a laser) is incident on a barrier into which are cut two narrow slits separated by a distance d . The width of the slits are small compared with the wavelength, so that the waves passing through the slits spread out (diffract) and illuminate the screen. (b) The interference pattern, consisting of bright and dark bands or fringes that would appear on the screen.



YOUNG'S DOUBLE SLIT EXPERIMENT:

Huygens wave theory of light was experimentally confirmed by a British scientist THOMAS YOUNG in 1801. Young's double slit experiment is shown in figure.

S is a monochromatic source of light which is a rectangular slit illuminated by monochromatic light. This light is divided into two parts by slits S_1 and S_2 . Now slits S_1 and S_2 act as coherent sources because they are obtained from single source. d is the distance between slits S_1 and S_2 . $D \gg d$ is the distance of screen and slits S_1 and S_2 . Point O is the central point of screen. Since the waves from S_1 and S_2 cover equal distance to point O, therefore, point O is a bright point due to constructive interference. Point O is called central maxima. Let us take any point P on the screen at a distance y from central point O. r_1 and r_2 are the distances of point P from S_1 and S_2 . The path difference between the waves arriving at P is ;

$$\text{Path Difference} = S_2P - S_1P$$

To calculate this path difference we draw a perpendicular S_1b on S_2P such that

$$S_1P = bP$$

From figure: $S_2P = S_2b + bP$

$$\Rightarrow S_2b = S_2P - bP$$

$$\Rightarrow S_2b = S_2P - S_1P \quad \therefore S_1P = bP$$

$$\begin{aligned} \Rightarrow \text{Path Difference} &= S_2b \quad \text{--- (1)} & \therefore \text{Path Difference} \\ &= S_2P - S_1P \end{aligned}$$

Now, from the right angled triangle ΔS_1bS_2 , we have:

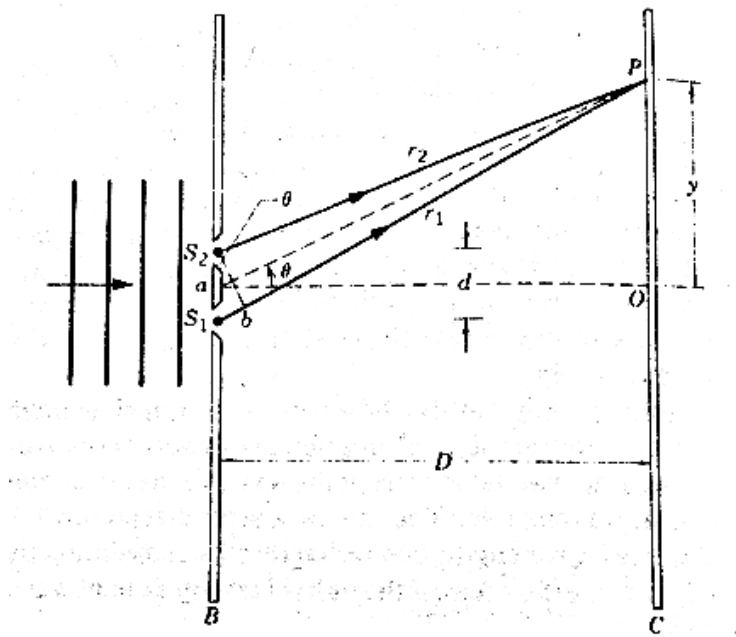
$$\frac{S_2b}{S_1S_2} = \sin \theta \quad \because S_1S_2 = d$$

$$\Rightarrow \frac{S_2b}{d} = \sin \theta$$

$$\Rightarrow S_2b = d \sin \theta$$

$$\text{Path Difference} = d \sin \theta \quad \text{--- (2)} \quad \therefore \text{by eq. (1)}$$

As $D \gg d$ so $\sin \theta \approx \tan \theta$. Therefore:



Rays from S_1 and S_2 combine at P. In actuality, $D \gg d$; the figure has been distorted for clarity. Point a is the midpoint between the slits.

$$\text{Path difference} = d \tan \theta$$

But from right triangle Δaop ; $\tan \theta = \frac{y}{D}$

$$\text{Path Difference} = d \frac{y}{D} \text{ --- (3)}$$

Condition for Constructive Interference

Now for point P to be a bright point the path difference should be equal to integral multiple of λ . i.e.,

$$\frac{dy}{D} = m\lambda, \quad m = 0, 1, 2, 3, \dots$$

$$\Rightarrow y = \frac{m\lambda D}{d}$$

where y is the distance of mth bright band from point O. For $m = 0$ we get central maxima.

Condition for Destructive Interference

Now for point P take a dark point, the path difference should be equal to odd multiple of $\frac{\lambda}{2}$.

$$\frac{dy}{D} = (2m + 1) \frac{\lambda}{2}$$

$$\Rightarrow y = (2m + 1) \frac{\lambda D}{2d}$$

where y is the distance of mth dark band from point O. For $m = 0$ we get 1st dark band.

Sample Problem 1. In double slit experiment with mercury vapor lamp, the strong green light of wavelength $\lambda = 546\text{nm}$ is visible. The slits are 0.12mm apart and the screen is 55cm away. What is the angular position of the first minimum? of the 10th maximum?.

Given Data: Wavelength $\lambda = 546\text{nm} = 546 \times 10^{-9}\text{m}$, Interslit Distance $d = 0.12\text{mm} = 0.12 \times 10^{-3}\text{m}$

Distance between Screens $D = 55\text{cm} = 55 \times 10^{-2}\text{m}$

(a) $\theta = ?$ for first minima. For minima we have: $d \sin \theta = (2m + 1) \frac{\lambda}{2}$

For first minima put $m = 0$: $d \sin \theta = \frac{\lambda}{2} \Rightarrow \sin \theta = \frac{\lambda}{2d} = \frac{546 \times 10^{-9}}{2 \times 0.12 \times 10^{-3}} = 2275 \times 10^{-6} = 0.0023$

$$\Rightarrow \theta = \sin^{-1} 0.0023 \quad \theta = 0.13^\circ$$

(b) $\theta = ?$ for 10th maxima. For maxima we have: $d \sin \theta = m\lambda$

For 10th maxima put $m = 10$: $d \sin \theta = 10\lambda \Rightarrow \sin \theta = \frac{10\lambda}{d} = \frac{10 \times 546 \times 10^{-9}}{0.12 \times 10^{-3}} = 4.55 \times 10^{-6}$
 $\Rightarrow \theta = \sin^{-1} 4.55 \times 10^{-6} = 2.6^\circ$

Sample Problem 2. What is the linear distance on screen C between the adjacent maximum and $m + 1$ of sample problem 1.

From sample problem 1:

Given Data: Wavelength $\lambda = 546\text{nm} = 546 \times 10^{-9}\text{m}$, Interslit Distance $d = 0.12\text{mm}$
 $= 0.12 \times 10^{-3}\text{m}$

Distance between Screens $D = 55\text{cm} = 55 \times 10^{-2}\text{m}$

To Determine: Distance between maxima m and $m + 1 = \Delta y = y_{m+1} - y_m = ?$

For m th maxima: $y_m = m \frac{\lambda D}{d}$, **For $(m+1)$ maxima:** $y_{m+1} = (m + 1) \frac{\lambda D}{d}$

Therefore, $\Delta y = y_{m+1} - y_m = (m + 1) \frac{\lambda D}{d} - m \frac{\lambda D}{d} = \frac{m\lambda D}{d} + \frac{\lambda D}{d} - m \frac{\lambda D}{d} = \frac{\lambda D}{d}$

$$\Rightarrow \Delta y = \frac{546 \times 10^{-9} \times 55 \times 10^{-2}}{0.12 \times 10^{-3}} = 2502.2 \times 10^{-6} = 2.5025 \times 10^{-3}\text{m} = 2.5 \times 10^{-3}\text{m}$$

$$= 2.5\text{mm}$$

Intensity in Young Double Slit Experiment

Consider two coherent sources S_1 and S_2 giving waves of same angular frequency ω . Let α be the distance between S_1 and S_2 .

Suppose the distance of observation point P on the screen from S_1 be r_1 and r_2 from S_2 . It is assumed that r_1 and r_2 are very large as compared to d . Let the waves reaching P due to coherent sources be given by:

$$\psi_1 = A_1(r_1) \cos(\omega t - kr_1)$$

$$\psi_2 = A_2(r_2) \cos(\omega t - kr_2)$$

where ψ_1 and ψ_2 are the displacements reaching P and $A_1(r_1)$, $A_2(r_2)$ are the amplitudes reaching there. The total displacement at P is obtained by their superposition:

$$\psi = \psi_1 + \psi_2$$

$$\Rightarrow \psi = A_1(r_1) \cos(\omega t - kr_1) + A_2(r_2) \cos(\omega t - kr_2)$$

$$\Rightarrow \psi = A_1(r_1) \cos \omega t \cos kr_1 + A_1(r_1) \sin \omega t \sin kr_1 + A_2(r_2) \cos \omega t \cos kr_2 + A_2(r_2) \sin \omega t \sin kr_2$$

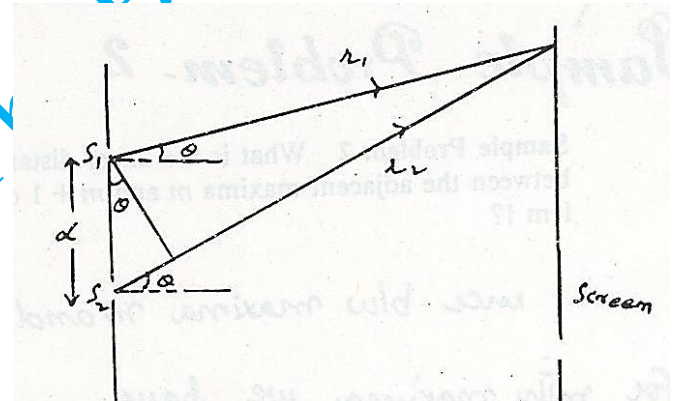
$$\Rightarrow \psi = (A_1(r_1) \cos kr_1 + A_2(r_2) \cos kr_2) \cos \omega t$$

$$+ (A_1(r_1) \sin kr_1 + A_2(r_2) \sin kr_2) \sin \omega t$$

$$\text{Let } A_1(r_1) \cos kr_1 + A_2(r_2) \cos kr_2 = A \cos \varphi \quad \text{--- (i)}$$

$$\text{And } A_1(r_1) \sin kr_1 + A_2(r_2) \sin kr_2 = A \sin \varphi \quad \text{--- (ii)}$$

$$\text{Then } \psi = A \cos \varphi \cos \omega t + A \sin \varphi \sin \omega t$$



$$\Rightarrow \psi = A[\cos \varphi \cos \omega t + \sin \varphi \sin \omega t]$$

$$\Rightarrow \psi = A \cos(\omega t - \varphi)$$

This is the equation of resultant wave at point P.

Phase Constant φ

The phase angle φ is given by dividing (2) by (1)

$$\tan \varphi = \frac{A_1(r_1) \sin kr_1 + A_2(r_2) \sin kr_2}{A_1(r_1) \cos kr_1 + A_2(r_2) \cos kr_2}$$

$$\Rightarrow \varphi = \tan^{-1} \left(\frac{A_1(r_1) \sin kr_1 + A_2(r_2) \sin kr_2}{A_1(r_1) \cos kr_1 + A_2(r_2) \cos kr_2} \right)$$

Amplitude of Resultant Wave

The amplitude A is given by squaring and adding (i) and (ii).

$$\begin{aligned} A^2 \cos^2 \varphi + A^2 \sin^2 \varphi &= A_1^2(r_1) \cos^2 kr_1 + A_2^2(r_2) \cos^2 kr_2 + 2A_1(r_1)A_2(r_2) \cos kr_1 \cos kr_2 \\ &\quad + A_1^2(r_1) \sin^2 kr_1 + A_2^2(r_2) \sin^2 kr_2 \\ &\quad + 2A_1(r_1)A_2(r_2) \sin kr_1 \sin kr_2 \\ \Rightarrow A^2(\cos^2 \varphi + \sin^2 \varphi) &= A_1^2(r_1)(\cos^2 kr_1 + \sin^2 kr_1) + A_2^2(r_2)(\cos^2 kr_2 + \sin^2 kr_2) \\ &\quad + 2A_1(r_1)A_2(r_2)[\cos kr_1 \cos kr_2 + \sin kr_1 \sin kr_2] \\ \Rightarrow A^2 &= A_1^2(r_1) + A_2^2(r_2) + 2A_1(r_1)A_2(r_2) \cos k(r_2 - r_1) \text{ --- (3)} \end{aligned}$$

Because $r_1 \approx r_2 \therefore A_1(r_1) = A_2(r_2) = A(r)$ where r

$$= \frac{r_1 + r_2}{2} \text{ is the mean distance of point P}$$

from S_1 and S_2 . Then equation (3) becomes

$$\begin{aligned} A^2 &= A^2(r) + A^2(r) + 2A^2(r) \cos k(r_2 - r_1) \\ \Rightarrow A^2 &= 2A^2(r) + 2A^2(r) \cos k(r_2 - r_1) \\ \Rightarrow A^2 &= 2A^2(r)[1 + \cos k(r_2 - r_1)] \because 1 + \cos \theta = 2 \cos^2 \frac{\theta}{2} \\ \Rightarrow A^2 &= 2A^2(r) \left[2 \cos^2 \frac{k(r_2 - r_1)}{2} \right] \\ \Rightarrow A^2 &= 4A^2(r) \cos^2 \frac{k(r_2 - r_1)}{2} \text{ --- (a)} \end{aligned}$$

Intensity of Resultant Wave

From figure, $r_2 - r_1 = d \sin \theta$ and $k = \frac{2\pi}{\lambda}$. Therefore,

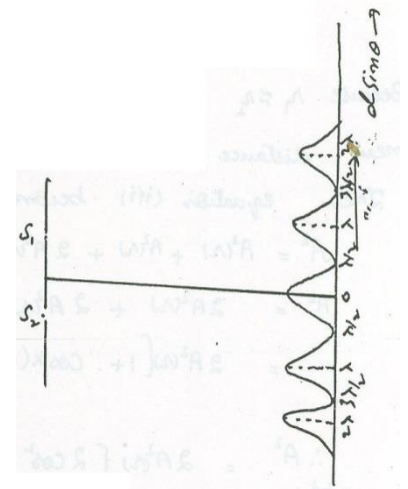
$$A^2 = 4A^2(r) \cos^2 \frac{kd \sin \theta}{2} = 4A^2(r) \cos^2 \frac{1}{2} \left[\frac{2\pi}{\lambda} d \sin \theta \right]$$

$$\Rightarrow A^2 = 4A^2(r) \cos^2 \left[\frac{\pi}{\lambda} d \sin \theta \right] \text{ --- (b)}$$

As the intensity of light is proportional to the square of the amplitude, so above equation gives the intensity of light at point P. Therefore,

$$I = I_{\max} \cos^2 \left(\frac{\pi d \sin \theta}{\lambda} \right)$$

where $I_{\max} = 4A^2(r)$ is the maximum intensity at point P for $\theta = 0^\circ$. This equation gives the intensity of a wave at a point P. Figure shows the plot of intensities as a function of $d \sin \theta$.

**Condition for Maxima**

The intensity at P will be maximum when

$$\frac{\pi d \sin \theta}{\lambda} = 0, \pi, 2\pi, 3\pi, \dots$$

$$\Rightarrow \frac{\pi d \sin \theta}{\lambda} = n\pi$$

$$\Rightarrow d \sin \theta = n\lambda \quad \text{where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow d \sin \theta = 0, \lambda, 2\lambda, 3\lambda, \dots$$

Condition for Minima

The intensity at P will be minimum when

$$\frac{\pi d \sin \theta}{\lambda} = (2n + 1) \frac{\pi}{2}$$

$$\Rightarrow d \sin \theta = (2n + 1) \frac{\lambda}{2} \quad \text{where } n = 0, 1, 2, 3, \dots$$

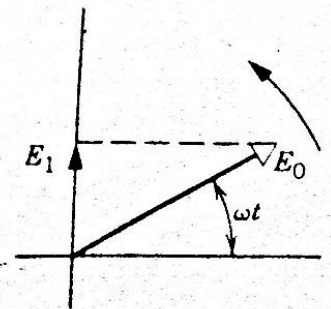
$$\Rightarrow d \sin \theta = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots$$

Adding of Electromagnetic Waves (Phasor Method)

We can obtain combined electric field of light in double slit experiment by algebraic method. But this method becomes very difficult when we add more than waves. So to overcome this problem the combined electric field is obtained by phasor method. A phasor is rotating vector which gives time varying quantities. A sinusoidal wave disturbance is represented by as

$$E_1 = E_0 \sin \omega t \quad \text{--- (1)}$$

This wave can be represented a rotating vector, called phasor. For this purpose, we take a phasor of magnitude E_0 which is rotating about the



(a)

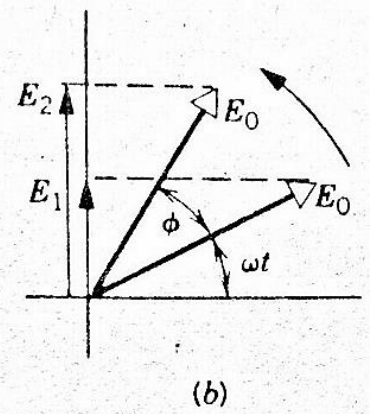
origin in anticlockwise direction with angular frequency ω . Then the disturbance E_1 in equation (1) is represented by the projection of phasor along y-axis [Fig. (a)].

Now consider a second wave of the same amplitude but having a phase difference ϕ given as:

$$E_2 = E_0 \sin \omega t \quad \text{--- (2)}$$

It is represented by the projection of 2nd phasor along y-axis as shown:

Now the sum $E = E_1 + E_2$ is obtained by adding the projection of phasors along y-axis. It can be explained more clearly if we redraw the phasors such that the tail of one coincide with the head of the other as shown in figure (b):



(b)

From figure, it is clear that sum $E = E_1 + E_2$ is equal to the projection of phasor of amplitude E_θ along y-axis. So, from figure (c), we can see that projection E can be written as:

$$E = E_\theta \sin(\omega t - \beta) \quad \text{--- (3)}$$

Now we have to determine E_θ and β of this resultant wave.

Determination of β

In figure (c), the three phasors E_0 , E_0 and E_θ form an

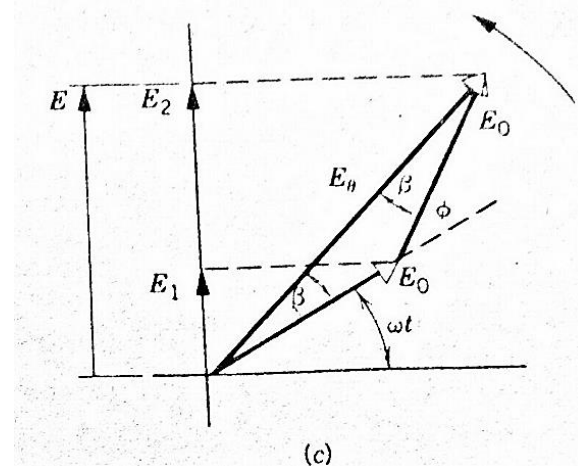
isosceles triangle. Since by mathematics, the sum of two interior angles of triangle is equal to the exterior angle:

$$\phi = \beta + \beta$$

$$\Rightarrow \phi = 2\beta \Rightarrow \beta = \frac{\phi}{2}$$

Determination of E_θ

In figure (c), the length of the base of the triangle can be described as:



(c)

$$E_{\theta} = E_o \cos \beta + E_o \cos \beta$$

$$\Rightarrow E_{\theta} = 2E_o \cos \beta \quad \text{--- -- -- --} \quad (4)$$

Where E_o is the amplitude of the resultant wave. The maximum possible value of the amplitude is $E_{\theta} = 2E_o$ i.e., twice the amplitude of individual wave.

Equation of Resultant Wave

By putting values in equation (3), we get:

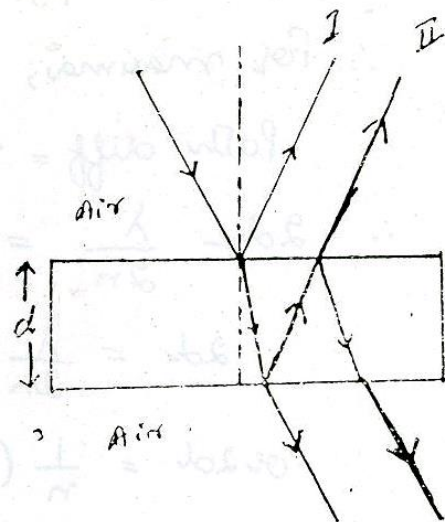
$$E = 2E_o \cos \beta \sin \left(\omega t - \frac{\phi}{2} \right)$$

This is the equation of resultant wave.

Interference in Thin Films

Consider a thin film of uniform thickness and refractive index n . Monochromatic light is made to fall on this film. Light is reflected from the upper and lower surfaces of the film. These rays I and II interfere to give interference fringes.

Ray reflected from the lower surface has a longer path than the ray reflected from the upper surface. For normal incidence the ray reflected from the bottom covers a distance $2d$ greater than the upper ray. In this time the ray reflected from the upper surface undergoes a phase. It is due to the reason that when reflection takes place at the boundary of a denser medium, the crest is reflected as trough and vice versa.



Equilant Path Differnece

$$= 2d - \frac{\lambda_n}{2} \quad \text{--- -- -- --} \quad (A)$$

Where λ_n is the wavelength in the medium of refractive index n . Now when light wave passes from one medium (air) to another medium (oil), both the speed and wavelength change but frequency remains constant.

$$f_{air} = f_{oil}$$

$$\Rightarrow \frac{c}{\lambda} = \frac{v}{\lambda_n} \quad \text{--- -- -- --} \quad (B)$$

where c = velocity of the light in the air.

v = velocity of the light in thin film.

λ = wavelength of the light in the air.

λ_n = wavelength of the light in thin film.

From equation (B), we can write:

$$\frac{\lambda}{\lambda_n} = \frac{c}{v} \Rightarrow \frac{\lambda}{\lambda_n} = n \quad \text{But } \frac{c}{v} = n \text{ (index of refraction)}$$

$$\Rightarrow \lambda_n = \frac{\lambda}{n}$$

Eq (A) becomes

$$\text{Equivalent Path Difference} = 2d - \frac{\lambda}{2n}$$

Condition for Maxima

For maxima:

$$\text{Path Difference} = m\lambda_n$$

$$\Rightarrow 2d - \frac{\lambda}{2n} = \frac{m\lambda}{2n}$$

$$\Rightarrow 2d = \frac{\lambda}{2n} + \frac{m\lambda}{2n} = \frac{1}{n} \left(\frac{\lambda}{2} + m\lambda \right)$$

$$\Rightarrow 2dn = \frac{\lambda}{2} (2m + 1) \quad \text{where } m = 0, 1, 2, 3 \dots \dots \dots$$

This is the condition for constructive interference.

Condition for Minima

Now for minima

$$\text{Path Difference} = (2m - 1) \frac{\lambda_n}{2}$$

$$\Rightarrow 2d - \frac{\lambda}{2n} = (2m - 1) \frac{\lambda}{2n}$$

Multiply by n on both sides, we get:

$$2an = (2m - 1) \frac{\lambda}{2} + \frac{\lambda}{2} = \frac{\lambda}{2} (2m - 1 + 1) = \frac{\lambda}{2} (2m)$$

$$2dn = m\lambda$$

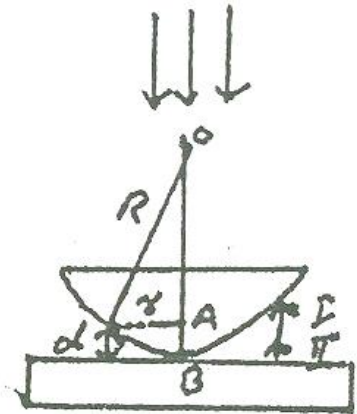
This is a condition of destructive interference.

Newton Rings

They are circular interference fringes. They consist of alternate bright and dark rings. These rings were first obtained by Newton. That is why they are known as Newton's rings. Newton's particles theory could not explain the formation of these rings.

Experimental Arrangement

Consider a plane convex lens placed on the glass with convex surface on the plate. Suppose light is made to fall normally on the plane surface of lens. Interference takes place b/w the rays reflected from upper and lower and upper surface of the air film from between the lens and the plate.



The path difference between two rays depends on thickness of air film at the point of incidence. It is equal to twice the thickness of air film at this point. Since the points of equal thickness lie in a circle. So the points of equal path difference lie in concentric circles. Thus the fringes obtained in this case are circular. These circular fringes are called Newton's rings. The ray of light reflected from the lower surface of the air film (upper surface of glass plate) undergoes a phase change of π or path difference of $\frac{\lambda}{2}$ due to its reflection from denser medium.

So for maxima i.e., for constructive interference:

$$2dn = \frac{\lambda}{2}(2m + 1) \quad \text{where } m \text{ is interger}$$

$$\Rightarrow 2d = \frac{\lambda}{2}(2m + 1) \quad \text{--- (1)} \quad \because n = 1 \text{ for air}$$

And for minima i.e., for destructive interference:

$$2d = m\lambda \quad \text{--- (2)}$$

Let R be the radius of curvature of the convex surface and r is the radius of the fringe. Then the thickness of air film is given by:

$$d = OB - OA$$

From Figure, $OB = R$ and $R^2 - r^2 = (OA)^2 \Rightarrow OA = \sqrt{R^2 - r^2}$. Therefore,

$$d = R - \sqrt{R^2 - r^2}$$

$$\Rightarrow d = R - (R^2 - r^2)^{\frac{1}{2}}$$

$$\Rightarrow d = R - R \left(1 - \frac{r^2}{R^2} \right)^{\frac{1}{2}}$$

$$\begin{aligned} \Rightarrow d &= R - R \left(1 - \frac{1}{2} \cdot \frac{r^2}{R^2} + \dots \right) \\ \Rightarrow d &= R - R \left(1 - \frac{1}{2} \cdot \frac{r^2}{R^2} + \text{negligible terms} \right) \\ \Rightarrow d &= R - R \left(1 - \frac{1}{2} \cdot \frac{r^2}{R^2} \right) \\ \Rightarrow d &= R - R + \frac{r^2}{2R} \\ \Rightarrow d &= \frac{r^2}{2R} \end{aligned}$$

For Maxima

Putting the value of d in eq. (1), we get:

$$\begin{aligned} 2 \left(\frac{r^2}{2R} \right) &= (2m + 1) \frac{\lambda}{2} \\ \Rightarrow r^2 &= (2m + 1) \frac{\lambda R}{2} \\ \Rightarrow r^2 &= \left(m + \frac{1}{2} \right) \lambda R \\ \Rightarrow r &= \sqrt{\left(m + \frac{1}{2} \right) \lambda R} \end{aligned}$$

This equation gives the radii of bright rings.

For Minima

Putting the value of d in eq. (2), we get:

$$\begin{aligned} 2 \left(\frac{r^2}{2R} \right) &= m\lambda \\ \Rightarrow r^2 &= m\lambda R \\ \Rightarrow r &= \sqrt{m\lambda R} \end{aligned}$$

This equation gives the radii of dark rings.

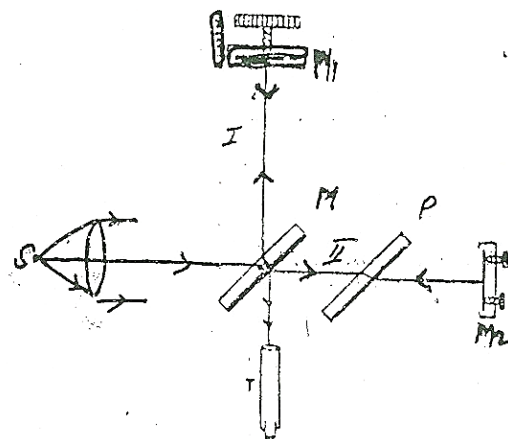
7. Michelson's Interferometer

It is a device used to produce interference and also makes certain measurements. For example it can be used to measure wavelength of light. It can be used to measure thickness of film.

Construction:

It consists of semi silvered glass plate M called splitter. This plate is semi silvered at the back. A glass plate P having the same thickness as that of M is also placed parallel to M. Two plane mirrors M_1 and M_2 are placed perpendicular to each other. The mirror M_1 is mounted on a carriage and can be moved parallel to itself with the help of micrometer screw.

The plane mirrors M_1 and M_2 are provided with leveling screws at their back with the help of which they can be made exactly perpendicular to direction of two beams. The interference band observed in the telescope T. Figure shows the arrangement.



Working:

Light from a monochromatic source S falls on a semi silvered glass plate M placed at an angle of $\frac{\pi}{4}$ to the incident light. This plate splits up the light into two equal parts due to reflection and transmission. The two parts I and II act as coherent sources because they are derived from the source S. Ray I is reflected and Ray II is transmitted. A glass plate P having the same thickness as that of M is placed in the path of Ray II because Ray I passes through the glass thrice while Ray II passes only once. So to keep the optical paths equal the plate P is placed in the path of Ray II. This plate is called compensating plate. Ray I after reflection from M_1 and transmission through M enters the telescope. The plane mirrors are placed perpendicular to the rays so that they retrace their path. In this way they superpose between M and T to produce interference.

Whether we observe a bright or dark band, it depends upon the path difference of two rays. Now suppose the path difference between two rays is zero i.e. the two rays cover equal distances before entering the telescope. Then they interfere constructively and we observe zeroth bright band. To change the path difference, the mirror M_1 is kept moveable and M_2 is fixed. The following measurements can be made with the interferometer.

(i) Measurements of wavelength (λ)

For measuring the wavelength, the source of light is monochromatic and M_1 is perpendicular to M_2 . So the fringes obtained are circular. Now we have the mirror M_1 , the fringes go on passing before the eye. Suppose we are observing zeroth bright band. To see the next bright band we have to produce a path difference of λ . So, we move the mirror M_1 through the distance of $\frac{\lambda}{2}$ because ray II has to cover it twice. So zeroth bright band is replaced by 1st bright band when M_1 is moved through the distance $\frac{\lambda}{2}$.

So by moving the mirror M_1 through the distance of $\frac{\lambda}{2}$ each time, we find that a bright band is replaced by the next bright band. Suppose n bright fringes pass before the eye when M_1 is moved through the distance d , then

$$d = n \frac{\lambda}{2}$$

$$\Rightarrow \lambda = \frac{2d}{n}$$

So knowing d and n , λ can be determined.

(ii) Measurements of Thickness

The plate whose thickness is required, is placed in the path of one of the rays. If t is the thickness and on the index of refraction, then path difference introduced = $(n - 1)t$

If M_1 is moved through a distance d then $d = (n - 1)t$. So knowing d and n we can determine the thickness t of the plate.

Sample Problem 7. Yellow light ($\lambda = 589\text{nm}$) illuminates a Michelson interferometer.

How many bright fringes will be counted as the mirror is moved through 1.00cm.

$$\lambda = 589\text{nm} = 5.89 \times 10^{-9}\text{m}, \quad m = ?, \quad P = 1\text{cm} = 10^{-2}\text{m}$$

As we know that

$$d = m \frac{\lambda}{2}$$

$$\Rightarrow m = \frac{2d}{\lambda} = \frac{2 \times 10^{-2}}{5.89 \times 10^{-9}} = 3.3956 \times 10^4 = 33956 \text{ fringes.}$$

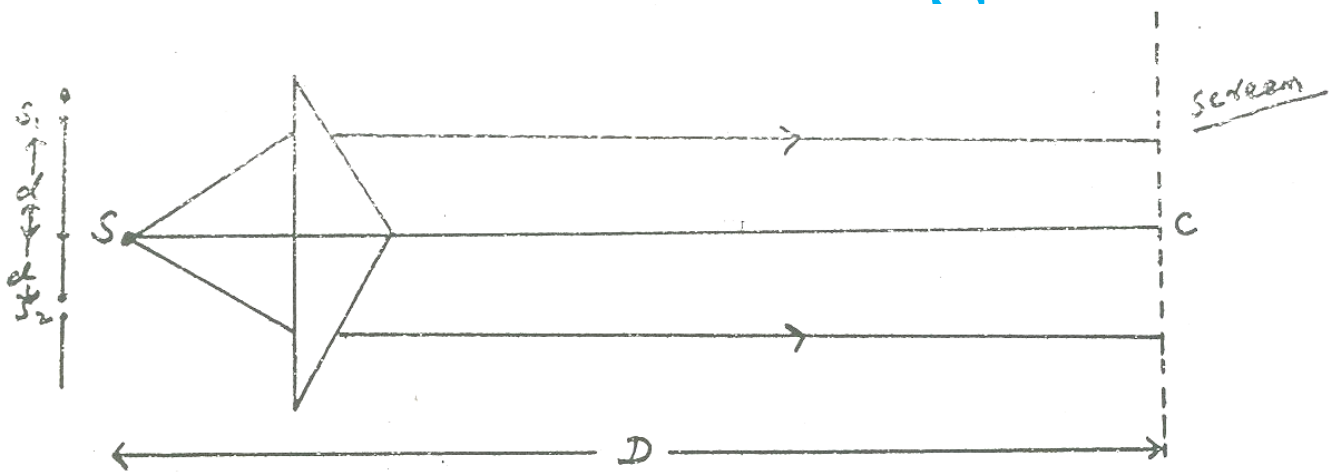
8. Fresnel's Biprism & its use to Determine Wavelength of Light:

A biprism is a prism whose one angle is little less than 180° and the other two angles are small and equal to each other.

Construction and Working

It is a combination of two glass prisms with their bases in contact i.e. it acts as two prisms placed base to base. It is the best apparatus to produce interference fringes.

Light from a narrow slit perpendicular to the plane of paper and illuminated by a monochromatic source is refracted from the biprism. Such that two virtual images S_1 and S_2 of the slit S are formed. So light appears to be coming from virtual images S_1 and S_2 . These virtual images act as coherent sources. If a screen is placed in front of them, interference fringes can be obtained.



Let $2d$ be the distance between the two coherent sources S_1 and S_2 . Suppose a screen is placed at a distance D from the slit S . Draw perpendicular SC on the screen. Then C is equidistant from S_1 and S_2 . Hence 'C' will be point of maximum intensity.

Let us consider any point P on the screen distant x from C . Join S_1P and S_2P . The two waves have a path difference $S_2P - S_1P$ in reaching 'P'. Now from figure:

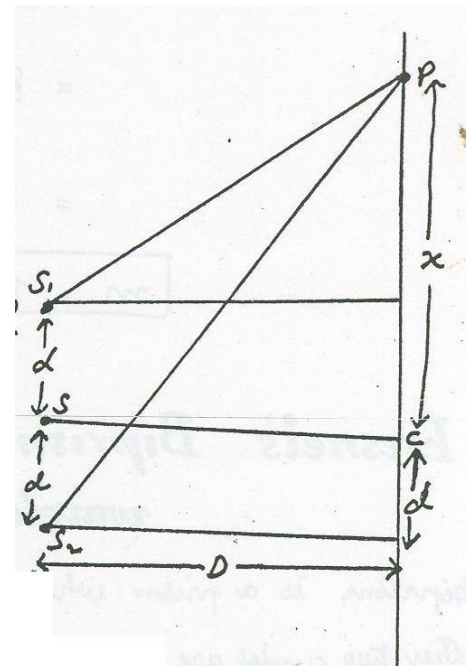
$$(S_2P)^2 = D^2 + (x + d)^2 \quad \text{--- (1)}$$

$$(S_1P)^2 = D^2 + (x - d)^2 \quad \text{--- (2)}$$

Subtracting (1) and (2):

$$\begin{aligned} (S_2P)^2 - (S_1P)^2 &= (x + d)^2 - (x - d)^2 \\ \Rightarrow x^2 + d^2 + 2xd - x^2 - d^2 + 2xd &= 4xd \\ \Rightarrow (S_2P - S_1P)(S_2P + S_1P) &= 4xd \end{aligned}$$

Now $S_2P \approx S_1P = D$ (for field approximation)



$$(S_2P - S_1P)(D + D) = 4xd$$

$$\Rightarrow (S_2P - S_1P)2D = 4xd$$

$$\Rightarrow S_2P - S_1P = \frac{4xd}{2D}$$

$$\Rightarrow \text{Path Difference} = \frac{2xd}{D}$$

Condition for maxima:

The point 'P' will be a bright band if path difference is integral multiple of λ :

$$\text{Path Difference} = n\lambda \text{ where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \frac{2xd}{D} = n\lambda \Rightarrow x = \frac{n\lambda D}{2d}$$

This is the distance of n th bright band from 'C'. The distance of $(n - 1)$ th bright band from 'C' is

$$x' = \frac{(n - 1)\lambda D}{2d}$$

Fringe width 'w':

The fringe width or fringe spacing w is defined as "The distance between two consecutive bright or dark bands." So 'w' is given as

$$w = x - x'$$

$$\Rightarrow w = \frac{n\lambda D}{2d} - \frac{(n - 1)\lambda D}{2d} = \frac{\lambda D}{2d} [n - (n - 1)]$$

$$\Rightarrow w = \frac{\lambda D}{2d} (n - n + 1)$$

$$\Rightarrow w = \frac{\lambda D}{2d}$$

$$\Rightarrow \lambda = \frac{2dw}{D}$$

So knowing, d and D , We can determine the unknown wavelength λ .

Diffraction

Diffraction and the Wave Theory of Light

Bending of light around the edges of an obstacle or an opening and spread into geometrical shadow of the obstacle or opening is known as diffraction of light waves.

The diffraction is the most prominent when the size of an opening or edges of obstacle is of the order of wavelength of waves. Since wavelength of light is very small (about 10^{-7}m), therefore light waves are diffracted by very tiny particles, sharp edges and very narrow opening.

Theoretical Explanation

Diffraction of light can be explained on the basis of Huygen principle. According to this principle:

Every point on a wavefront can be considered as a secondary source.

- When a wavefront of monochromatic light reaches an opening then a part of it falls on a slit. This part of wavefront will act as a secondary source and wavelets originating from this source also reach in the shadow region where light is not incident directly. This is in fact called diffraction of light.
- In the slit or opening, there will be infinite points which act as point sources and all of them will be infinite points which act as point sources and all of those will be phase coherent. Therefore diffraction of light can also be regarded as superposition of infinite phase coherent waves.

Comparison of Interference and Diffraction

Similarities

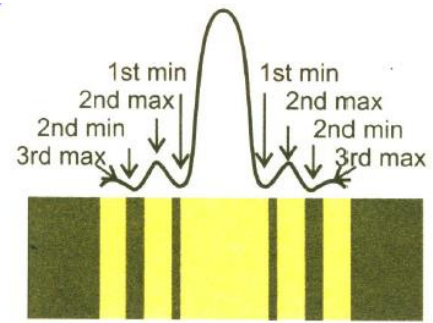
1. Both are typical wave phenomenon and occur due to superposition of phase coherent waves.
2. Minima and maxima of intensity are observed due to both interference and diffraction.

Differences

1. Interference occurs due to superposition of two phase coherent waves, while diffraction occurs due to superposition of infinite number of phase coherent waves
2. Large number of maxima and minima can be observed in interference but only a few maxima and minima are observed in diffraction.
3. In interference, maxima and minima are of equal widths and maxima have equal brightness (intensity). But in diffraction, maxima and minima are not of equal size and also their brightness (intensity) is different.

Diffraction by a Single Slit

Consider the diffraction pattern of plane waves of light of wavelength λ that are diffracted by a single long, narrow slit of width a . When the diffracted light reaches viewing screen C , waves from different points within the slit undergo interference and produce a diffraction pattern of bright and dark fringes on the screen.



Central maxima in single slit diffraction

The central bright fringe in diffraction pattern due to the reason that the Huygens wavelets from all points in the slit travel about the same distance to reach the center of the pattern and thus are in phase and interfere constructively to give central bright fringe.

First minima in single slit diffraction

Consider the first dark fringe, at point P_1 on screen. To explain it, we divide the slit into two zones of equal widths $\frac{a}{2}$ and the rays are extended to point P_1 from upper boundary of each zone. The light ray r_1 and a r_2 cancel each other when they arrive at P_1 .

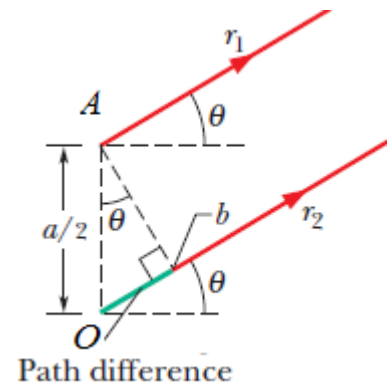
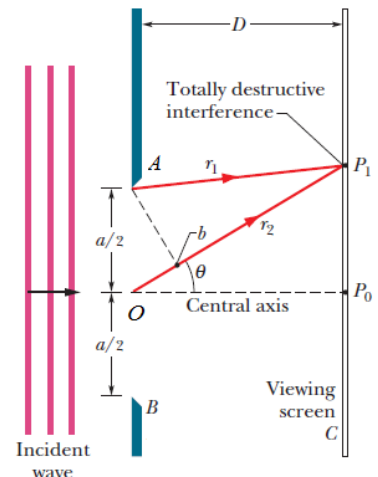
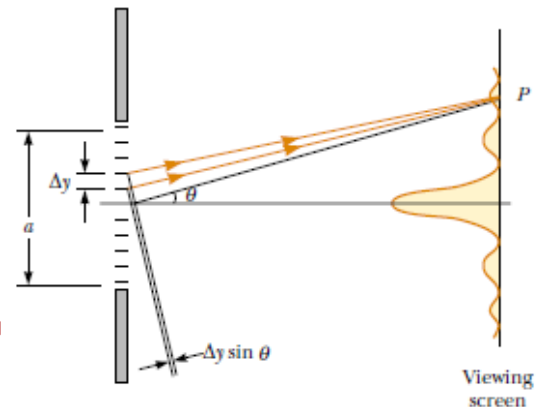
The wavelets of the pair of rays r_1 and r_2 are in phase within the slit because they originate from the same wavefront passing through the slit. However, to produce the first dark fringe they must be out of phase by $\frac{\lambda}{2}$ when they reach P_1 .

In order to determine the path difference graphically between to rays, draw perpendicular from point A on ray r_2 at point b . From figure:

$$\begin{aligned}
 |OP_1| &= |Ob| + |bP_1| \\
 \Rightarrow r_2 &= |Ob| + r_1 \\
 \Rightarrow r_2 - r_1 &= |Ob| \\
 \Rightarrow \text{Path Difference} &= |Ob| \quad \text{--- (1)}
 \end{aligned}$$

In right triangle ΔAOb :

$$\begin{aligned}
 \sin \theta &= \frac{|Ob|}{|OA|} \\
 \Rightarrow |Ob| &= |OA| \sin \theta
 \end{aligned}$$



$$\Rightarrow |Ob| = \frac{a}{2} \sin \theta$$

Equation (1) takes the form:

$$\text{Path Difference} = \frac{a}{2} \sin \theta \quad \text{--- (2)}$$

For first minima:

$$\text{Path Difference} = \frac{\lambda}{2} \quad \text{--- (3)}$$

Comparing equation (1) and (2), we have:

$$\frac{a}{2} \sin \theta = \frac{\lambda}{2}$$

$$\Rightarrow a \sin \theta = \lambda$$

Second minima in single slit diffraction

In order to determine the condition for second dark fringe at point P_2 on screen C, we divide the slit into four zones of equal widths $\frac{a}{4}$ and the rays are extended to point P_2 from upper boundary of each zone. The light ray r_1, r_2, r_3 and r_4 cancel each other when they arrive at P_2 . For this the “path difference between r_1 and r_2 ”, “path difference between r_2 and r_3 ” and “path difference between r_3 and r_4 ” must be equal to $\frac{\lambda}{2}$.

Graphically, the path difference between two consecutive rays will be

$\frac{a}{4} \sin \theta$. Therefore:

$$\frac{a}{4} \sin \theta = \frac{\lambda}{2}$$

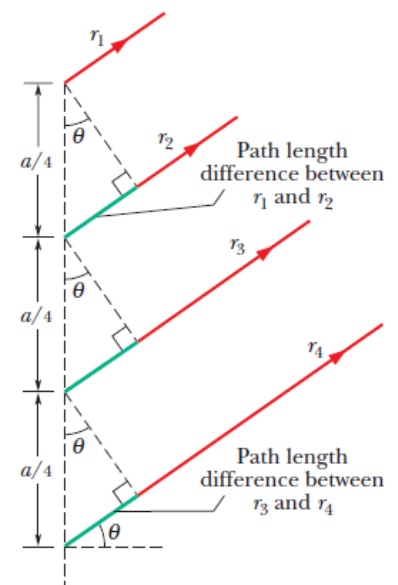
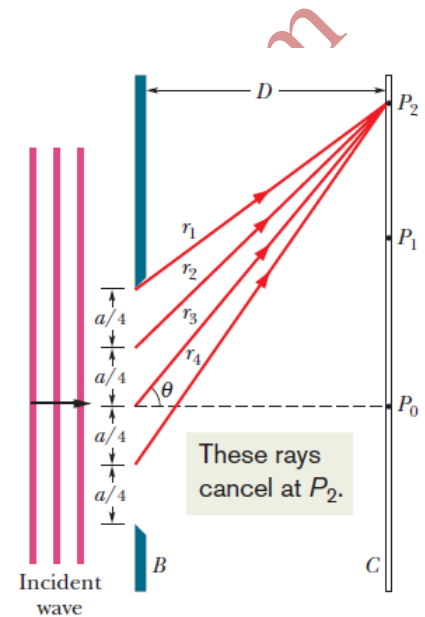
$$\Rightarrow a \sin \theta = 2\lambda$$

Generalization for All Minima

In the similar way, we could now continue to locate dark fringes in the diffraction pattern by splitting up the slit into more zones of equal width. We would find that the dark fringes above and below the central axis can be located with the general equation:

$$a \sin \theta = m\lambda$$

where m is integer. The above equation is called diffraction equation.



Intensity in Single-Slit Diffraction, Qualitative Analysis

In order to find out an expression for the intensity I of the pattern as a function of θ , we divide the slit into N zones of equal widths Δx . Consider the resultant at an arbitrary point P on the screen makes an angle θ to the central axis.

Initially, we shall determine E_θ of the electric component of the resultant wave at P . Then, the intensity of the light at P is determined by using fact that E_θ proportional to the square of that amplitude.

To find E_θ , we need the phase relationships among the arriving wavelets. The wavelets arriving at point P have different phases because they travel different distances to reach P . The phase difference between wavelets from adjacent zones is given by:

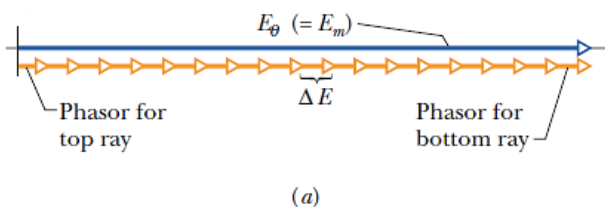
$$(\text{Phase Difference}) = \frac{2\pi}{\lambda} (\text{Path Difference})$$

$$\Rightarrow \Delta\phi = \frac{2\pi}{\lambda} (\Delta x \sin \theta) \quad \text{--- (1)}$$

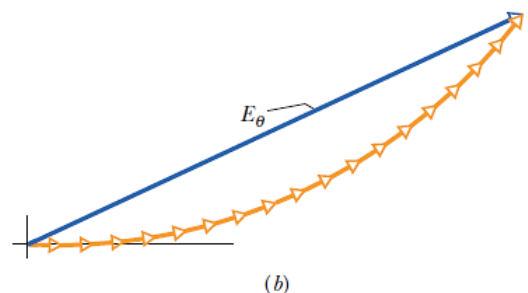
We assume that the wavelets arriving at P all have the same amplitude ΔE . To find the amplitude E_θ of the resultant wave at P , we add the amplitudes ΔE via phasors. To do this, we construct a diagram of N phasors, one corresponding to the wavelet from each zone in the slit.

Central Maximum

For point P_0 at $\theta = 0$, Eq. (1) tells us that the phase difference $\Delta\phi$ between the wavelets is zero, i.e., all wavelets arrive in phase at point P_0 . Figure (a) is the corresponding phasor diagram. All adjacent phasors represent wavelets from adjacent zones and are arranged head to tail. The amplitude E_θ of the net wave at P_0 is the vector sum of these phasors. This arrangement of the phasors turns out to be the one that gives the greatest value for the amplitude E_θ . We call this value E_m ; that is, E_m is the value of E_θ for $\theta = 0$.



We next consider a point P that is at a small angle θ to the central axis. Figure (b) shows the corresponding phasor diagram. The phasors are arranged head to tail, but now there is an angle $\Delta\phi$ between adjacent phasors. The amplitude E_θ at this new point is still the vector sum of the

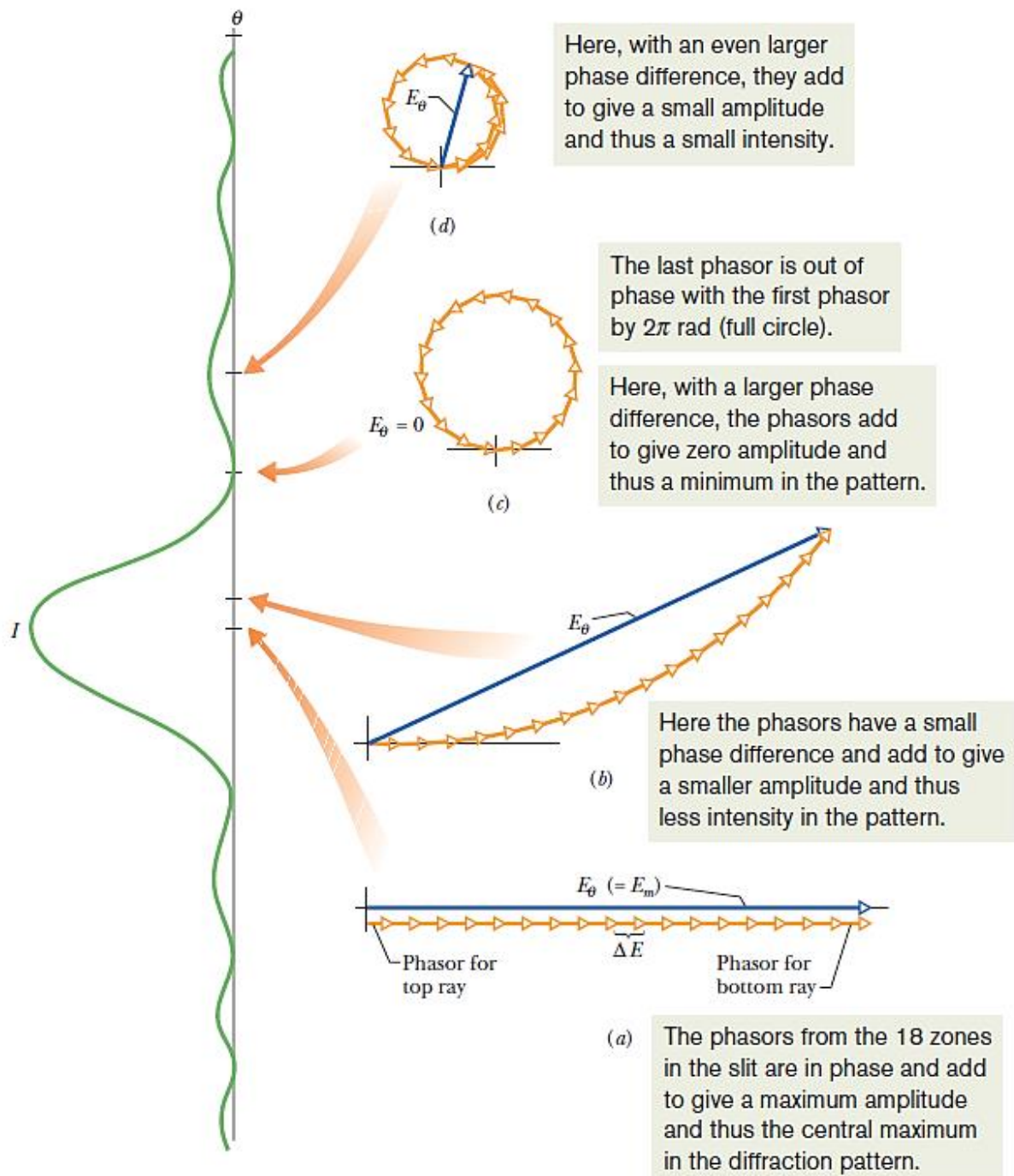


First Minimum

(c)

(d)

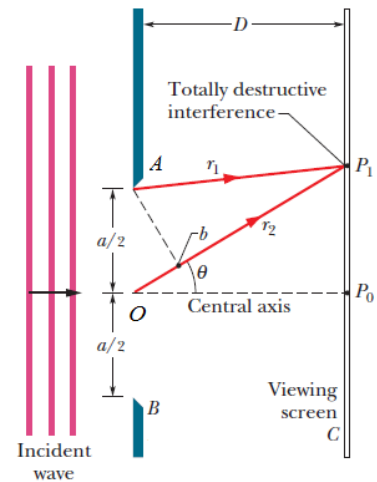
If we increase u a bit more, the resulting shrinkage of the coil decreases E_θ , which means that the intensity also decreases. When θ is increased enough, the head of the last phasor again meets the tail of the first phasor. We have then reached the second minimum.

Qualitative Analysis of Single Slit Diffraction (Overview)

Single Slit Diffraction, Quantitative Analysis

A slit is usually a rectangular aperture of very small width as compare to its length. Suppose parallel rays of monochromatic light of wavelength λ are normally incident on a slit of width a as shown in the figure.

The plane wavefront will reach at slit on all points simultaneously. According to Huygen's Principle, every point of unobstructed point of a wavefront will act as a secondary source and all of them will be phase coherent. The rays or wave trains diffracted parallel to each other can be focused at a point on screen with the help of convex lens. For this purpose, position of the lens is so adjusted that the screen will coincide its focal plane.



- (i)** We first consider an element of length 'dl' at the center O of slit.

If dE_0 be electric field component of light wave reaching point P from this element 'dl' at instant t, then

$$dE_o = \frac{A dl}{r} \sin(\omega t - kr)$$

where r is distance covered by waves in reaching point P, k is wave number and A is proportionality constant.

- (ii) Now consider another element of length dl at a distance $l = \frac{a}{2}$ below the center O. Let dE_- be electric field component at point P due to the waves from this element at instant t , then

$$dE_- = \frac{A dl}{r} \sin[\omega t - k(r + \Delta)]$$

Where $\Delta = \lambda \sin \theta = \frac{a}{2} \sin \theta$ and θ is the angle parallel rays diffracted from slit.

- (iii) Similarly if dE_+ is the electric field component due to waves reaching at point P from another length element dl at a distance $l = \frac{a}{2}$ above center O at instant t , then

$$dE_+ = \frac{A dl}{r} \sin[\omega t - k(r - \Delta)]$$

If dE is the net electric field due to both elements then

$$dE = dE_+ + dE_-$$

$$\Rightarrow dE = \frac{A dl}{r} \sin[\omega t - k(r - \Delta)] + \frac{A dl}{r} \sin[\omega t - k(r + \Delta)]$$

$$\Rightarrow dE = \frac{A dl}{r} [\sin(\omega t - kr + k\Delta) + \sin(\omega t - kr - k\Delta)]$$

$$\Rightarrow dE = \frac{A dl}{r} \cdot 2 \sin\left(\frac{\omega t - kr + k\Delta + \omega t - kr - k\Delta}{2}\right) \cos\left(\frac{\omega t - kr + k\Delta - \omega t + kr + k\Delta}{2}\right)$$

$$\Rightarrow dE = \frac{2A dl}{r} \sin\left(\frac{2\omega t - 2kr}{2}\right) \cos\left(\frac{2k\Delta}{2}\right)$$

$$\Rightarrow dE = \frac{2A dl}{r} \sin(\omega t - kr) \cos(k\Delta)$$

$$\Rightarrow dE = \frac{2A}{r} \sin(\omega t - kr) \cos(kl \sin \theta) dl$$

The net electric field component “E” at point P due to all waves originating from slit can be obtained by integrating above expression from 0 to $\frac{a}{2}$

$$E = \int_0^{\frac{a}{2}} dE = \int_0^{\frac{a}{2}} \frac{2A}{r} \sin(\omega t - kr) \cos(kl \sin \theta) dl$$

$$\Rightarrow E = \frac{2A}{r} \sin(\omega t - kr) \int_0^{\frac{a}{2}} \cos(kl \sin \theta) dl$$

$$\Rightarrow E = \frac{2A}{r} \sin(\omega t - kr) \left| \frac{\sin(kl \sin \theta)}{k \sin \theta} \right|_0^{\frac{a}{2}}$$

$$\Rightarrow E = \frac{2A \sin(\omega t - kr)}{r (k \sin \theta) a} \sin\left(k \frac{a}{2} \sin \theta\right) a$$

$$\Rightarrow E = \frac{Aa \sin\left(\frac{1}{2} ka \sin \theta\right)}{r \frac{1}{2} ka \sin \theta} \sin(\omega t - kr)$$

If E_θ is the amplitude of resultant electric field component, then

$$E_\theta = \frac{Aa}{r} \frac{\sin\left(\frac{1}{2} ka \sin \theta\right)}{\frac{1}{2} ka \sin \theta} \quad \text{--- (1)}$$

Let

$$\frac{1}{2} ka \sin \theta = \alpha \quad \text{--- (2)}$$

Equation (1) takes the form:

$$E_\theta = \frac{Aa \sin \alpha}{r \alpha}$$

If I_θ is resultant intensity at point “P” then

$$I_\theta \propto E_\theta^2$$

$$\Rightarrow I_\theta \propto \left(\frac{Aa \sin \alpha}{r \alpha} \right)^2$$

The diffraction pattern can be observed with a circular aperture by focusing parallel diffracted rays at a point on a screen with the help of a convex lens. Position of lens is so adjusted that the screen coincides with its focal plane. The diffraction pattern consists of a central maxima of circular shape surrounded by minima and secondary maxima in the form of circular rings. The intensity of secondary maxima decreases with the increase in their order and therefore few bright rings are observed around the central maxima.

$$\sin \theta = 1.22 \frac{\lambda}{D}$$

$$\Rightarrow \theta = \sin^{-1} \left(1.22 \frac{\lambda}{D} \right)$$

The above relation shows that if diameter of aperture is reduced, the width of central maxima will increase. Hence if an object is a point, its image formed by the lens will not be a point image due to diffraction because the central maxima will have finite width.

Resolving Power and Rayleigh Criteria

Images of two point object are said to be just resolved if the separation between them is such that the principal maximum of image of one falls on first minimum of image of the other and vice versa.

When the images are just resolved, the angular separation between them is called resolving power of lens and minimum angle of resolution denoted by θ_p . Thus it is given by:

$$\theta_R = \sin^{-1} \left(1.22 \frac{\lambda}{D} \right)$$

$$\sin \theta_R = 1.22 \frac{\lambda}{D}$$

But θ_R is very small, therefore $\sin \theta_R \approx \theta_R$

$$\Rightarrow \theta_R = 1.22 \frac{\lambda}{D}$$

Smaller the value of θ_R , greater will be the resolving power.

Double Slit Interference and Diffraction Combined

Suppose that parallel rays of monochromatic light of wavelength λ are normally incident on an arrangement of two slits each of width 'a'. Let 'd' be the separation between their centers, then plane wavefronts will reach simultaneously on all parts of two slits as shown in the figure.

According to Huygen Principle, every point on the unobstacated part of wavefront will act as a secondary source and all such sources will be phase coherent. The rays of light diffracted parallel to each other can be focused at a point on the screen with the help of convex lens. For this purpose, the position of lens is so adjusted that screen coincides with its focal plane.

Since every point on the two slits can be considered as a secondary source, therefore there will be superposition of infinite number of phase coherent waves at a point (say P) on screen. This will result in diffraction. At the same time, each slit can also be treated as a single source, then there will be superposition of two coherent waves which will result in interference. Thus both interference and diffraction can be studied in this arrangement.

If there were only diffraction, then their resultant intensity at P would be:

$$I_\theta = I_{m,diff} \left(\frac{\sin \alpha}{\alpha} \right)^2 \quad \text{--- (1)}$$

where

$$\alpha = \frac{\pi a \sin \theta}{\lambda} \quad \text{--- (2)}$$

If there were only interference then resultant at P would be:

$$I_\theta = I_{m,int} \cos^2 \beta \quad \text{--- (3)}$$

Where

$$\beta = \frac{\pi d \sin \theta}{\lambda} \quad \text{--- (4)}$$

Since both interference and diffraction are taking place in this case, therefore combining equation (1) and (3), given by:

$$I_\theta = I_m \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos^2 \beta \quad \text{--- (5)}$$

Condition for Maxima and Minima due to Diffraction

If $\theta = 0$

$$\Rightarrow \alpha = 0$$

$$\lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} = 1$$
$$\sin^2 \alpha = 1$$

$$\Rightarrow \alpha = (2m + 1) \frac{\pi}{2}$$

where $m = \pm 1, \pm 2, \pm 3 \dots$

$$\frac{\pi a \sin \theta}{\lambda} = (2m + 1) \frac{\pi}{2}$$

$$\Rightarrow a \sin \theta = (2m + 1) \frac{\lambda}{2}$$

$$\sin^2 \alpha = 1$$

$$\Rightarrow \alpha = m\pi$$

Where $m = \pm 1, \pm 2, \pm 3 \dots$

$$\frac{\pi a \sin \theta}{\lambda} = m\pi$$

$$\Rightarrow a \sin \theta = m\lambda$$

Condition of Maxima and Minima due to Interference**Maxima**

They will be observed at those points on screen where

$$\cos^2 \beta = 1$$

$$\Rightarrow \alpha = n\pi$$

$$\text{where } n = \pm 1, \pm 2, \pm 3 \dots$$

Putting value in (ii), we have:

$$\frac{\pi a \sin \theta}{\lambda} = n\pi$$

$$\Rightarrow a \sin \theta = n\lambda$$

Minima

They will be observed at those points on screen where

$$\cos^2 \beta = 0$$

$$\Rightarrow \beta = (2n + 1) \frac{\pi}{2}$$

$$\text{where } n = \pm 1, \pm 2, \pm 3 \dots$$

Putting value in (ii), we have:

$$\frac{\pi a \sin \theta}{\lambda} = (2n + 1) \frac{\pi}{2}$$

$$\Rightarrow a \sin \theta = (2n + 1) \frac{\lambda}{2}$$

Only Interference by Double Slit

If we want to observe only interference pattern by double slit arrangement, then following condition must be satisfied $a \ll d$ i.e., the width of slit is very very small as compare to the separation between slits. In this case quite a large number of interference maxima will be contained in principle diffraction maxima and interference maxima in the middle region will have equal intensity.

UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2017

Physics

Paper: A

Time Allowed: 3 Hours

Maximum Marks: 50

Note: Attempt any five questions in all, selecting at least two questions from each part. All questions carry equal marks.

SECTION-I

Q # 1. (a) What is special theory of relativity? State its basic postulates. Also apply it to calculate relativistic momentum of a system of two particles.

(b) The most energetic proton detected in the cosmic rays coming to earth from space had kinetic energy of $3 \times 10^{20} \text{ eV}$. What were the proton's Lorentz factor γ and speed v (both relative to the ground based detector)?

Q # 2. (a) Describe three Kepler's laws about the motions of planets and satellites in detail.

(b) A satellite, orbiting at an altitude of two earth's radii above its surface, launches an equipment canister of mass m towards the earth's center with a speed of 525 m/s. With what speed do the canister enter the earth's atmosphere (a distance of $h = 100 \text{ km}$ above its surface).

Q # 3. (a) State and prove the Bernoulli's equation for an ideal fluid.

(b) Castor oil of density $0.96 \times 10^3 \text{ kg/m}^3$ at room temperature is forced through a pipe of circular cross section by a pump that maintains a gauge pressure of 950 Pa. The pipe has a diameter of 2.6 cm and a length of 65 cm. The castor oils emerging from the free end of the pipe at atmospheric pressure is collected. After 90 s, a total of 1.23 kg has been collected. What is the coefficient of viscosity of the castor oil at this temperature?

Q # 4. (a) What is a vector field? Explain curl of a vector field, give its physical significance and hence prove that $\text{Curl } \vec{V} = \vec{\nabla} \times \vec{V}$.

(b) Prove that if the vector is the gradient of a scalar function then its line integral round a closed curve is zero.

Q # 5. (a) Define center of mass of a system of many particles. Find velocity and acceleration of center of mass and show that in the absence of external force, center of mass moves with uniform velocity.

(b) A canon whose mass M is 1300 kg fires a 72 kg ball in a horizontal direction with a speed v of 55 m/s relative to the canon. The canon is mounted so that it can recoil freely. (i) What is the velocity of the recoiling canon with respect to earth? (ii) What is the initial velocity of the ball with respect to the earth?

SECTION-II

Q # 6. (a) Prove that simple harmonic motion can be described as the projection of uniform circular motion along the diameter of a circle. Find equations of displacement, velocity and acceleration of projection. Also prove that circular motion can be regarded as combination of two identical simple harmonic motions.

(b) A pendulum is formed by pivoting a long thin rod of length L and mass m about a pivot on the rod that is at distance d above the center of rod, (i) find small amplitude period of this pendulum in terms of d , L , m , and g (ii) Show that the period has a minimum value when $d = 0.289L$.

Q # 7. (a) What is Doppler effect? Find expressions for apparent frequencies when (i) Observer is in motion and source is at rest. (ii) Source is in motion and observer is at rest.

(b) A sound wave of intensity $1.6 \text{ } \mu\text{W/cm}^2$ passes through a surface of area 4.70 cm^2 . How much energy passes through the surface in one hour?

Q # 8. (a) What is plane polarization of light? Explain polarization of light by reflection and derive an expression of Brewster law.

b) Light traveling in water with n_i is incident on plane of a glass plate with $n_g = 1.53$. At what polarizing angle reflected light is completely polarized.

Q # 9. (a) What is diffraction grating and grating element? Derive an expression for its dispersion and resolving power.

(b) Calculate approximately the relative intensities of the secondary maxima in the single slit Fraunhofer diffraction pattern.

Q # 10. Write note on any two of the following:

- Traveling waves
- Michelson's interferometer
- Rocket equation

UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2016

Physics

Paper: A

Time Allowed: 3 Hours

Maximum Marks: 50

Note: Attempt any five questions in all, selecting at least two questions from each part. All questions carry equal marks.

SECTION-I

Q # 1. (a) What are direction cosines? Prove that $l^2 + m^2 + n^2 = 1$.

(b) Find the Cartesian cosines of (2, -1, 2)

Q # 2. (a) What is friction? Discuss in detail static and kinetic friction. Which is greater?

(b) What is the greatest acceleration that can be generated by runner if the coefficient of static friction between shoes and road is 0.95.

Q # 3. (a) What is rotational K.E. Derive its formula?

(b) Derive Rotational inertia of hollow cylinder.

Q # 4. (a) Define pressure of a fluid. Derive an expression for pressure gradient between the point inside a fluid and point at its surface. Determine variation in pressure in atmosphere.

(b) A stream of water from a faucet necks down as it falls. The cross section area A_0 is 1.2 cm^2 and that of A is 0.35 cm^2 . The two levels are separated by a vertical distance $h = 35 \text{ mm}$. At what rate does water flow from the tap.

Q # 5. (a) Write short notes on any two of the following.

- a) Relativity of length by the consideration of special theory of relativity.
- b) Oblique collisions
- c) Artificial satellite.

SECTION-II

Q # 6. (a) What is torsional oscillator? Show that its motion is SHM and find its time period

(b) A uniform disc is pivoted at its rim. Find its period for small oscillations and the length of equivalent simple pendulum

Q # 7. (a) Compute wave speed using mechanical analysis, also calculate transverse velocity of the particle.

(b) A vibrator sets the string into motion at a frequency of 120 Hz. The string has a length of $l = 1.2 \text{ m}$ and linear density 1.2 g/m . To what value, must the tension be adjusted to give four loops?

Q # 8. (a) Explain the interference in thin film.

(b) A water film with $n = 1.33$ in air is 320 nm thick. If it is illuminated with white light at normal incidence, what color will it appear to be in reflected light?

Q # 9. (a) Discuss circular aperture diffraction.

(b) What is dispersion and resolving power?

UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2015

Physics

Paper: A

Time Allowed: 3 Hours

Maximum Marks: 50

Note: Attempt any five questions in all, selecting at least two questions from each part. All questions carry equal marks.

SECTION-I

Q # 1. (a) State and prove Gauss's Divergence theorem.

(b) Prove that the magnitude of vector product of two vectors is equal to the area of parallelogram.

Q # 2. (a) What is conical pendulum? Calculate its period of revolution.

(b) A conical pendulum is formed by attaching a 53g pebble to 1.4 m string. The pebble swings around in a circle of radius 25cm: (i) What is the period of the pebble? (ii) What is acceleration? (iii) What is the tension in string?

Q # 3. (a) Calculate rotational inertia of solid cylinder about an axis passing through center and perpendicular to its axis of symmetry.

(b) A disk spins on horizontal shaft mounted in bearings with angular speed of $\omega_1 = 84 \text{ rad/s}$. The entire disk and shaft assembly are placed on turn table rotating about vertical axis at $\omega_2 = 43 \text{ rad/s}$ counter clock wise as we viewed it from above. Describe the rotation of disk as seen by observer in room.

Q # 4. (a) What is the equation of continuity? Prove it. What result you get from it?

(b) Crew members to escape from a damaged submarine 112 m below the surface. How much force must be applied to pop-out hatch which is 1.22 m by 0.590 m to push it out?

Q # 5. Write short note on any two of the following:

- Rocket Equation
- Mosely's Law
- Pseudo Forces

SECTION-II

Q # 6. (a) Derive wave equation.

(b) Calculate the speed of a transverse wave in a string of length 2.15 m and a mass 62.5 g under the tension of 487 N.

Q # 7. (a) What are standing waves? Prove that the frequency of standing waves is quantized.

(b) Speed of a wave in a string is 172 ms^{-1} when tension is 123 N. To what value must the tension be increased in order to increase the wave speed to 180 ms^{-1} .

Q # 8. (a) What is diffraction? Explain single slit diffraction in detail and hence derive diffraction equation.

(b) A slit of width "a" is illuminated by white light. For what value "a" does the first minima for red light having wavelength of 650 nm fall $\theta=15^\circ$.

Q # 9. (a) What is meant by polarization of light? How you can get a plane polarized light by polarizing sheet. Show that in a circular polarization, the average intensity of light is proportional to the square of maximum amplitude of light waves?

(b) We wish to see a glass plate with $n=1.5$ in air as a polarizer. Find the polarizing angle and the angle of refraction.

UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2014

Physics

Paper: A

Time Allowed: 3 Hours

Maximum Marks: 50

Note: Attempt any five questions in all, selecting at least two questions from each part. All questions carry equal marks.

SECTION-I

Q # 1. (a) Explain Gradient of a scalar and show that for a scalar function ϕ : $\text{Grad } \phi = \vec{\nabla} \phi$

(b) A particle moves along curve $x = e^{2t}, y = \cos 2t, z = \sin 2t$ where t is the time, determine its velocity and acceleration at any time.

Q # 2 (a). What is conical pendulum? Show that the period of motion of: conical pendulum does not depend on mass.

(b) A block is at rest on an incline plane making an angle θ with the horizontal, as the angle of inclination is raised, it is found that slipping just begins at an angle of inclination $\theta = 15^\circ$. What is the coefficient of static friction between block and incline?

Q # 3 (a). Define the parallel axis theorem. Find the rotational inertia of a solid rod of length L and mass M about an axis which is perpendicular to length and passes through its center.

(b) The angular speed of an automobile engine is increased from 1170 rev/min to 2880 rev/min in 12.6 s. Find the angular acceleration in rev/min^2 . How many revolutions does the engine make during this time?

Q # 4 (a). Find the Gravitational force on a point mass when it is outside and inside a uniformly dense spherical shell.

(b) It is desired to place a communication satellite into orbit so that it remain fixed above a given spot on the equator of the rotating earth, what is the height above the earth of such an orbit? $M_E = 5.98 \times 10^{24} \text{ kg}$.

Q # 5 (a). What are postulates of special theory of relativity, write note on:

- Time Dilation
- Length Contraction

(b) The mean lifetime of muons stopped in a lead block in the laboratory is measured to be $2.20 \mu\text{s}$. The mean life time of high speed muons in a burst of cosmic rays observed from the earth is measured to be $16.0 \mu\text{s}$. Find the speed of those cosmic ray muons.

Q # 6 (a). Write short note on two of the following:

- Work Energy theorem
- Pseudo Forces
- Interference from Thin Films

SECTION-II

Q # 7. (a) Define Simple Harmonic Motion. Write down simple harmonic equation and solve it for the displacement, time period and acceleration.

(b) In an electric shaver, the blade moves back and forth over a distance of 2 mm. The motion is simple harmonic with frequency 120 Hz. Find (a) amplitude, (b) maximum blade speed, (c) maximum acceleration of blade.

Q # 8. (a) Define travelling waves. Derive the expression for velocity of a wave in a string by mechanical analysis.

(b) What are the three lowest frequencies for standing waves on a wire 9.88 m long having mass of 0.107 kg, which is stretched under a tension of 236 N.

Q # 9. (a) What are Beats? Show analytically that number of beats per second is equal to the difference of frequencies of two oscillators.

(b) A string of violin is a little too taut. Four beats per second are heard when it is sounded together with a tuning fork that is vibrating actually at the pitch of concert A (440 Hz). What is the period of the violin string vibration?

Q # 10 (a). What is meant by diffraction of light? Derive condition for maxima and minima of diffraction due to single slit.

(b) A converging lens is of 32 mm in diameter. (a) What angular separation must two distant point objects have to satisfy Rayleigh's criterion? Assume $\lambda = 550 \text{ nm}$.

UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2013

Physics

Paper: A

Time Allowed: 3 Hours

Maximum Marks: 50

Note: Attempt any five questions in all, selecting at least two questions from each part. All questions carry equal marks.

SECTION-I

Q # 1. (a) State and prove Stoke's theorem.

(b) Evaluate the following integral for close surface:

$$\oint_S \vec{r} \cdot \hat{n} ds$$

Q # 2 (a). Calculate rotational inertia of a disc about an axis passing through a center and perpendicular to its plane.

(b). Which is greater, angular momentum of earth associated with the rotation about its own axis or angular momentum of earth associated with its orbital motion around sun. The distance between sun and earth is $1.5 \times 10^{11} m$.

Q # 3 (a). What is elasticity? Discuss in detail Young's Modulus and Bulk Modulus.

(b). In a hurricane air density 1.2 kg/m^3 is blowing over the roof of a house at a speed of 110 km/h . (a) What is the pressure difference between inside and outside that tends to lift the roof. (b) What should be the lifting force on a roof of area 93 m^2 .

Q # 4 (a). What is Lorentz Transformation? Derive it mathematically. When Lorentz transformation reduces to Galilean transformation.

(b) Find speed parameter β and Lorentz factor γ for a particle whose K.E. = 10 MeV if particle is electron.

Q # 5 (a). Discuss the elastic collision in two dimensions. How will you find the velocities of the two particles after collision having mass m_1 and m_2 . The initial velocity of the first particle is V_1 and second particle of mass m_2 is at rest. The first particle scattered at an angle θ and second particle makes an angle ϕ along the original direction.

(b) A gas molecule having speed of 322 ms^{-1} collides elastically with another molecule of the same mass which is initially at rest. After collision the first molecule moves at an angle of 30° to its initial direction. Find the speed of each molecule after the collision.

Q # 6. Write a short note on any two of the following:

- (i) Equation of continuity of a fluid
- (ii) Physical Pendulum
- (iii) $E = mc^2$

SECTION-II

Q # 7 (a). Define interference of waves. Derive the formula for the condition of constructive interference.

(b) A string can vibrate with the frequency of 440 Hz and has a length of 0.34 m . What are the three longest wavelengths of the resonance of the string?

Q # 8 (a). What is Michelson Interferometer? Explain its construction, working and uses.

(b) In a double slit experiment with blue light of wavelength 512 nm the slits are 1.2 mm apart and screen is 5.4 m from the slits. How far apart are the bright fringes as seen on the screen.

Q # 9 (a). What is torsional pendulum? Prove that its motion is SHM. Find its time period.

(b) The period of disc of radius 10.2 cm executing small oscillations about a pivot at its rim is measured to be 0.784 s . Find the value of g .

Q # 10 (a). What is diffraction grating? Explain how grating diffract light. Find its resolving power.

(b) A grating has 9600 lines uniformly spread over a width of 3 cm and is illuminated by light from mercury vapor discharge. (i) What is expected dispersion in the third order in the vicinity of the intense green line of wave length $\lambda = 546 \text{ nm}$. (ii) What is the resolving power of this grating in the fifth order?

UNIVERSITY OF SARGODHA

B.A. / B.Sc. 1st Annual Exam 2012

Physics

Paper: A

Time Allowed: 3 Hours

Maximum Marks: 50

Note: Attempt any five questions in all, selecting at least two questions from each part. All questions carry equal marks.

SECTION-I

Q # 1 (a). Explain Divergence of a vector field. Show that $\text{div } \vec{V} = \vec{\nabla} \cdot \vec{V}$.

(b). If $\theta(x, y, z) = 3x^2y - y^3z^2$, find $\text{grad } \theta$ at point (1, -2, 1).

Q # 2 (a). What is rotor? Find an expression for tangential speed of a man on the wall of the rotor.

(b). A circular curve of highway is designed for traffic moving at 60 km/h. If the radius of curve is 150 m, what is the correct angle of banking of the road?

Q # 3 (a). Discuss inelastic collision in one dimension of two particles which stick together after collision in Lab's frame.

(b). Two skaters A and B mass 83kg and 55 kg respectively collide and embrace in a completely inelastic collision.

Before collision skater A was moving east with velocity 6.4 Km/h and skater B with velocity 8.8 km/h towards north.

What is the velocity of couple after impact?

Q # 4 (a). Show that the force exerted by a uniform thin spherical sheet of mass M on a point mass 'm' when it is outside, is the same as if all the mass of the spherical sheet were concentrated at its center.

(b) What is the gravitational potential energy of the Moon-Earth system, relative to the potential energy at infinite separation?

Q # 5 (a). What is theory of relativity? Define its two basic postulates. Prove that $E = mc^2$, where symbols have usual meanings.

(b) What is the momentum of a photon moving at speed of $v = 0.86c$?

Q # 6. Write a note on any two of the following:

- (i) Escape velocity (integrating techniques)
- (ii) Rocket Equation
- (iii) Bernoulli equation
- (iv) work-energy Theorem

SECTION-II

Q # 7 (a). What is physical pendulum? Write down its equation of motion and find its time period. Also find the length of equivalent simple pendulum.

(b) Discuss briefly the Lissajous patterns.

Q # 8 (a). What is meant by interference of light? Describe the Young's double slit experiment and find the conditions of constructive and destructive interference.

(b) The double slit arrangement is illuminated by light of wavelength $\lambda = 546 \text{ nm}$, the slit are 0.12 mm apart, and the Screen on which the interference pattern appears is 55 cm away. What is angular position of the first minimum? What is the linear distance on screen between adjacent maximum m and m+1.

Q # 9 (a). Derive an equation of motion of travelling waves in one dimension (on a string).

(b) In an arrangement, a vibrator sets the string into motion at a frequency of 120 Hz. The string has a length $L = 1.2 \text{ m}$, and its linear mass density is 1.6 g/m . To what value must the tension be adjusted to obtain the pattern of motion having four loops?

Q # 10 (a). What is polarization? How the polarized light can be produced by reflection?

(b). Two polarizing sheets have their polarizing directions parallel so that the intensity I_m of the transmitted light is a maximum. Through what angle must either sheet be turned if the intensity is to drop by one half?

Mass of moon $m = 7.36 \times 10^{22} \text{ kg}$, distance between Earth and Moon $d = 3.82 \times 10^8 \text{ m}$.